|INCH-POUND|
MIL-M-28787/369
29 March 1991

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

CONVERTER, ANALOG/DIGITAL, KEY CODE CKA

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic 12-bit analog to digital converter module. This specification covers one class of module, class I as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specification and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

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3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 4.8515 failures per million operating hours for class I, when predicted in accordance with MIL-STD-1389.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be CKA.

3.6.2 Size. The module size shall be 1B Format A.

3.6.3 Weight. The module shall have a maximum weight of 60 grams.

3.6.4 Contact pins. The module shall have 80 contact pins positioned A1 through A40 and B1 through B40.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A, and a heat dissipation fin on increment B.

3.6.6 Depth. The module shall have no additional depth requirements.

3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

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TABLE I. Module electrical requirements.

Test cond	Parameter	Ref test table	Test type 1/-3/	T_A = 25 ± 5°C 4/		T_A = 0 ± 5°C 4/		T_A = 60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation, frame									
	initial	II	S	10	INF	10	INF	10	INF	MOhm
	end-of-life	II	S	1	INF	1	INF	1	INF	MOhm
A2-A18	Isolation, pin									
	initial	II	X	10	INF	10	INF	10	INF	MOhm
	end-of-life	II	X	1	INF	1	INF	1	INF	MOhm
B1-B16	Continuity	III	X	0 0	1 0	0.0	1.0	0.0	1.0	Ohm
B17-B31	Continuity	III	Q	0 0	0.1	0.0	0 1	0.0	0.1	Ohm
C	Power supply stress	IV	Q	---- Module shall pass all remaining tests ---						--
D1-D5	Breakdown current	V	S	-1	50	-1	50	-1	50	uA
E1-E12	Short circuit current	VI	S	-40	1	-40	1	-40	1	mA
F1,F4	Power supply current	VII	X	1	25	1	25	1	25	mA
F2,F5	Power supply current	VII	X	1	50	1	50	1	50	mA
F3,F6	Power supply current	VII	X	1	45	1	45	1	45	mA
G1-G3	Capacitance, filter	VIII	S	12	18	12	18	12	18	uF
G4-G9	Capacitance, input	VIII	Q	2	25	2	25	2	25	pF
H1-H3	Filter inductance	IX	S	30	50	30	50	30	50	uH
J	10 V divider	X	X	4 9	5.1	4.9	5.1	4.9	5.1	V
K1-K5	Input current, high	XI	X	-1	50	-1	50	-1	50	uA
L1-L5	Input current, low	XII	X	-50	1	-50	1	-50	1	uA
M1-M13	Output voltage, high	XIII	X	2.4	4 5	2.4	4.5	2.4	4.5	V
N1-N13	Output voltage, low	XIV	X	0 0	0 4	0.0	0.4	0.0	0.4	V

- See footnotes at end of table

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter	Ref test table	Test type 1/-3/	T_A = 25 ± 5°C 4/		T_A = 0 ± 5°C 4/		T_A = 60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
P1-P24	Leakage current	XV	X	-40	40	-40	40	-40	40	uA
Q	Reference voltage	XVI	X	9 9	10.1	9.9	10.1	9.9	10.1	V
R1	Conversion time	XVII	X	10	28	10	28	10	28	uS
R2	Conversion time	XVII	X	15	40	15	40	15	40	uS
S1-S12	Operating modes	XVIII	X	2 4	5.0	2 4	5.0	2.4	5.0	V
S13-S16	Operating modes	XVIII	X	0.0	0.4	0.0	0 4	0.0	0.4	V
S17-S24	Operating modes	XVIII	X	-40.0	40.0	-40 0	40.0	-40.0	40.0	uA
U1-U18	Supply sensitivity	XIX	Q	-1	1	-1	1	-1	1	LSB
V1-V4096	Missing code check	XX	X	0	0	0	0	0	0	codes
W1	ADC offset voltage	XXI	X	-1.5	2 5	-1.5	2.5	-2.0	3.0	LSB
W2	ADC offset voltage	XXI	X	-4.5	3.5	-4.5	3 5	-5.5	4.5	LSB
X	ADC gain error	XXII	X	-0 3	0.3	-0 3	0.3	-0.4	0.4	per-cent FSR
Y1	ADC f.s. endpoint	XXIII	X	4.9664	5.0264	4.9664	5.0264	4.9664	5.0264	V
Y2	ADC zero error	XXIII	X	-5 0264	-4.9664	-5.0264	-4.9664	-5.0264	-4.9664	V
Y3-Y60	ADC linearity error	XXIII	X	-0 5	0 5	-0.5	0.5	-1.0	1.0	LSB
AA1	ADC transition time	XXIV	S	0	450	0	450	0	450	nS
AA2-AA7	ADC transition time	XXIV	S	0	250	0	250	0	250	nS
AA8-AA13	ADC transition time	XXIV	Q	25	INF	25	INF	25	INF	nS
AA14-AA19	ADC transition time	XXIV	S	200	1000	200	1000	200	1000	nS
AA20-AA25	ADC transition time	XXIV	Q	0	250	0	250	0	250	nS
AA26	ADC transition time	XXIV	S	0	650	0	650	0	650	nS
AA27-AA32	ADC transition time	XXIV	S	0	250	0	250	0	250	nS
AA33-AA38	ADC transition time	XXIV	Q	25	INF	25	INF	25	INF	nS
BB1, BB2	ADC transition error	XXV	X	-0.5	0.5	-0.5	0.5	-0.5	0.5	LSB

- See footnotes at end of table

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter	Ref test table	Test type 1/-3/	T_A = 25 ± 5°C 4/		T_A = 0 ± 5°C 4/		T_A = 60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
CC	S/H input impedance	XXVI	X	1	INF	1	INF	1	INF	MOhm
EE	S/H acquisition time	XXVII	X	0.0	1 5	0 0	1.5	0.0	1.5	uS
FF	S/H hold droop rate	XXVIII	X	-0.5	0 5	-0.5	0.5	-0.5	0 5	V/S
GG1	I/A offset voltage	XXIX	X	-3.25	3.25	-4 5	4 5	-5.0	5.0	mV
GG2	I/A offset voltage	XXIX	X	-0.11	0 11	-0.16	0 16	-0.18	0 18	mV
HH1, HH2	I/A bias current	XXX	X	-50 5	50 5	-53 0	53 0	-54.0	54.0	nA
HH3 -	I/A offset current	XXX	X	-35.5	35 5	-38.0	38.0	-39.0	39 0	nA
JJ1	I/A PSS (G=1)	XXXI	S	75	INF	75	INF	75	INF	dB
JJ2	I/A PSS (G=10)	XXXI	S	95	INF	95	INF	95	INF	dB
JJ3	I/A PSS (G=100)	XXXI	S	105	INF	105	INF	105	INF	dB
JJ4	I/A PSS (G=1000)	XXXI	S	110	INF	110	INF	110	INF	dB
kk1	I/A CMRR (G=1)	XXXII	S	70	INF	70	INF	70	INF	dBV
KK2	I/A CMRR (G=10)	XXXII	S	90	INF	90	INF	90	INF	dBV
KK3	I/A CMRR (G=100)	XXXII	S	100	INF	100	INF	100	INF	dBV
kk4	I/A CMRR (G=1000)	XXXII	S	110	INF	110	INF	110	INF	dBV
LL1	I/A output swing	XXXIII	X	9.5	11 0	9.5	11 0	9.5	11.0	V
LL2	I/A output swing	XXXIII	X	-11 0	-9 5	-11 0	-9.5	-11.0	-9.5	V
MM1-MM3	I/A settling time	XXXIV	Q	0	30	0	30	0	30	uS
MM4	I/A settling time	XXXIV	Q	0	150	0	150	0	150	uS
MM5	I/A settling time	XXXIV	Q	0	22	0	22	0	22	uS
NN1-NN3	I/A input resistance	XXXV	Q	10	INF	10	INF	10	INF	MOhm
PP1	I/A gain accuracy	XXXVI	X	0.9985	1 0015	0.9984	1.0015	0.9984	1.0015	V/V
PP2	I/A gain accuracy	XXXVI	X	9.975	10.025	9.974	10.026	9.974	10.026	V/V
PP3	I/A gain accuracy	XXXVI	X	99.5	100 5	99.5	100 5	99.4	100.6	V/V
PP4	I/A gain accuracy	XXXVI	X	980.0	1020.0	977.0	1023.0	976.0	1024.0	V/V

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter	Ref test table	Test type 1/-3/	T_A = 25 ± 5°C 4/		T_A = 0 ± 5°C 4/		T_A = 60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
QQ1	I/A slew rate X1	XXXVII	X	3.0	INF	3 0	INF	3.0	INF	V/uS
QQ2	I/A slew rate X10	XXXVII	X	3.0	INF	3 0	INF	3.0	INF	V/uS
QQ3	I/A slew rate X100	XXXVII	X	3.0	INF	3.0	INF	3.0	INF	V/uS
QQ4	I/A slew rate X1000	XXXVII	X	0.5	INF	0.5	INF	0.5	INF	V/uS

1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/369-1. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram and pin assignments shall conform to figure 1.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 through 3 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25 degrees C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. The module shall be capable of conforming to the requirements in table I for power supply voltages of 5.0 V dc ± 10 percent, +15 V dc ± 10 percent and -15 V dc ± 10 percent, except that the limits for output short circuit current apply only for power supply voltages of 5.5 V dc ± 1 percent, +16.5 V dc ± 1 percent and -16.5 V dc ± 1 percent.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Caution. Do not insert or remove the module from the test fixture with power applied

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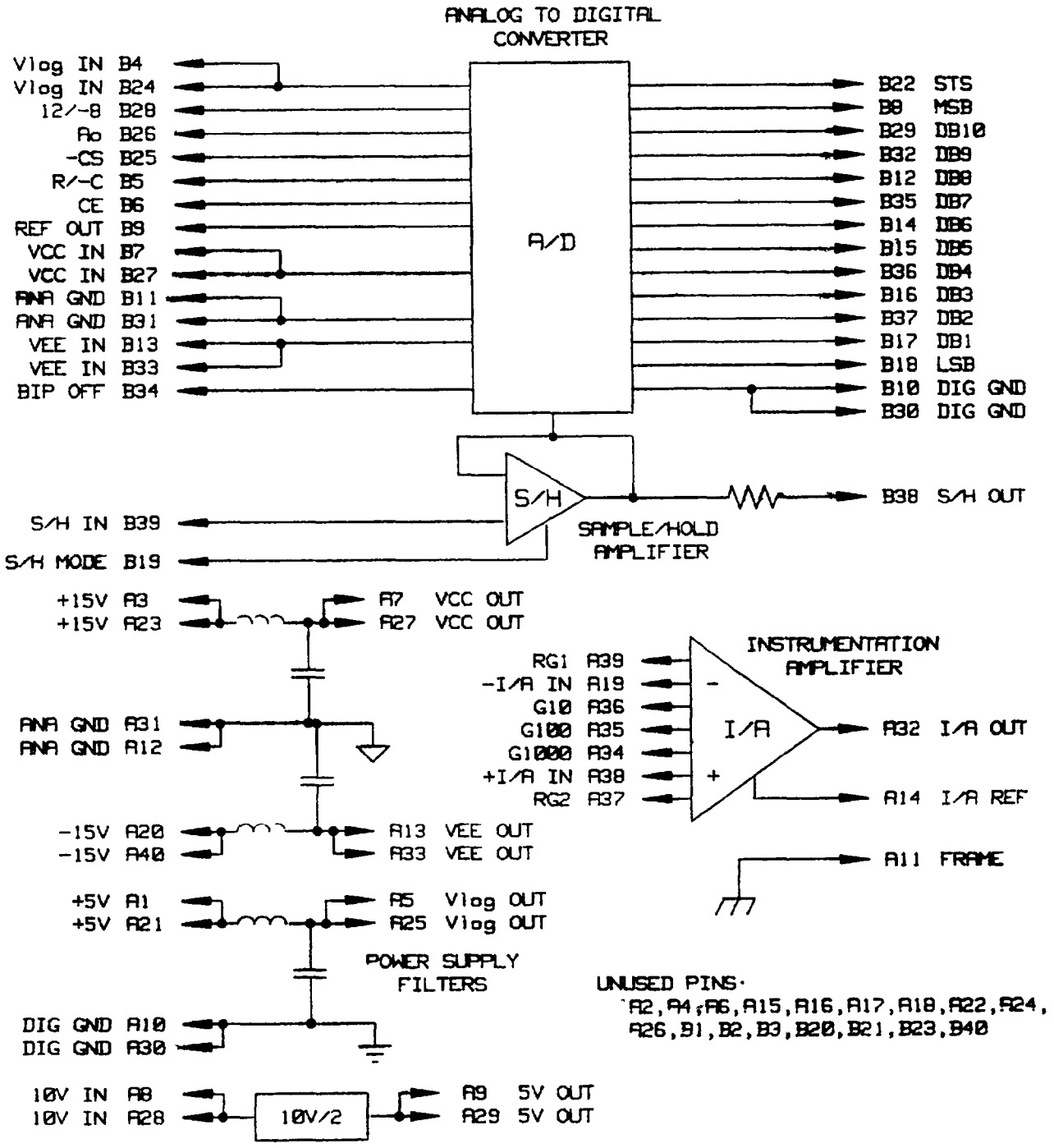


FIGURE 1. CKA functional block diagram and pin assignments.

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4.4.2 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.3 Order of testing. Test conditions A through QQ shall be performed in any order with the following exceptions. Test conditions A, C, D and E are stress tests and shall be performed only at 25 ± 5 degrees C. If test conditions A, C, D or E are part of a sequence of tests, the order of testing shall be as follows: A, C, D and E where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through QQ in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25 degrees C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, F, G1 through G3 and H through QQ shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 3. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test conditions G1 through G3, shall be performed using the initial limits at 25 ± 5 degrees C ambient.

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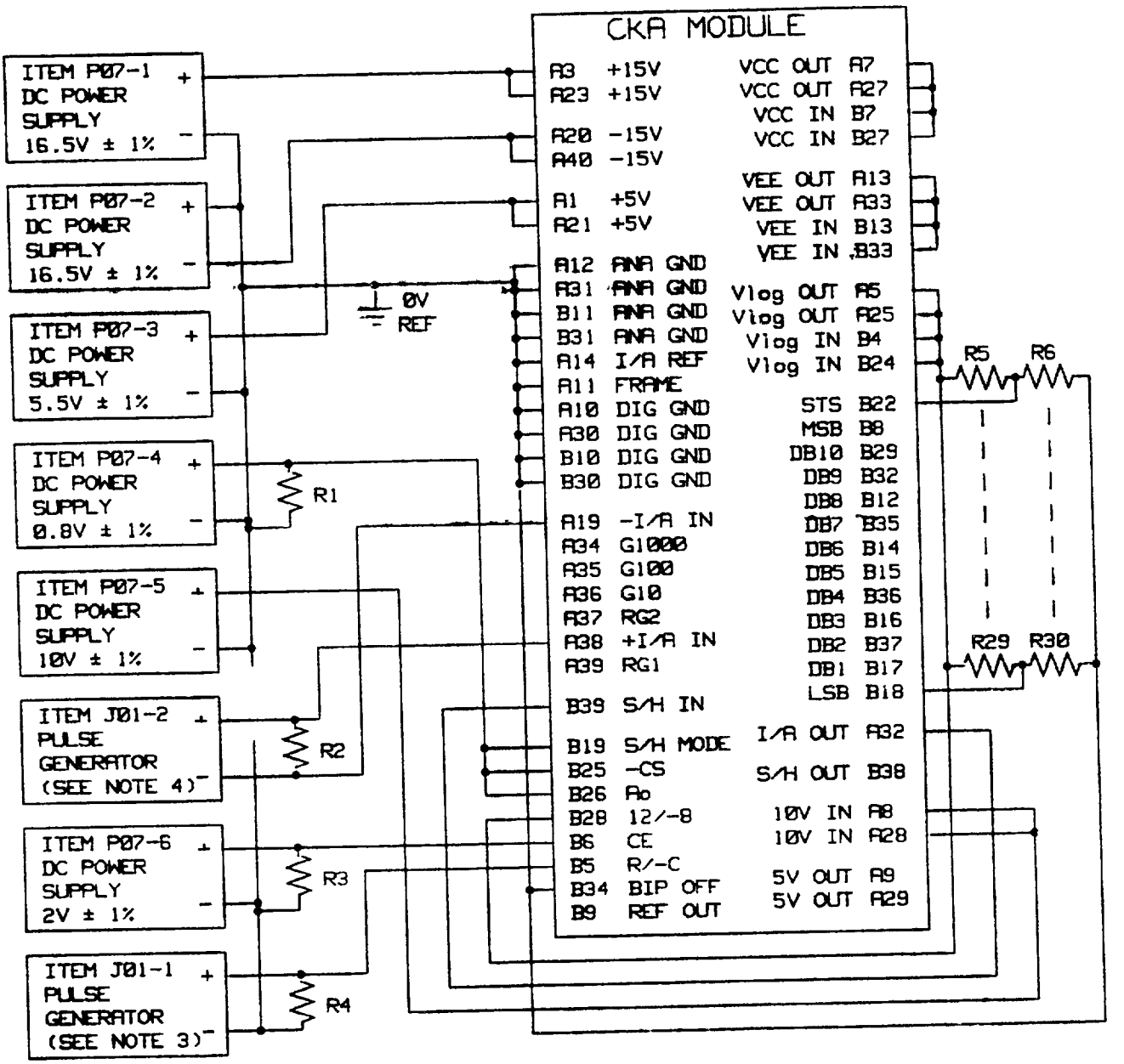


FIGURE 2. Operating temperature, component temperature, and life test circuit.

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NOTES:

1. R1 = R2 = R3 = R4 = 51 Ohm ± 5 percent, 1 W.

2. R5 through R30 = 3.01 kOhm ± 5 percent, 1/4 W

3. Item J01-1 setup:
   Frequency = 1 kHz ± 10 percent.
   Pulse width = 1 uS ± 10 percent.
   High level = 4V ± 1 V.
   Low level = 0.2V ± 0.1 V.
   $t_R = t_F$ = 50 nS ± 10 percent.

4. Item J01-2 setup
   Frequency = 500 Hz ± 10 percent.
   Duty cycle = 50 percent.
   High level = 10.0V ± 0.2 V.
   Low level = 0.0V ± 0.2 V.
   $t_R = t_F$ = 10 uS ± 50 percent.

5. Item J01-2 shall be synchronized to item J01-1.

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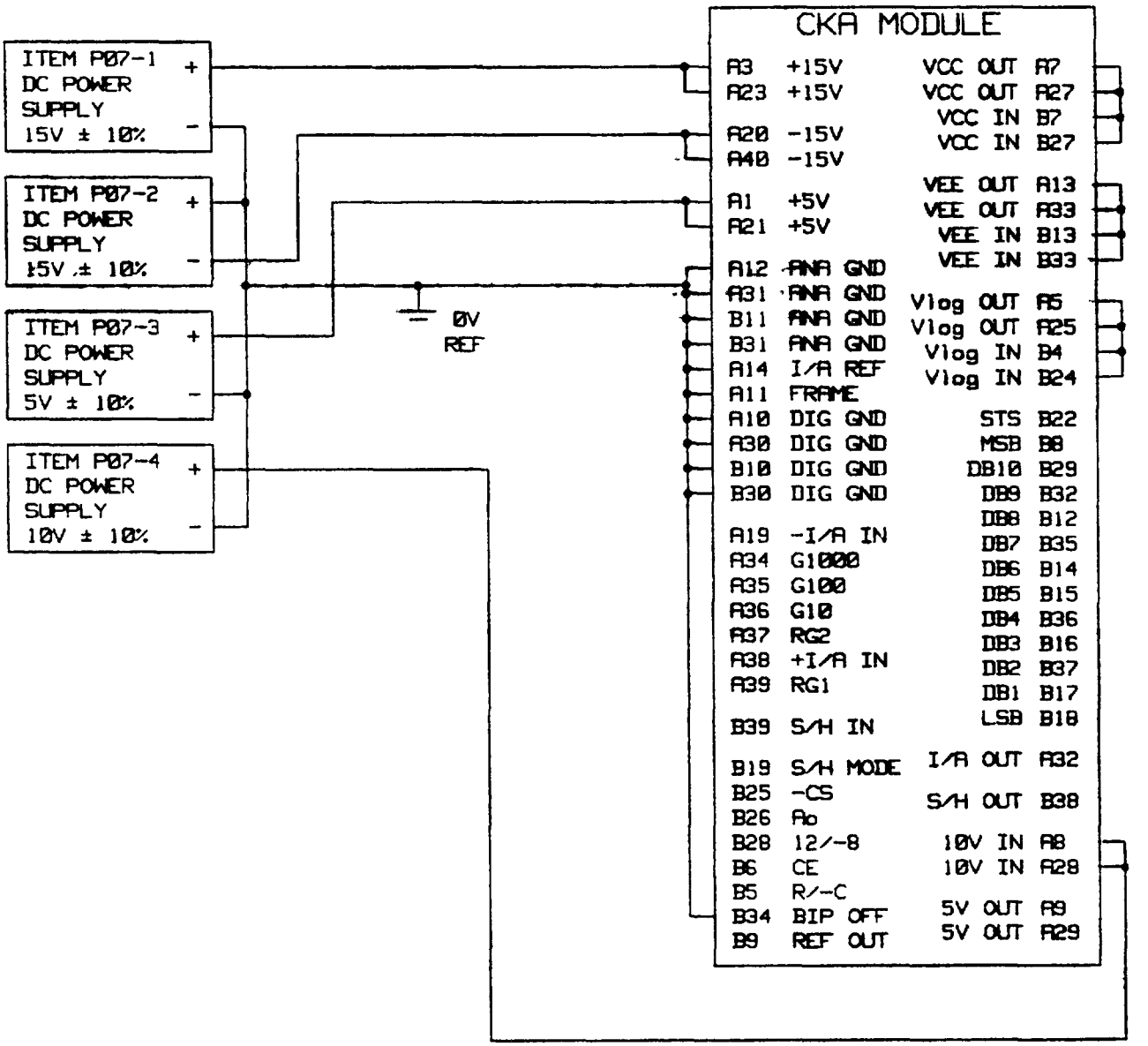


FIGURE 3. Vibration test circuit.

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4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests B, F, G1 through G3 and H through QQ shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5 degrees C, electrical tests B, F, G1 through G3 and H through QQ shall be performed in the order specified in 4.4.2 using the 25 ± 5 degrees C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5 degrees C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Abbreviations used in this specification are defined as follows:

(a) ADC = analog to digital converter.
(b) CMRR = common mode rejection ratio.
(c) FSR = full-scale reading.
(d) I/A = instrumentation amplifier.
(e) LSB = least significant bit.
(f) MSB = most significant bit.
(g) PSS = power supply sensitivity.
(h) S/H = sample and hold amplifier.
(i) tDD = access time from CE.
(j) tDDR = data access time.
(k) tDS = status delay from R/-C.
(l) tDSC = status delay from CE.
(m) tHD = data valid after CE low.
(n) tHDR = data valid after R/-C low.
(o) tHL = output float delay.
(p) tHS = status delay after data valid.
(q) TTL = transistor-to-transistor logic.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the 0 degrees C to 60 degrees C end-of-life parameters in table I.

Power dissipation	1.265 W max.
Supply voltages	+15 V dc ± 10 percent, 22 mA max. -15 V dc ± 10 percent, 47 mA max. +5 V dc ± 10 percent, 40 mA max.
Absolute maximum voltage supply ratings	±16.5 V dc +7 V dc
Power supply capacitance	15 uF ± 25 percent

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Analog input characteristics	
Input voltage ranges	
I/A	±10 V
Sample/hold	
Bipolar mode	±5 V
Unipolar mode	0 to +10 V
Input impedance	
I/A	10000 MOhm
Sample/hold	1 MOhm min.
Digital input characteristics	
Input capacitance	2 to 15 pF
Input current	
Low level	-50 uA to +1 uA
High level	-1 uA to +50 uA
Input voltage	
Low level	-0.5 V to +0.8 V
High level	+2.0 V to +5.5 V
Digital output characteristics	
Output voltage	
Low level	+0.4 V max. at 1.6 mA
High level	+2.4 V min. at 0.5 mA
Output leakage current	
Low level	±40 uA
High level	±40 uA
Internal reference voltage	+10 V dc ± 0.1 V dc
Conversion time	
8-bit	10 uS to 28 uS
12-bit	15 uS to 40 uS
Power supply rejection	
ADC	±0.2 LSB/percent change
I/A	110 dB max. (A=1000)
Output offset voltage	
ADC	-2 LSB to +3 LSB
I/A	±3 mV
Gain error	
ADC	±0 3 percent FSR
I/A	±0.5 percent FSR (A=100)

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Linearity error	
ADC	±1 LSB
I/A	0.01 percent FSR
I/A	
Output voltage swing	±10 V min.
Slew rate	3 V/uS min. (A=1 to 100)
Settling time	30 uS max. (A=1 to 100)

6.4 Module function. The module contains a 12-bit successive approximation ADC with a sample and hold circuit for throughput rate improvement, and an optional instrumentation amplifier for improved common mode rejection.

6.4.1 Analog-to-digital converter. The ADC contains all the functions necessary for interfacing to most popular microprocessors with an 8-, 12- or 16-bit data bus without external buffers or peripheral interface controllers. Multiple-mode three-state output buffers connect directly to the data bus while the read and convert commands are taken from the control bus. The 12 bits of output data can be read either as one 12-bit word or as two 8-bit bytes; one with 8 data bits, the other with 4 data bits and four trailing zeroes. The formatted data is left justified. This means that the data represents the analog input as a fraction of full scale, ranging from 0 to 4095/4096. This implies a binary point to the left of the MSB. The CKA module is pin programmable for either bipolar or unipolar operation with a linear input-output relationship for a bipolar analog input voltage range (pin B39) of ±5 V or for a unipolar analog input voltage range of 0 V to +10 V.

6.4.2 Clock and reference. The converter contains a high precision reference and clock to ensure full rated performance without external circuitry or clock signals.

6.4.3 Control signals. The ADC contains the necessary logic to provide conversion initiation and data read operations from signals commonly available in microprocessor systems. Logic inputs are TTL compatible with the exception of the 12/-8 input. This input must be hardwired either to Vlog or digital ground. The control signals CE, -CS and R/-C control the operation of the converter. The state of R/-C when CE and -CS are both asserted determines whether a data read (R/-C = 1) or data convert (R/-C = 0) is in progress. Control inputs Ao and

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12/-8 control conversion length and data format. If a conversion is started with Ao low, a full 12-bit conversion cycle is initiated. If Ao is high during a convert start, a shorter 8-bit conversion cycle results. During a data read operation, Ao determines whether the three-state buffers containing the 8 MSBs of the conversion results (Ao = 0) or the 4 LSBs (Ao = 1) are enabled. The 12/-8 pin determines whether the output data is to be organized as two 8-bit words (12/-8 tied to digital ground) or a single 12-bit word (12/-8 tied to Vlog). The 12/-8 pin is not TTL compatible and must be hardwired either to Vlog or digital ground. In the 8-bit mode, the byte addressed when Ao is high contains the 4 LSBs from the conversion followed by 4 trailing zeroes. This allows the data lines to be overlapped for direct interface to 8-bit buses without the need for external buffers.

6.4.4 Sample and hold. The sample and hold amplifier, connected as a unity gain non-inverting amplifier, is committed to the ADC input. The sample and hold enables the ADC to convert higher frequency analog signals without a loss in accuracy, and also serves to buffer the analog input of the ADC.

6.4.5 Instrumentation amplifier. The instrumentation amplifier has pin programmable fixed gains of 1, 10, 100 and 1000 and permits the use of differential input signals. It has low linearity error, high common mode rejection, low offset voltage drift, and low noise. It has an output slew rate of 3 V/uS and settles to within 0.01 percent of its final value in 30 uS when programmed for a gain of 100 or less (150 uS for a gain of 1000).

6.4.6 Filters. LC filters are provided at all power supply inputs for added power supply ripple rejection and to allow operation of the ADC with switching type power supplies.

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6.4.1 Functional description of I/O pins.

Signal name	Pin number	I/O	Description
RG1 RG2	A39 A37	I I	I/A programmable gain inputs, connected to A34-A36.
G10 G100 G1000	A36 A35 A34	I I I	I/A preset programmable gain inputs can set gains of 10, 100 or 1000.
+I/A -I/A	A38 A19	I I	I/A signal inputs.
I/A REF	A14	I	I/A reference ground.
I/A OUT	A32	O	I/A analog output.
S/H IN	B39	I	S/H signal input.
S/H MODE	B19	I	S/H mode control - 0 - sample. 1 - hold.
S/H OUT	B38	O	S/H signal output test point.
12/-8	B28	I	ADC data mode select - 0 - output data organized as 2 8-bit words. 1 - output data organized as 1 12-bit word.
-CS	B25	I	ADC chip select - must be low for a conversion to start or to read data output. Low-going transition may be used to start a conversion.
CE	B6	I	ADC chip enable - must be high for a conversion to start or to read data output.

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
R/-C	B5	I	ADC read/convert - 0 - start conversion. 1 - read output data.
Ao	B26	I	ADC byte address - selects conversion mode. 0 - 12-bit. 1 - 8-bit. In read mode selects output format. 0 - 8 MSBs. 1 - 4 LSBs and 4 trailing zeroes in a left justified format.
BIP OFF	B34	I	ADC bipolar offset. If connected to REF OUT, it puts ADC in bipolar mode. If connected to analog ground, it puts ADC in unipolar mode.
STS	B22	O	ADC status output - 0 - conversion complete. 1 - conversion in progress.
REF OUT	B9	O	Reference output - connect to BIP OFF for bipolar operation.
LSB DB1 DB2 DB3 DB4 DB5 DB6 DB7 DB8 DB9 DB10 MSB	B18 B17 B37 B16 B36 B15 B14 B35 B12 B32 B29 B8	O	ADC digital output bits. The range is from MSB which is the most significant data bit to LSB, the least significant data bit.

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6.5 Description of tests. The following tables and associated figures describe the tests to be performed:

(a) Isolation test - table II, figure 4.

(b) Continuity test - table III, figure 5.

(c) Power supply stress test - table IV, figure 6.

(d) Breakdown current test - table V, figure 13.

(e) Short circuit current test - table VI, figure 7.

(f) Power supply current test - table VII, figures 8 and 9.

(g) Filter and input capacitance test - table VIII, figure 10.

(h) Filter inductance test - table IX, figure 11.

(i) 10V divider test - table X, figure 12.

(j) Input current, high level, test - table XI, figure 13.

(k) Input current, low level, test - table XII, figure 13.

(l) Output voltage, high level, test - table XIII, figures 14, 15 and 16.

(m) Output voltage, low level, test - table XIV, figures 14, 15 and 16.

(n) Leakage current test - table XV, figure 17.

(o) Reference voltage test - table XVI, figure 18.

(p) Conversion time test - table XVII, figure 19.

(q) Operating modes test - table XVIII, figure 20.

(r) Supply sensitivity test - table XIX, figures 21 and 22.

(s) Missing code check test - table XX, figures 21 and 23.

(t) ADC offset voltage test - table XXI, figures 21, 22 and 24.

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(u) ADC gain error test - table XXII, figures 21, 22 and 24.

(v) ADC linearity error test - table XXIII, figures 21, 24 and 25.

(w) ADC transition time test - table XXIV, figures 26 and 27.

(x) ADC transition error test - table XXV, figures 21, 22 and 28.

(y) S/H input impedance test - table XXVI, figure 29.

(z) S/H acquisition time test - table XXVII, figures 30 and 31

(aa) S/H droop rate test - table XXVIII, figures 32 and 33.

(bb) I/A offset voltage test - table XXIX, figure 34.

(cc) I/A input bias and offset current test - table XXX, figure 35.

(dd) I/A power supply sensitivity test - table XXXI, figure 36.

(ee) I/A common mode rejection ratio test - table XXXII, figure 37.

(ff) I/A output swing test - table XXXIII, figure 38.

(gg) I/A settling time test - table XXXIV, figures 39 and 40.

(hh) I/A input resistance test - table XXXV, figure 41.

(ii) I/A gain accuracy test - table XXXVI, figure 42.

(jj) I/A slew rate - table XXXVII, figures 43 and 44.

Custodians:	Review activities:	Preparing activity:
Army - ER	Army - AT	Navy - SH
Navy - SH	Navy - AS, MC	
Air Force - 85	Air Force - 13, 19	(Project 5963-0062)
	DLA - ES	
		Agent:
		NW

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TABLE II. Isolation test.

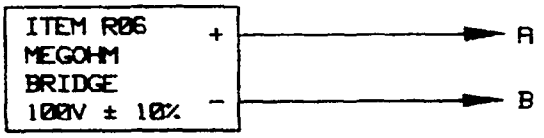
Test cond	Parameter	Connect		25 ± 5°C Initial limits		Units	Fig
		A to pin	B to pin	Min	Max		
A1	Frame to all other pins	A1-A10,A12-A40, B1-B40	A11	10	INF	MOhm	4
A2	Used to unused pins	A1,A3,A5,A7-A14, A19-A21,A23,A25, A27-A40,B4-B19, B22,B24-B39	A2,A4,A6,A15-A18,A22, A24,A26,B1-B3,B20,B21, B23,B40				
A3	Unused to unused pins	A2	A4,A6,A15-A18,A22,A24, A26,B1-B3,B20,B21,B23, B40				
A4	Unused to unused pins	A4	A6,A15-A18,A22,A24, A26,B1-B3,B20,B21,B23, B40				
A5	Unused to unused pins	A6	A15-A18,A22,A24,A26, B1-B3,B20,B21,B23,B40				
A6	Unused to unused pins	A15	A16-A18,A22,A24,A26, B1-B3,B20,B21,B23,B40				
A7	Unused to unused pins	A16	A17,A18,A22,A24,A26, B1-B3,B20,B21,B23,B40				
A8	Unused to unused pins	A17	A18,A22,A24,A26,B1-B3, B20,B21,B23,B40				
A9	Unused to unused pins	A18	A22,A24,A26,B1-B3,B20, B21,B23,B40				
A10	Unused to unused pins	A22	A24,A26,B1-B3,B20,B21, B23,B40				
A11	Unused to unused pins	A24	A26,B1-B3,B20,B21,B23, B40				
A12	Unused to unused pins	A26	B1-B3,B20,B21,B23,B40				
A13	Unused to unused pins	B1	B2,B3,B20,B21,B23,B40	10	INF	MOhm	4

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TABLE II. Isolation test - Continued.

Test cond	Parameter	Connect		25 ± 5°C Initial limits		Units	Fig
		A to pin	B to pin	Min	Max		
A14	Unused to unused pins	B2	B3,B20,B21,B23,B40	10	INF	MOhm	4
A15	Unused to unused pins	B3	B20,B21,B23,B40	↓	↓	↓	↓
A16	Unused to unused pins	B20	B21,B23,B40				
A17	Unused to unused pins	B21	B23,B40				
A18	Unused to unused pins	B23	B40	10	INF	MOhm	4

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CKA MODULE			
A3	+15V	VCC OUT	A7
A23	+15V	VCC OUT	A27
		VCC IN	B7
A20	-15V	VCC IN	B27
A40	-15V		
		VEE OUT	A13
A1	+5V	VEE OUT	A33
A21	+5V	VEE IN	B13
		VEE IN	B33
A12	ANA GND		
A31	ANA GND	Vlog OUT	A5
B11	ANA GND	Vlog OUT	A25
B31	ANA GND	Vlog IN	B4
A14	I/A REF	Vlog IN	B24
A11	FRAME		
A10	DIG GND	STS	B22
A30	DIG GND	MSB	B8
B10	DIG GND	DB10	B29
B30	DIG GND	DB9	B32
		DB8	B12
A19	-I/A IN	DB7	B35
A34	G1000	DB6	B14
A35	G100	DB5	B15
A36	G10	DB4	B36
A37	RG2	DB3	B16
A38	+I/A IN	DB2	B37
A39	RG1	DB1	B17
		LSB	B18
B39	S/H IN		
B19	S/H MODE	I/A OUT	A32
B25	-CS	S/H OUT	B38
B26	Ao		
B28	12/-8	10V IN	A8
B6	CE	10V IN	A28
B5	R/-C		
B34	BIP OFF	5V OUT	A9
B9	REF OUT	5V OUT	A29

NOTE:

1 Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge to 'DISCHARGE' before connecting or disconnecting the test leads.

FIGURE 4. Isolation test circuit for table II.

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TABLE III. Continuity test.

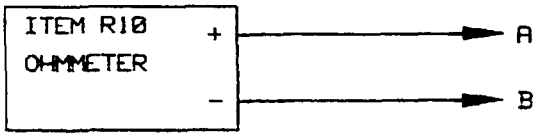
Test cond	Parameter	Connect		25 ± 5°C Initial limits		Units	Fig
		A to pin	B to pin	Min	Max		
B1	Continuity, +15V	A3	A23	0 0	1.0	Ohm	5
B2	Continuity, analog ground	A31	A12	↓	↓	↓	↓
B3	Continuity, -15V	A20	A40				
B4	Continuity, VCC OUT	A7	A27				
B5	Continuity, VEE OUT	A13	A33				
B6	Continuity, +5V	A1	A21				
B7	Continuity, Vlog OUT	A5	A25				
B8	Continuity, digital ground	A10	A30				
B9	Continuity, 10V IN	A8	A28				
B10	Continuity, 5V OUT	A9	A29				
B11	Continuity, Vlog IN	B4	B24				
B12	Continuity, VCC IN	B7	B27				
B13	Continuity, VEE IN	B13	B33				
B14	Continuity, analog ground	B11	B31				
B15	Continuity, digital ground	B10	B30				
B16	Continuity, frame pin to frame	A11	Frame	0 0	1 0	Ohm	5
B17	Continuity, +15V	A3	A23	0 0	0.1	Ohm	5
B18	Continuity, analog ground	A31	A12	↓	↓	↓	↓
B19	Continuity, -15V	A20	A40				
B20	Continuity, VCC OUT	A7	A27				
B21	Continuity, VEE OUT	A13	A33	0 0	0.1	Ohm	5

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TABLE III. Continuity test - Continued.

Test cond	Parameter	Connect		25 ± 5°C Initial limits		Units	Fig
		A to pin	B to pin	Min	Max		
B22	Continuity, +5V	A1	A21	0.0	0.1	Ohm	5
B23	Continuity, Vlog OUT	A5	A25	↓	↓	↓	↓
B24	Continuity, digital ground	A10	A30				
B25	Continuity, 10V IN	A8	A28				
B26	Continuity, 5V OUT	A9	A29				
B27	Continuity, Vlog IN	B4	B24				
B28	Continuity, VCC IN	B7	B27				
B29	Continuity, VEE IN	B13	B33				
B30	Continuity, analog ground	B11	B31				
B31	Continuity, digital ground	B10	B30	0.0	0.1	Ohm	5

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CKA MODULE			
A3	+15V	VCC OUT	A7
A23	+15V	VCC OUT	A27
		VCC IN	B7
A20	-15V	VCC IN	B27
A40	-15V		
		VEE OUT	A13
A1	+5V	VEE OUT	A33
A21	+5V	VEE IN	B13
		VEE IN	B33
A12	ANA GND		
A31	ANA GND	Vlog OUT	A5
B11	ANA GND	Vlog OUT	A25
B31	ANA GND	Vlog IN	B4
A14	I/A REF	Vlog IN	B24
A11	FRAME		
A10	DIG GND	STS	B22
A30	DIG GND	MSB	B8
B10	DIG GND	DB10	B29
B30	DIG GND	DB9	B32
		DB8	B12
A19	-I/A IN	DB7	B35
A34	G1000	DB6	B14
A35	G100	DB5	B15
A36	G10	DB4	B36
A37	RG2	DB3	B16
A38	+I/A IN	DB2	B37
A39	RG1	DB1	B17
		LSB	B18
B39	S/H IN		
B19	S/H MODE	I/A OUT	A32
B25	-CS	S/H OUT	B38
B26	Ao		
B28	12/-8	10V IN	A8
B6	CE	10V IN	A28
B5	R/-C		
B34	BIP OFF	5V OUT	A9
B9	REF OUT	5V OUT	A29

NOTE

1. For test condition B16, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the ohmmeter.

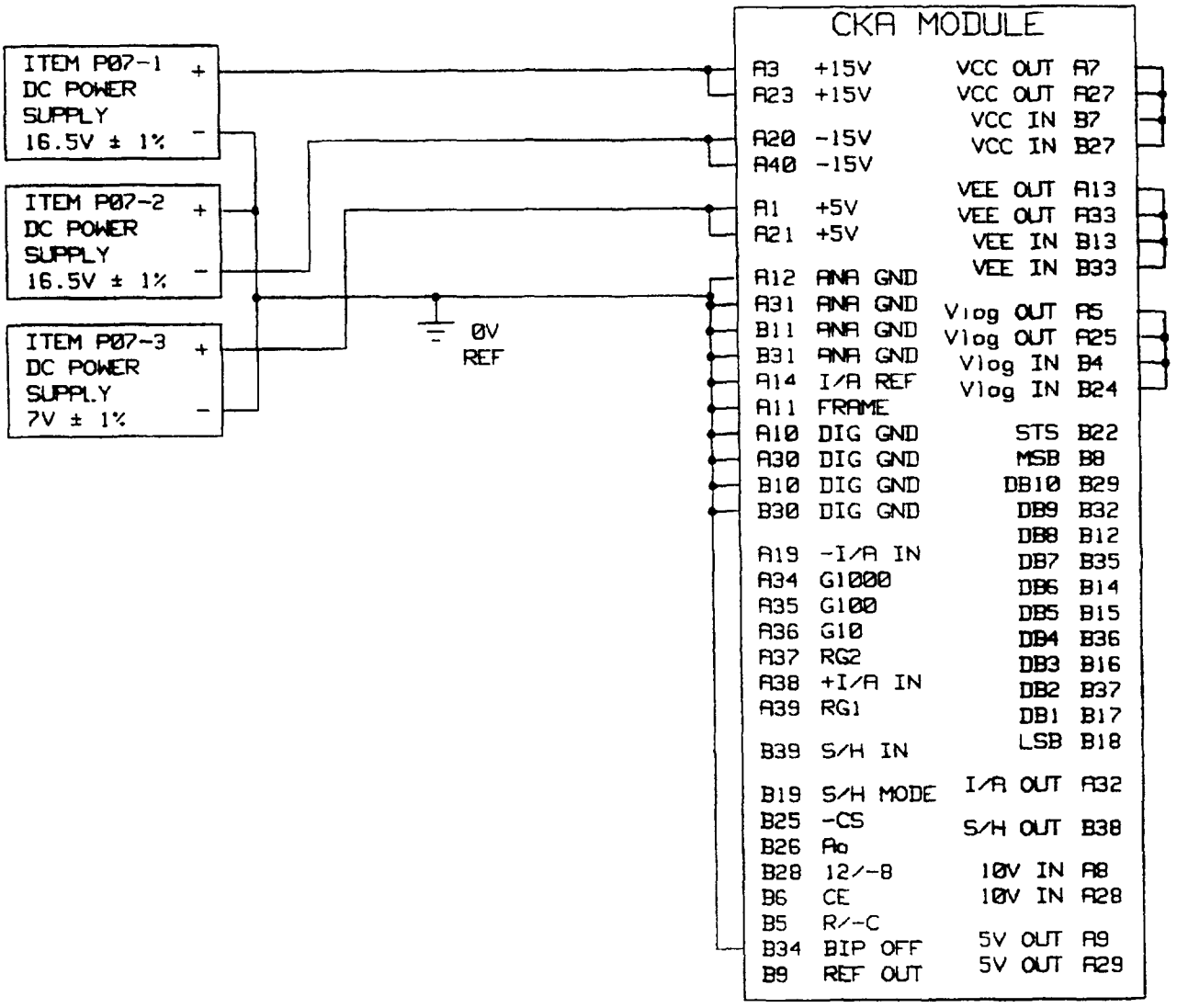
FIGURE 5. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Parameter	25 ± 5°C Initial limits	Units	Fig
C	Power supply stress	Module shall pass all remaining tests.	---	6

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NOTE

1. Test shall be maintained for a minimum of 5 seconds

FIGURE 6 Power supply stress test circuit for table IV.

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TABLE V. Breakdown current test.

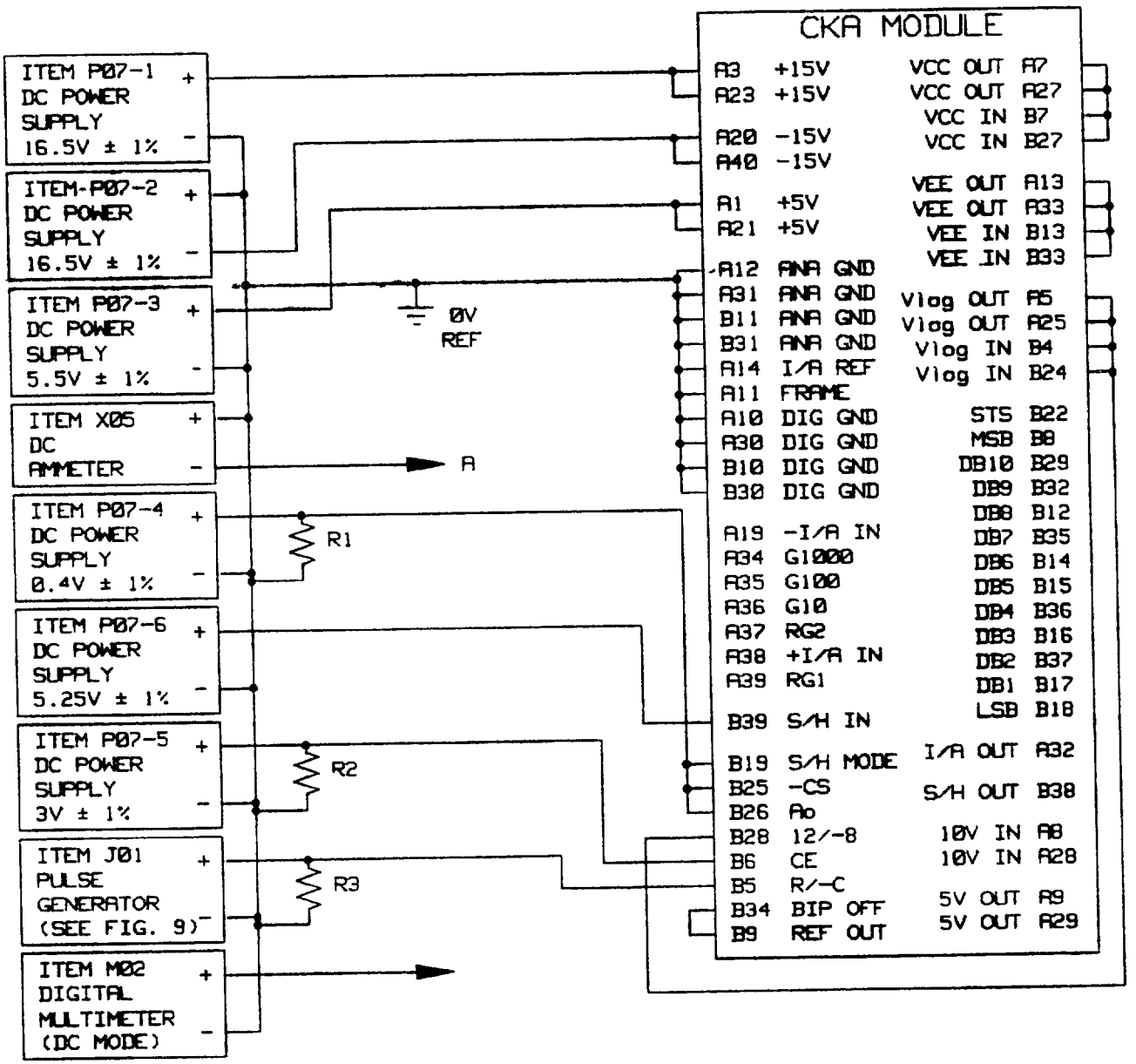
Test cond	Parameter	Connect		Item P07-4 voltage (V) ± 0.1 V	25 ± 5°C Initial limits		Units	Fig
		A to pin	B to pin		Min	Max		
D1	S/H MODE	B19	B5,B6,B25,B26,B28	5.5	-1	50	uA	13
D2	CE	B6	B5,B19,B25,B26,B28	↓	↓	↓	↓	↓
D3	R/-C	B5	B6,B19,B25,B26,B28					
D4	Ao	B26	B5,B6,B19,B25,B28					
D5	-CS	B25	B5,B6,B19,B26,B28	5.5	-1	50	uA	13

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TABLE VI. Short circuit current test.

Test cond	Parameter	Connect A to pin	25 ± 5° C Initial limits		Units	Fig
			Min	Max		
E1	LSB output short circuit current	B18	-40	1	mA	7
E2	DB1 output short circuit current	B17	↓	↓	↓	↓
E3	DB2 output short circuit current	B37				
E4	DB3 output short circuit current	B16				
E5	DB4 output short circuit current	B36				
E6	DB5 output short circuit current	B15				
E7	DB6 output short circuit current	B14				
E8	DB7 output short circuit current	B35				
E9	DB8 output short circuit current	B12				
E10	DB9 output short circuit current	B32				
E11	DB10 output short circuit current	B29				
E12	MSB output short circuit current	B8	-40	1	mA	7

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NOTES:

1. R1 = R2 = R3 = 51 Ohm ± 5 percent, 1 W.
2. Verify that the digital output code word is 1111 1111 1111 by using item M02 for the appropriate test condition.
3. When performing this test, only one output shall be tested at a time. The duration of the test shall not exceed one second

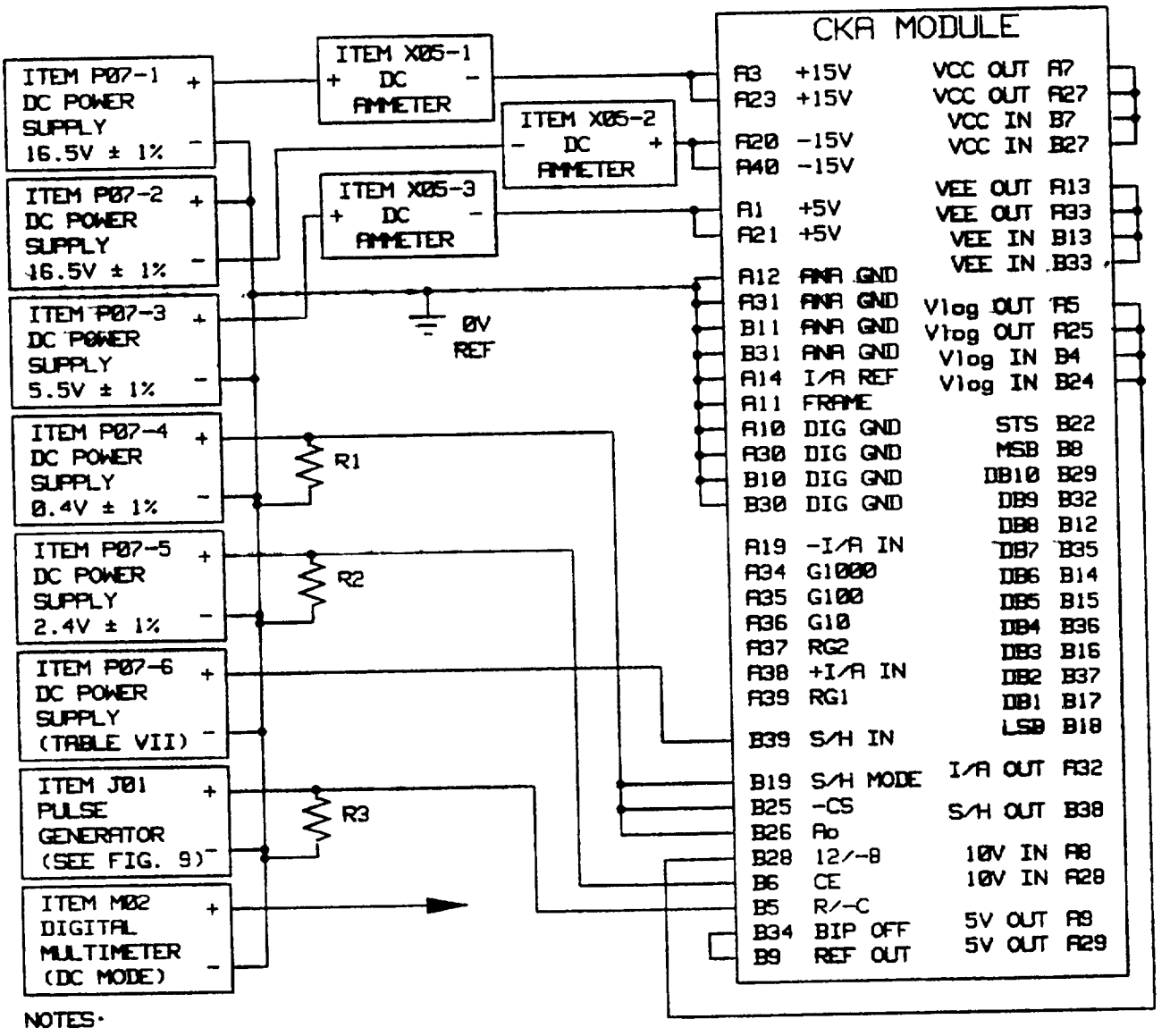
FIGURE 7. Short circuit current test circuit for table VI

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TABLE VII. Power supply current test.

Test cond	Parameter	Ammeter to be used	Item P07-6 voltage (V) ± 52 mV	25 ± 5°C Initial limits		Units	Figs
				Min	Max		
F1	+15V power supply current	X05-1	-5.25	1	25	mA	8,9
F2	-15V power supply current	X05-2	-5.25	1	50	mA	8,9
F3	+5V power supply current	X05-3	-5.25	1	45	mA	8,9
F4	+15V power supply current	X05-1	5.25	1	25	mA	8,9
F5	-15V power supply current	X05-2	5.25	1	50	mA	8,9
F6	+5V power supply current	X05-3	5.25	1	45	mA	8,9

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NOTES:

1. R1 = R2 = R3 = 51 Ohm ± 5 percent, 1 W.

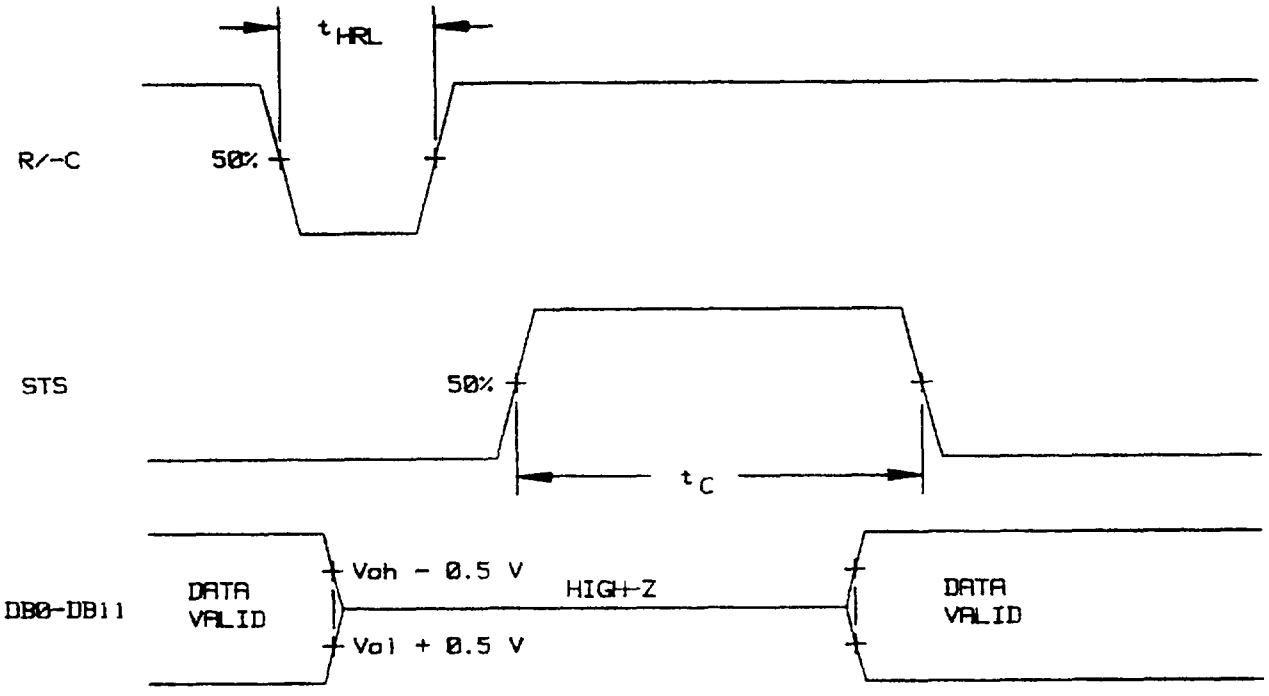
2. Since items X05-1, X05-2, and X05-3 are not used simultaneously, only one item X05 is required.

3. Verify that the output code word is as specified below by using item M02 for the appropriate test condition.

4. Code word = 0000 0000 0000 for F1-F3 and 1111 1111 1111 for F4-F6.

FIGURE 8. Power supply current test circuit for table VII.

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NOTE:

1 R/-C input:
Single pulse.
t_{HRL} = 350 nS minimum.
Rise time = fall time = 50 nS maximum
High level = 2.0 V min, 5.5 V max.
Low level = -0.5 V min, 0.8 V max.

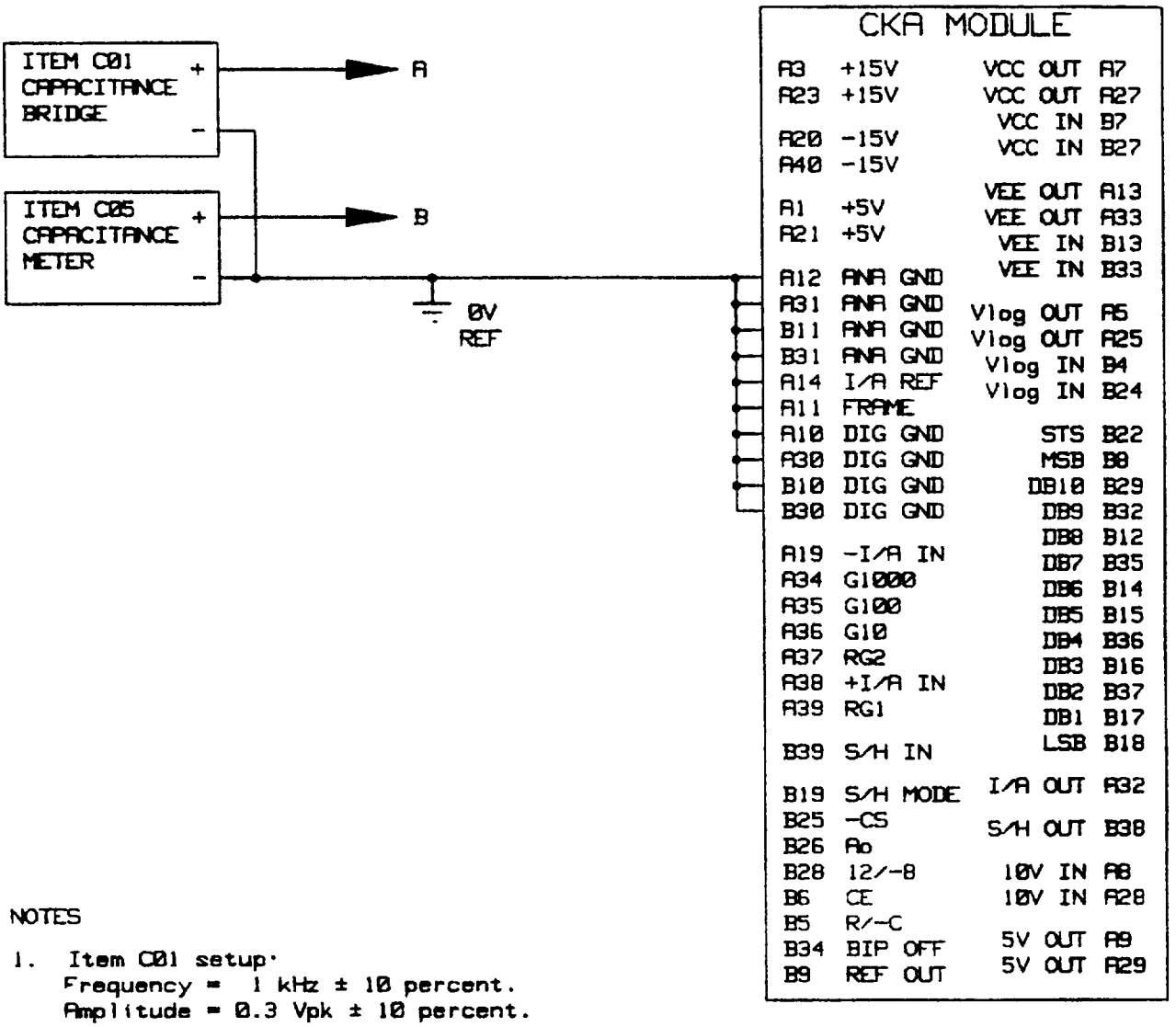
FIGURE 9. Timing diagram.

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TABLE VIII. Filter and input capacitance test.

Test cond	Parameter	Connect		25 ± 5°C Initial limits		Units	Fig
		A to pin	B to pin	Min	Max		
G1	+15V capacitance	A7	Open	12	18	uF	10
G2	-15V capacitance	A13	Open	12	18	uF	10
G3	+5V capacitance	A5	Open	12	18	uF	10
G4	S/H MODE input capacitance	Open	B39	2	25	pF	10
G5	12/-8 input capacitance	↓	B28	↓	↓	↓	↓
G6	-CS input capacitance		B25				
G7	Ao input capacitance		B26				
G8	R/-C input capacitance		B5				
G9	CE input capacitance	Open	B6	2	25	pF	10

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NOTES

1. Item C01 setup:
   Frequency = 1 kHz ± 10 percent.
   Amplitude = 0.3 Vpk ± 10 percent.

2. Null out the effect of the test leads prior to connecting to the module.

3. Do not attach test wires to any of the module pins other than the pin under test and the ground pins. Stray capacitance may affect the readings.

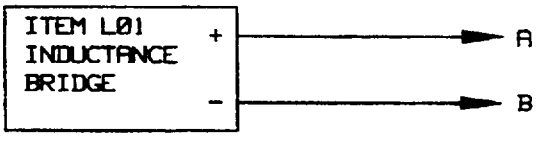
FIGURE 10. Filter and input capacitance test circuit for table VIII.

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TABLE IX. Filter inductance test.

Test cond	Parameter	Connect		25 ± 5°C Initial limits		Units	Fig
		A to pin	B to pin	Min	Max		
H1	+15V inductance	A3	A7	30	50	uHy	11
H2	-15V inductance	A20	A13	30	50	uHy	11
H3	+5V inductance	A1	A5	30	50	uHy	11

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CKA MODULE			
A3	+15V	VCC OUT	A7
A23	+15V	VCC OUT	A27
		VCC IN	B7
A20	-15V	VCC IN	B27
A40	-15V		
		VEE OUT	A13
A1	+5V	VEE OUT	A33
A21	+5V	VEE IN	B13
		VEE IN	B33
A12	ANA GND		
A31	ANA GND	Vlog OUT	A5
B11	ANA GND	Vlog OUT	A25
B31	ANA GND	Vlog IN	B4
A14	I/A REF	Vlog IN	B24
A11	FRAME		
A10	DIG GND	STS	B22
A30	DIG GND	MSB	B8
B10	DIG GND	DB10	B29
B30	DIG GND	DB9	B32
		DB8	B12
A19	-I/A IN	DB7	B35
A34	G1000	DB6	B14
A35	G100	DB5	B15
A36	G10	DB4	B36
A37	RG2	DB3	B16
A38	+I/A IN	DB2	B37
A39	RG1	DB1	B17
		LSB	B18
B39	S/H IN		
		I/A OUT	A32
B19	S/H MODE		
B25	-CS	S/H OUT	B38
B26	Ao		
B28	12/-8	10V IN	A8
B6	CE	10V IN	A28
B5	R/-C		
B34	BIP OFF	5V OUT	A9
B9	REF OUT	5V OUT	A29

NOTE

1 Do not attach test wires to any of the module pins other than the pins under test and the ground pins. Stray inductance may affect the readings.

FIGURE 11 Power supply filter inductance test circuit for table IX.

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TABLE X. 10 V divider test.

Test cond	Parameter	25 ± 5°C Initial limits		Units	Fig
		Min	Max		
J	+10 V divider	4.9	5.1	V	12

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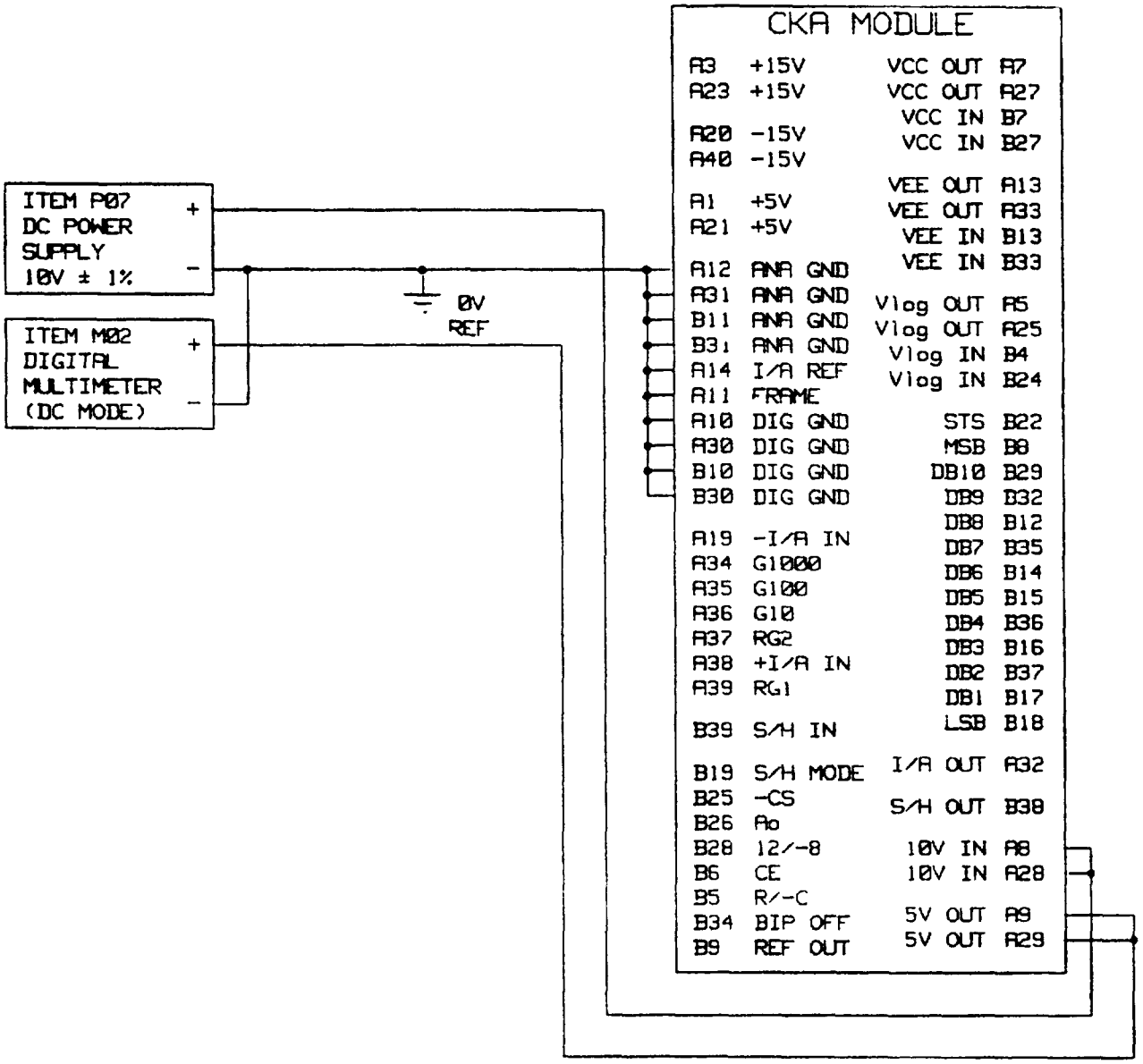


FIGURE 12. 10V divider test circuit for table X.

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TABLE XI. Input current, high level, test.

Test cond	Parameter	Connect		Item P07-4 voltage (V) ± 0.1 V	25 ± 5°C Initial limits		Units	Fig
		A to pin	B to pin		Min	Max		
K1	S/H MODE input current	B19	B5,B6,B25,B26,B28	2.4	-1	50	uA	13
K2	CE input current	B6	B5,B19,B25,B26,B28	↓	↓	↓	↓	↓
K3	R/-C input current	B5	B6,B19,B25,B26,B28					
K4	Ao input current	B26	B5,B6,B19,B25,B28					
K5	-CS input current	B25	B5,B6,B19,B26,B28	2.4	-1	50	uA	13

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TABLE XII. Input current, low level, test.

Test cond	Parameter	Connect		Item P07-4 voltage (V) ± 0.1 V	25 ± 5°C Initial limits		Units	Fig
		A to pin	B to pin		Min	Max		
L1	S/H MODE input current	B19	B5,B6,B25,B26,B28	0.0	-50	1	uA	13
L2	CE input current	B6	B5,B19,B25,B26,B28	↓	↓	↓	↓	↓
L3	R/-C input current	B5	B6,B19,B25,B26,B28					
L4	Ao input current	B26	B5,B6,B19,B25,B28					
L5	-CS input current	B25	B5,B6,B19,B26,B28	0.0	-50	1	uA	13

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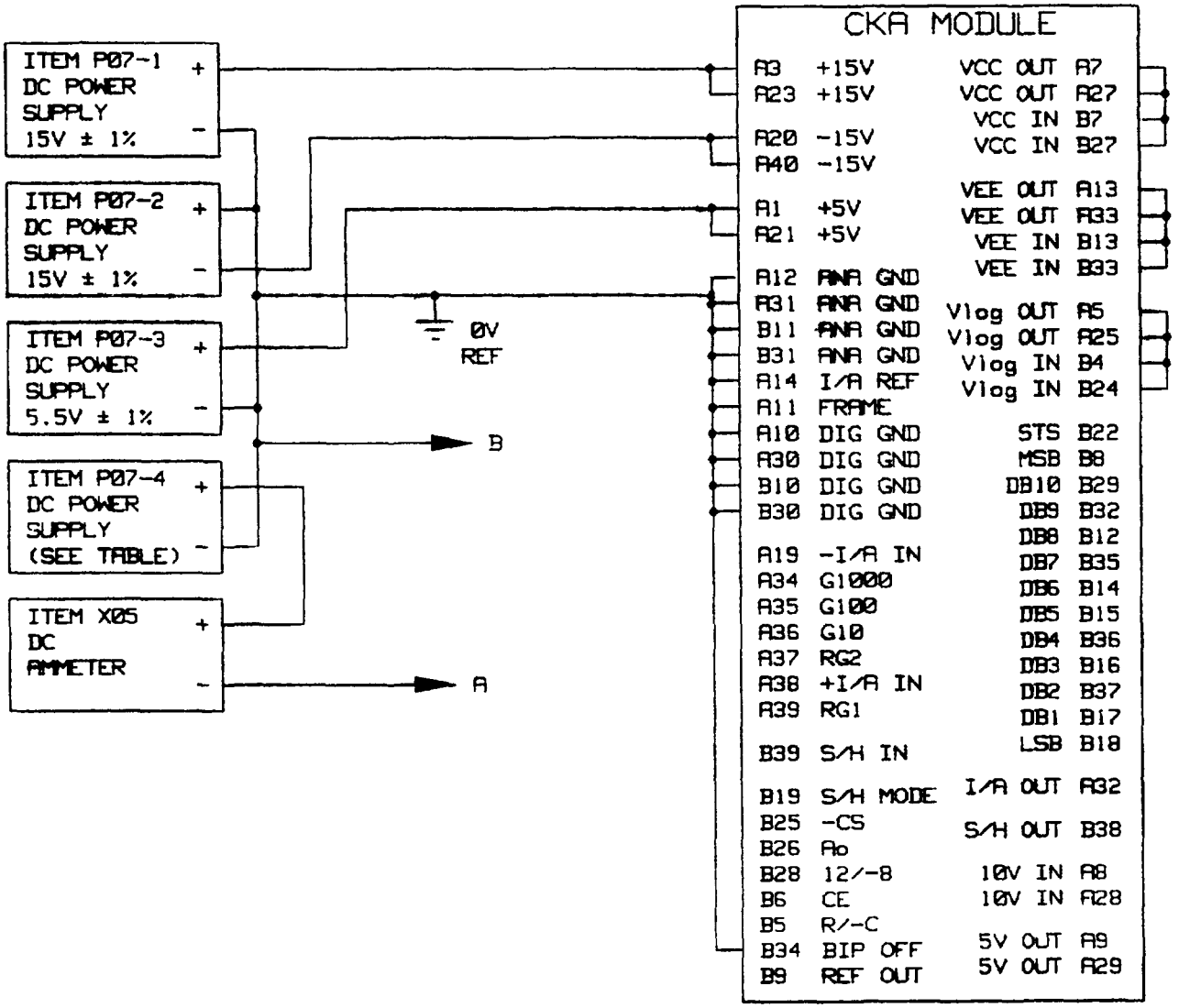


FIGURE 13. Breakdown and input current, high and low level, test circuit for tables V, XI, and XII.

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TABLE XIII. Output voltage, high level, test.

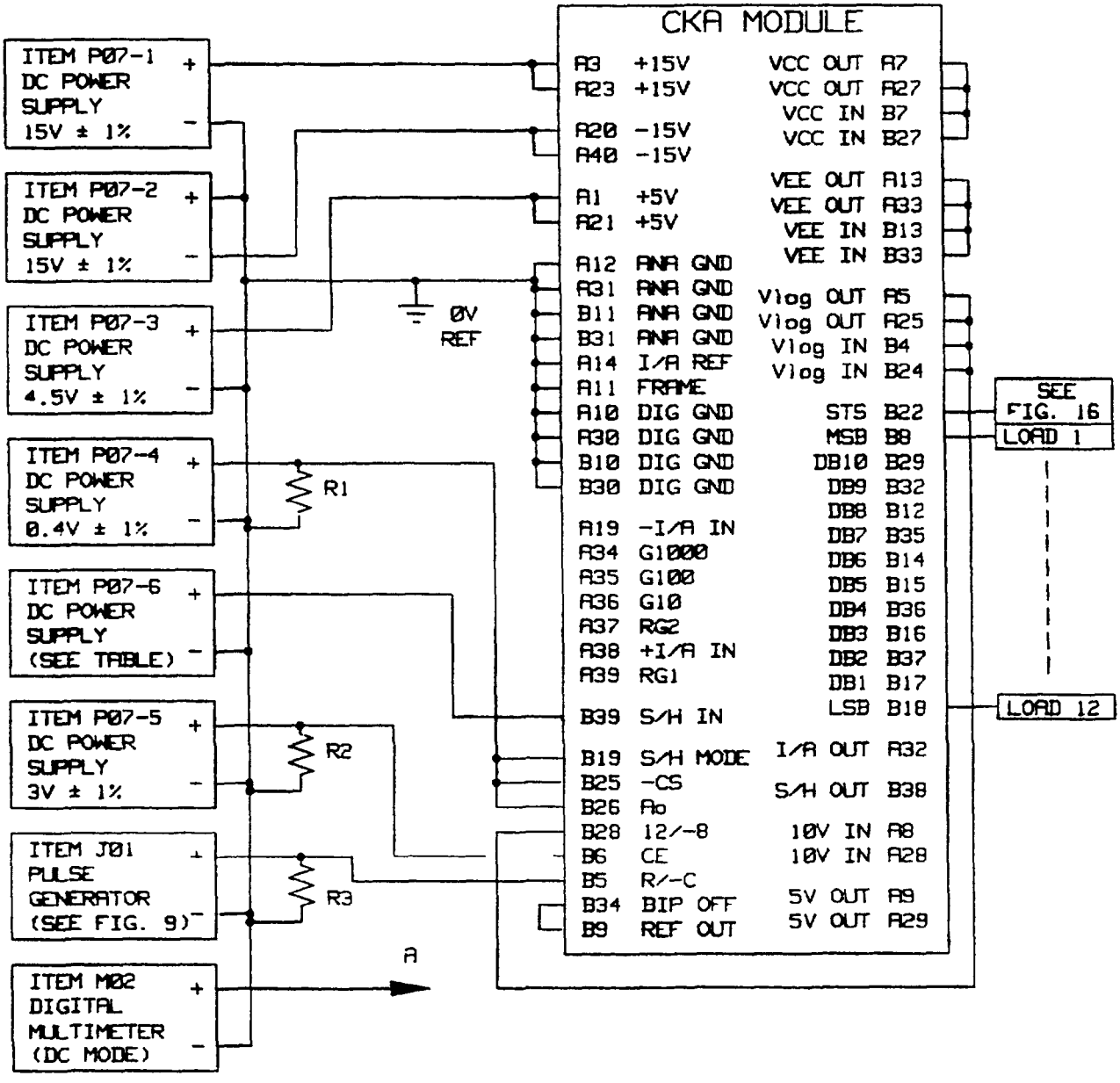
Test cond	Parameter	Connect A to pin	Item P07-6 voltage (V) ± 52 mV	25 ± 5°C Initial limits		Units	Figs
				Min	Max		
M1	LSB output voltage	B18	5.25	2.4	4.5	V	14, 15, 16
M2	DB1 output voltage	B17					
M3	DB2 output voltage	B37					
M4	DB3 output voltage	B16					
M5	DB4 output voltage	B36					
M6	DB5 output voltage	B15					
M7	DB6 output voltage	B14					
M8	DB7 output voltage	B35					
M9	DB8 output voltage	B12					
M10	DB9 output voltage	B32					
M11	DB10 output voltage	B29					
M12	MSB output voltage	B8					
M13	STS output voltage	Point B	5.25	2.4	4.5	V	14, 15, 16

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TABLE XIV. Output voltage, low level, test.

Test cond	Parameter	Connect A to pin	Item P07-6 voltage (V) ± 52 mV	25 ± 5°C Initial limits		Units	Figs
				Min	Max		
N1	LSB output voltage	B18	-5.25	0.0	0.4	V	14, 15, 16
N2	DB1 output voltage	B17	↓	↓	↓	↓	↓
N3	DB2 output voltage	B37					
N4	DB3 output voltage	B16					
N5	DB4 output voltage	B36					
N6	DB5 output voltage	B15					
N7	DB6 output voltage	B14					
N8	DB7 output voltage	B35					
N9	DB8 output voltage	B12					
N10	DB9 output voltage	B32					
N11	DB10 output voltage	B29					
N12	MSB output voltage	B8	↓	↓	↓	↓	↓
N13	STS output voltage	Point B	-5.25	0 0	0.4	V	14, 15, 16

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:09Z
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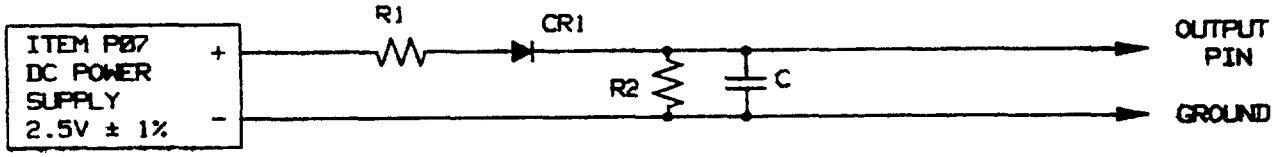


NOTES

1. R1 = R2 = R3 = 51 Ohm ± 5 percent, 1 W.
2. Loads 1 through 12 are as shown on figure 15.

FIGURE 14. Output voltage, high and low level, test circuit for tables XIII and XIV.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

   (a) C = 150 ± 15 pF (including probe and parasitic capacitance) for sample dynamic tests.

   (b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

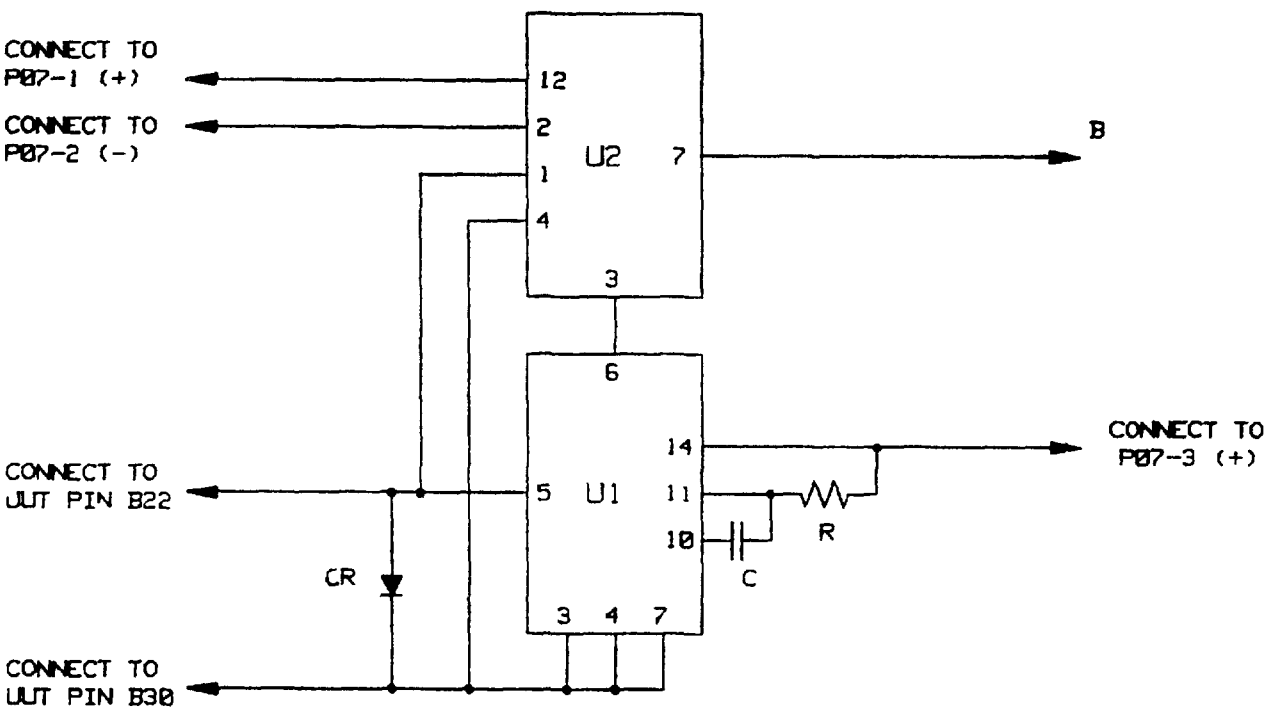
   (c) C not required for static DC tests.

3. Select R1 value for a current flow of 1.6 mA ± 5 percent out of the output pin with the output pin held at 0.4 V.

4. R2 = 4.8 kOhm ± 5 percent, 1/4 W.

FIGURE 15. Logic load circuit.

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NOTES

1. C = 820 pF ± 10 percent (M39014/01-1236 or equivalent)
2. CR = 1N5291 or equivalent.
3 R = 10 kOhm ± 5 percent, 1/4 W.
4. U1 = 74121 or equivalent.
5. U2 = SHC-85 or equivalent.

FIGURE 16. STS logic "1" voltage level test circuit.

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TABLE XV. Leakage current test.

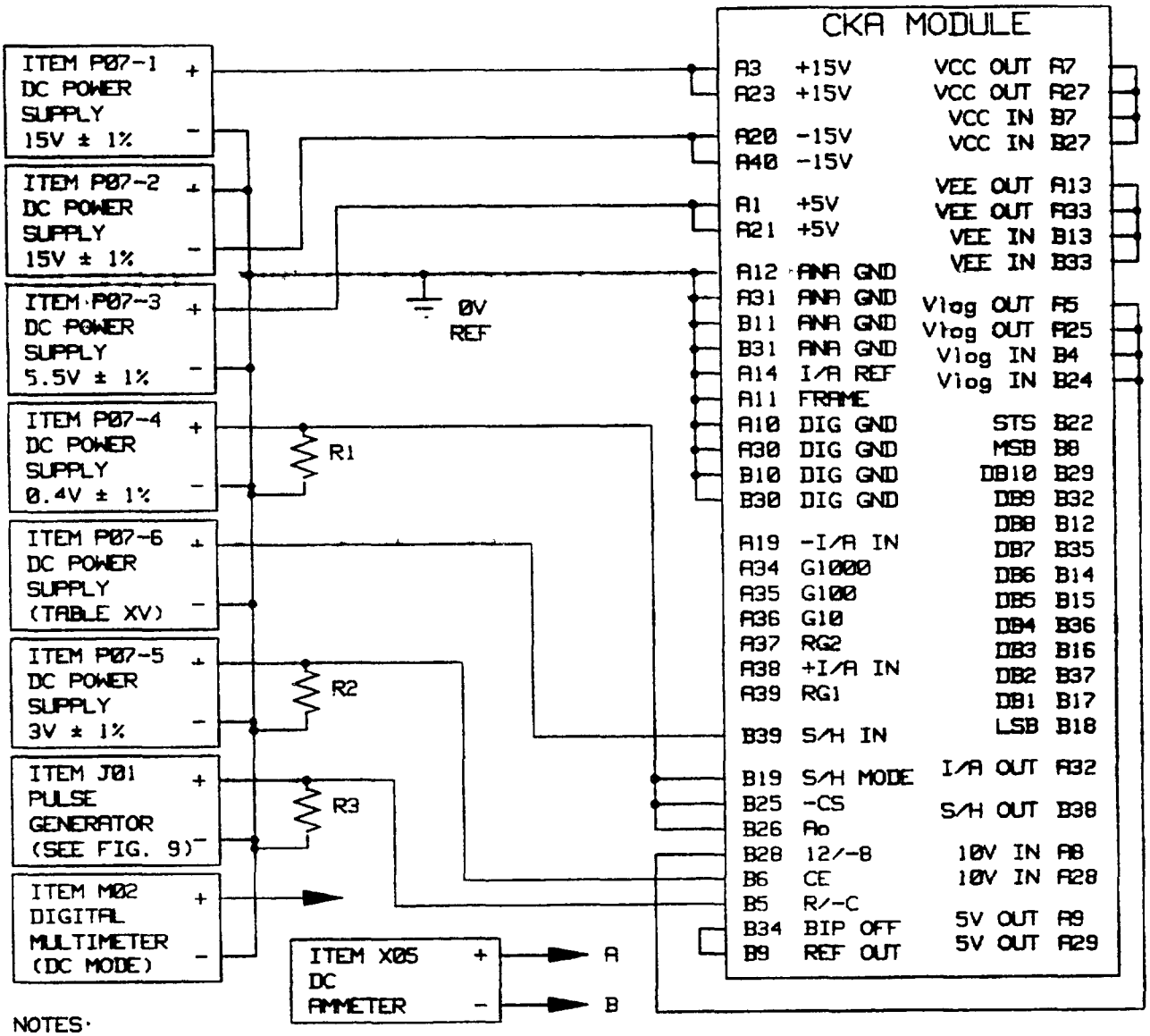
Test cond	Parameter	Connect		Item P07-6 voltage (V) ± 52 mV	25 ± 5°C Initial limits		Units	Fig
		A to pin	B to pin		Min	Max		
P1	LSB output leakage, low	B18	B30	5.25	-40	40	uA	17
P2	DB1 output leakage, low	B17	↓	↓	↓	↓	↓	↓
P3	DB2 output leakage, low	B37						
P4	DB3 output leakage, low	B16						
P5	DB4 output leakage, low	B36						
P6	DB5 output leakage, low	B15						
P7	DB6 output leakage, low	B14						
P8	DB7 output leakage, low	B35						
P9	DB8 output leakage, low	B12						
P10	DB9 output leakage, low	B32						
P11	DB10 output leakage, low	B29	↓	↓				
P12	MSB output leakage, low	B8	B30	5.25				
P13	LSB output leakage, high	B24	B18	-5.25				
P14	DB1 output leakage, high	↓	B17	↓				
P15	DB2 output leakage, high		B37					
P16	DB3 output leakage, high		B16					
P17	DB4 output leakage, high		B36					
P18	DB5 output leakage, high		B15					
P19	DB6 output leakage, high		B14					
P20	DB7 output leakage, high	↓	B35	↓	↓	↓	↓	↓
P21	DB8 output leakage, high	B24	B12	-5.25	-40	40	uA	17

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TABLE XV. Leakage current test - Continued.

Test cond	Parameter	Connect		Item P07-6 voltage (V) ± 52 mV	25 ± 5°C Initial limits		Units	Fig
		A to pin	B to pin		Min	Max		
P22	DB9 output leakage, high	B24	B32	-5.25	-40	40	uA	17
P23	DB10 output leakage, high	B24	B29	-5.25	-40	40	uA	17
P24	MSB output leakage, high	B24	B8	-5.25	-40	40	uA	17

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NOTES:

1. R1 = R2 = R3 = 51 Ohm ± 5 percent, 1W.

2. Item J01 setup
   Single pulse output.
   tHRL = 400 ± 50 nS
   High level = 4.0 ± 0.5 V.
   Low level = 0.2 ± 0.2 V.

3 Verify the appropriate output level with item M02.

FIGURE 17. Leakage current test circuit for table XV.

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TABLE XVI. Reference voltage test.

Test cond	Parameter	25 ± 5°C Initial limits		Units	Fig
		Min	Max		
Q	Internal reference voltage	9.9	10.1	V	18

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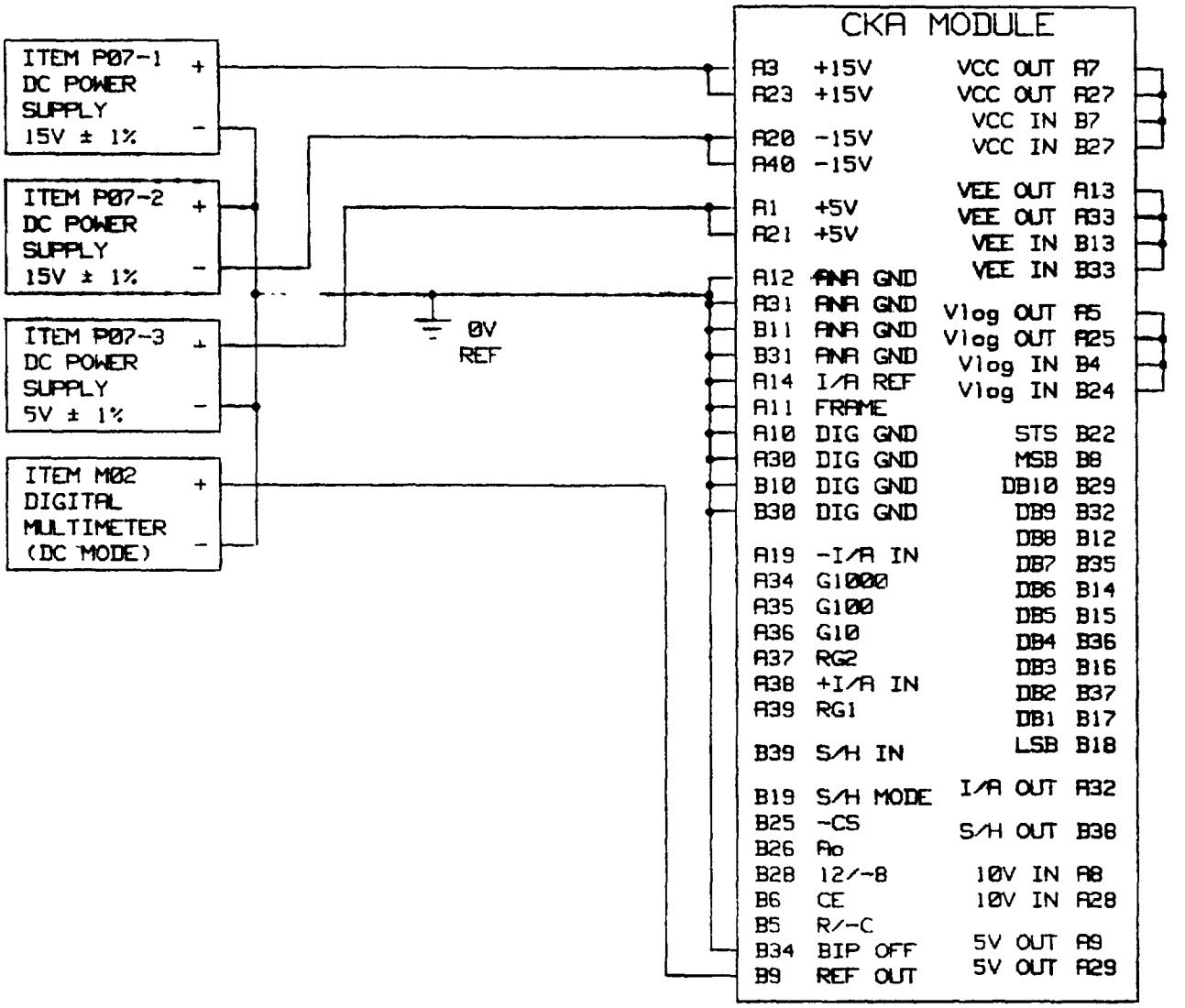


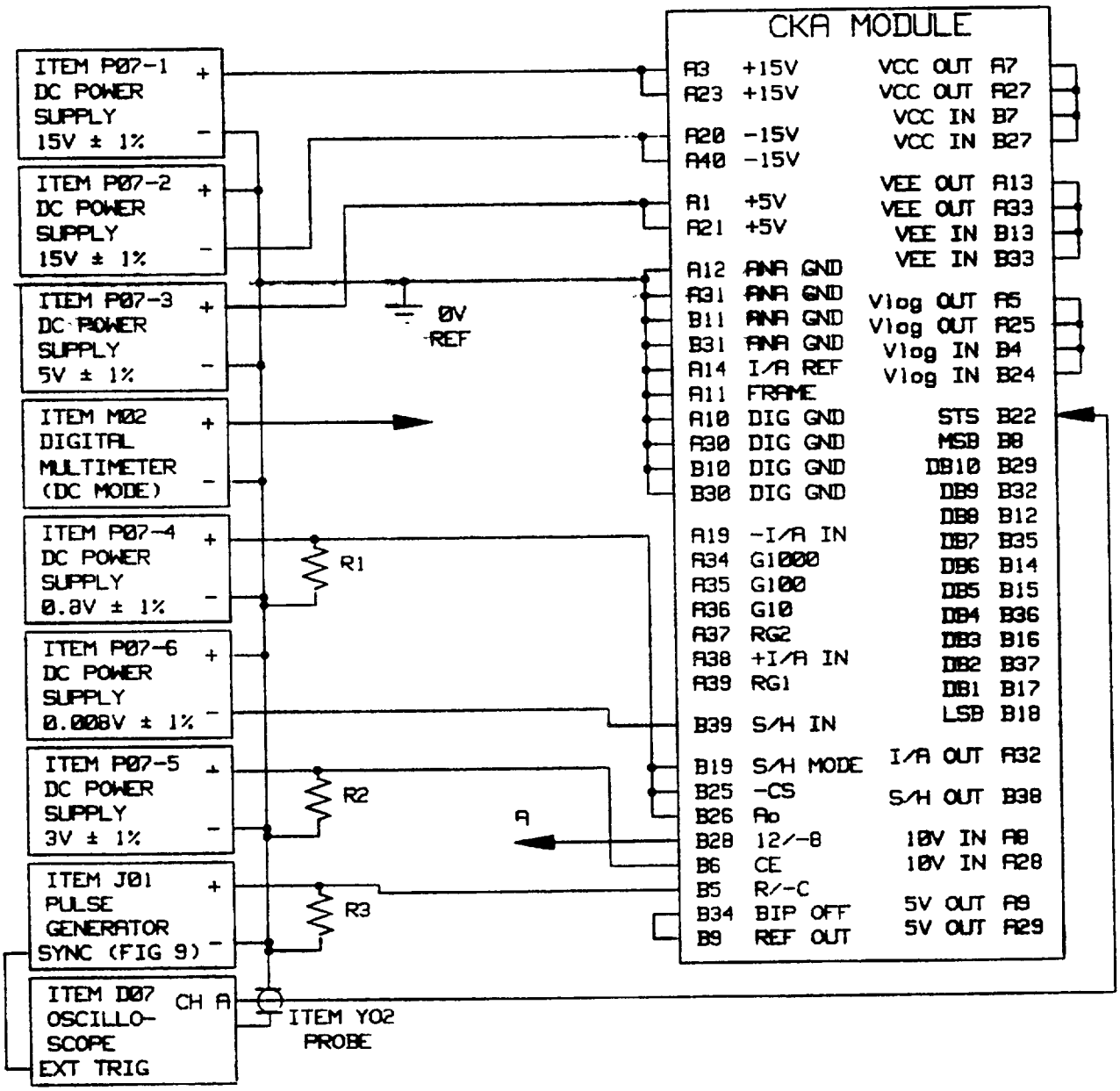
FIGURE 18. Reference voltage test circuit for table XVI.

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TABLE XVII. Conversion time test.

Test cond	Parameter	Connect A to pin	Digital output code word	25 ± 5°C Initial limits		Units	Fig
				Min	Max		
R1	Conversion time, 8-bit	B30	0000 0000 0000	10	28	uS	19
R2	Conversion time, 12-bit	B24	0000 0000 0000	15	40	uS	19

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NOTES:

1. R1 = R2 = R3 = 51 Ohm ± 5 percent, 1 W.

2. Use item M02 to verify that the digital ouput code word is as specified in table XVII.

FIGURE 19. Conversion time test circuit for table XVII.

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TABLE XVIII. Operating modes test.

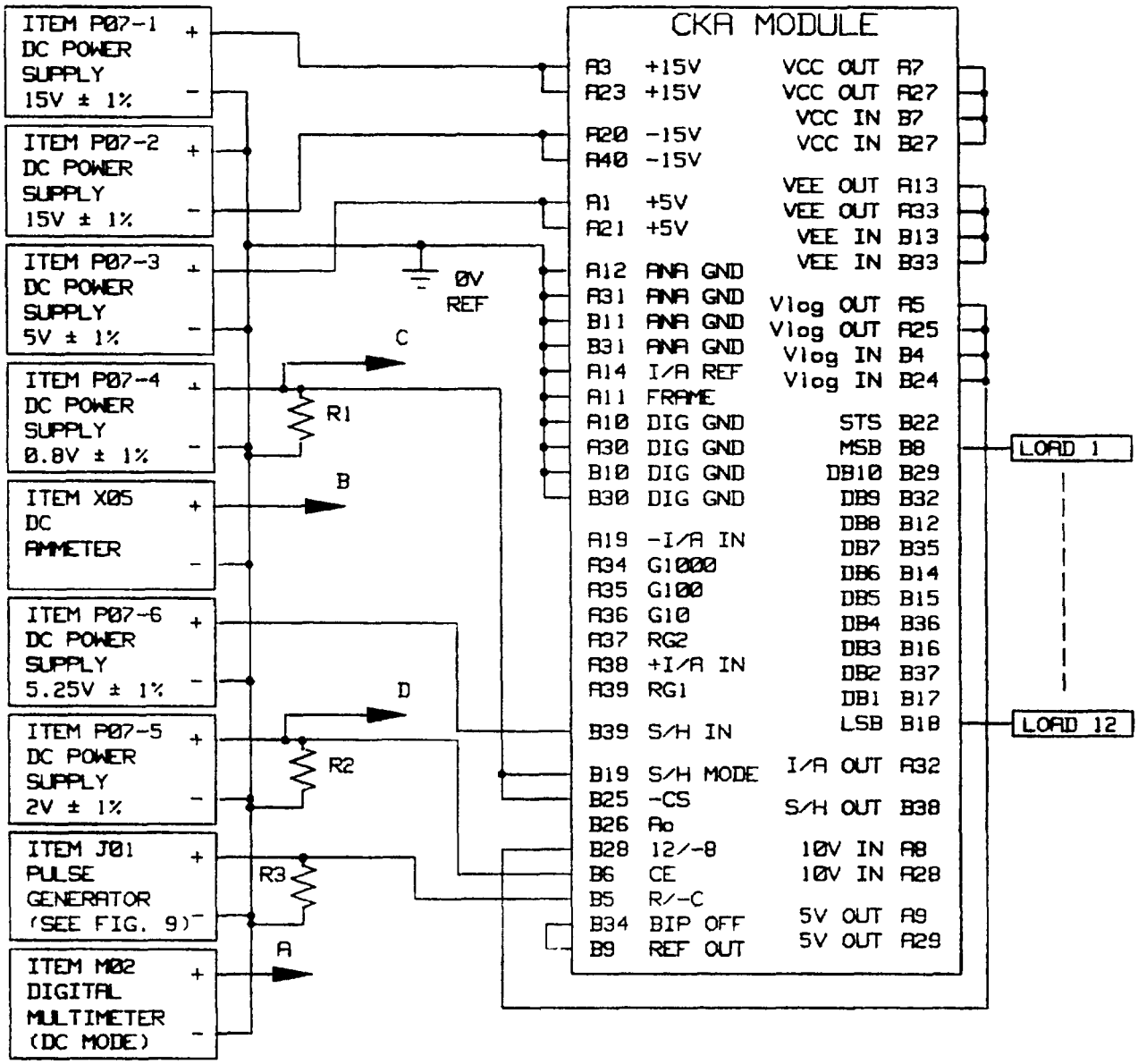
Test cond	Parameter	Connect				25 ± 5°C Initial limits		Units	Fig
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S1	MSB (hi byte)	B8	Open	B26	Open	2.4	5.0	V	20
S2	DB10 (hi byte)	B29	↓	↓	↓	↓	↓	↓	↓
S3	DB9 (hi byte)	B32							
S4	DB8 (hi byte)	B12							
S5	DB7 (hi byte)	B35							
S6	DB6 (hi byte)	B14							
S7	DB5 (hi byte)	B15							
S8	DB4 (hi byte)	B36		B26	Open				
S9	DB3 (lo byte)	B16		Open	B26				
S10	DB2 (lo byte)	B37		↓	↓				
S11	DB1 (lo byte)	B17							
S12	LSB (lo byte)	B18				2.4	5.0	V	20
S13	DB7 (mid nybble)	B35				0 0	0.4	V	20
S14	DB6 (mid nybble)	B14				↓	↓	↓	↓
S15	DB5 (mid nybble)	B15	↓						
S16	DB4 (mid nybble)	B36	Open			0.0	0.4	V	20
S17	MSB (hi nybble, hi-Z)	Open	B8			-40.0	40.0	uA	20
S18	DB10 (hi nybble, hi-Z)	↓	B29			↓	↓	↓	↓
S19	DB9 (hi nybble, hi-Z)		B32	↓	↓				
S20	DB8 (hi nybble, hi-Z)	↓	B12	Open	B26				
S21	DB3 (hi nybble, hi-Z)	Open	B16	B26	Open	-40 0	40.0	uA	20

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TABLE XVIII Operating modes test - Continued.

Test cond	Parameter	Connect				25 ± 5°C Initial limits		Units	Fig
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S22	DB2 (lo nybble)	Open	B37	B26	Open	-40.0	40.0	uA	20
S23	DB1 (lo nybble)	Open	B17	B26	Open	-40.0	40.0	uA	20
S24	LSB (lo nybble)	Open	B18	B26	Open	-40.0	40.0	uA	20

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NOTES:

1. Warning: Changing B26 during the read cycle may result in damage to the module; therefore, make sure that B6 is low or B25 is high before B26 is changed.

2 R1 = R2 = R3 = 51 Ohm ± 5 percent, 1 W.

3 Loads 1 through 12 are as shown on figure 15

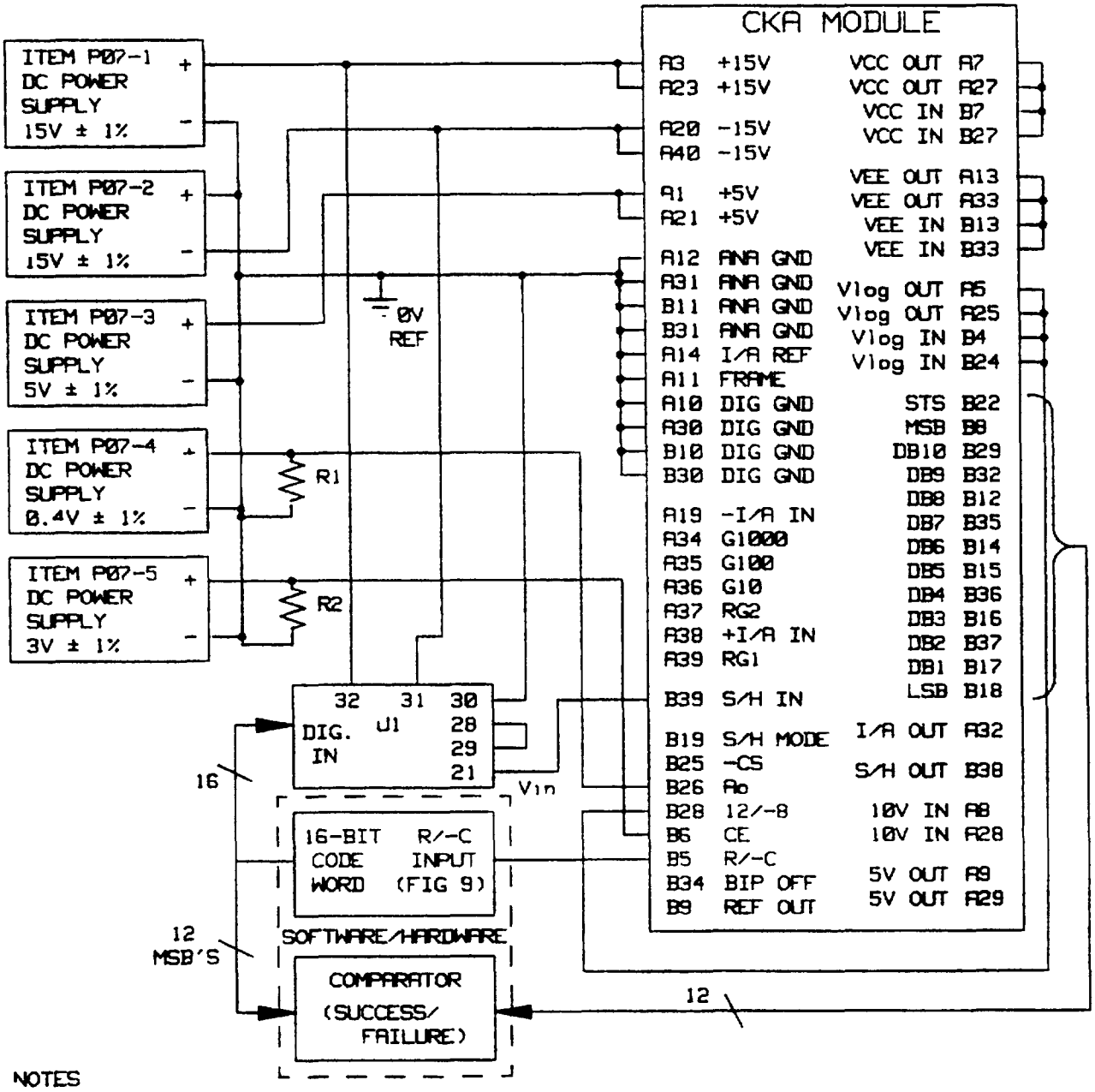
FIGURE 20 Operating modes test circuit for table XVIII.

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TABLE XIX. Supply sensitivity test.

Test cond	Parameter	Supply voltages (V) ± 0.1 V			Digital input code word	Digital output code word	25 ± 5°C Initial limits		Units	Figs
		P07-1	P07-2	P07-3	MSB LSB	MSB LSB	Min	Max		
U1	+PSS1 VCC	+15.0	-15.0V	+5.0	0000 0000 0000	0000 0000 0000	-1	1	LSB	21,22
U2	+PSS1 VCC	+13.5	↓	↓	0000 0000 0000	0000 0000 0000	↓	↓	↓	↓
U3	+PSS1 VCC	+16.5	↓	↓	0000 0000 0000	0000 0000 0000	↓	↓	↓	↓
U4	+PSS1 VCC	+15.0	↓	↓	1111 1111 1111	1111 1111 1111	↓	↓	↓	↓
U5	+PSS1 VCC	+13.5	↓	↓	1111 1111 1111	1111 1111 1111	↓	↓	↓	↓
U6	+PSS1 VCC	+16.5	↓	↓	1111 1111 1111	1111 1111 1111	↓	↓	↓	↓
U7	+PSS2 VCC	+15.0	↓	+5.0	0000 0000 0000	0000 0000 0000	↓	↓	↓	↓
U8	+PSS2 VCC	↓	↓	+4.5	0000 0000 0000	0000 0000 0000	↓	↓	↓	↓
U9	+PSS2 VCC	↓	↓	+5.5	0000 0000 0000	0000 0000 0000	↓	↓	↓	↓
U10	+PSS2 VCC	↓	↓	+5.0	1111 1111 1111	1111 1111 1111	↓	↓	↓	↓
U11	+PSS2 VCC	↓	↓	+4.5	1111 1111 1111	1111 1111 1111	↓	↓	↓	↓
U12	+PSS2 VCC	↓	↓	+5.5	1111 1111 1111	1111 1111 1111	↓	↓	↓	↓
U13	-PSS3 VEE	↓	-15.0	+5.0	0000 0000 0000	0000 0000 0000	↓	↓	↓	↓
U14	-PSS3 VEE	↓	-13.5	↓	0000 0000 0000	0000 0000 0000	↓	↓	↓	↓
U15	-PSS3 VEE	↓	-16.5	↓	0000 0000 0000	0000 0000 0000	↓	↓	↓	↓
U16	-PSS3 VEE	↓	-15.0	↓	1111 1111 1111	1111 1111 1111	↓	↓	↓	↓
U17	-PSS3 VEE	↓	-13.5	↓	1111 1111 1111	1111 1111 1111	↓	↓	↓	↓
U18	-PSS3 VEE	+15.0	-16.5	+5.0	1111 1111 1111	1111 1111 1111	-1	1	LSB	21,22

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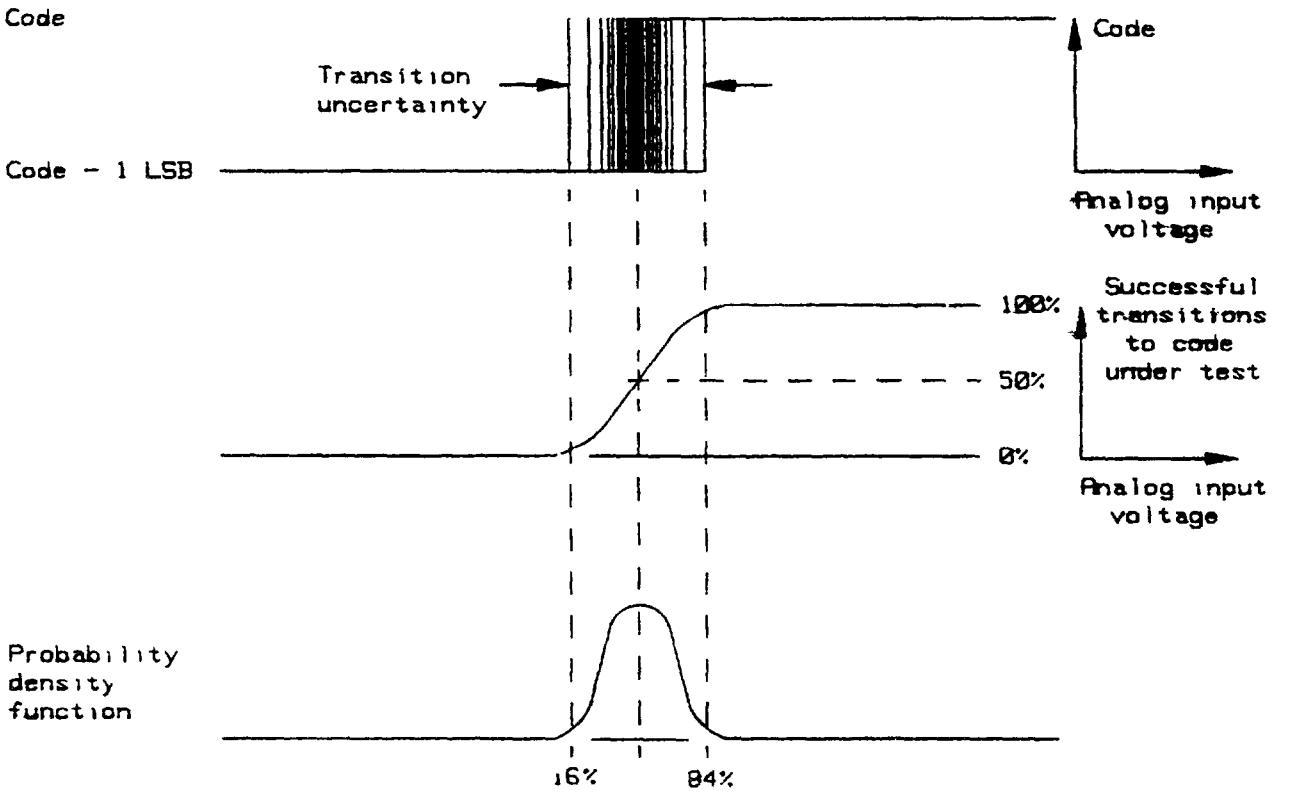


NOTES

1. R1 = R2 = R3 = 51 Ohm ± 5 percent, 1 W.
2. Vin = Vzs + ((Vfs-Vzs)/4094)*(12-bit code) ± TR.

FIGURE 21. Test circuit for tables XIX, XX, XXI, XXII, XXIII, and XXV.

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NOTES:

1. Linearity testing is performed using the end-point straight line method. Actual transition levels are determined statistically by making 100 conversions at a particular analog input and determining the percentage of failures and successes. The criterion for a transition is the input at which the success/fail rate is 50/50 (see figure above).

2. This technique is used industry-wide and is based on the assumption that any noise affecting the ADC is random and, therefore, the transition levels will have a purely Gaussian distribution.

FIGURE 22. Transition uncertainty.

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TABLE XX. Missing code check test.

Test cond	Parameter	Digital input code word MSB LSB	Digital code word (decimal)	Digital output code word MSB LSB	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
V1-V4096	Missing code check	0000 0000 0000 through 1111 1111 1111	N = 0 through N = 4095	0000 0000 0000 through 1111 1111 1111	0	0	Codes	21,23

NOTES

1 Vin = Vzs + ((Vfs - Vzs)/4094)*N ± TR

2. Vzs = zero scale voltage (0 to 1 transition).

3 Vfs = full scale voltage (4094 to 4095 transition).

4. N = input code word in decimal form.

5. TR = voltage difference between the ideal transition voltage and the actual transition voltage (0.5 LSB max).

6 Transition voltage = analog input voltage required to cause a specific output transition between adjacent code words

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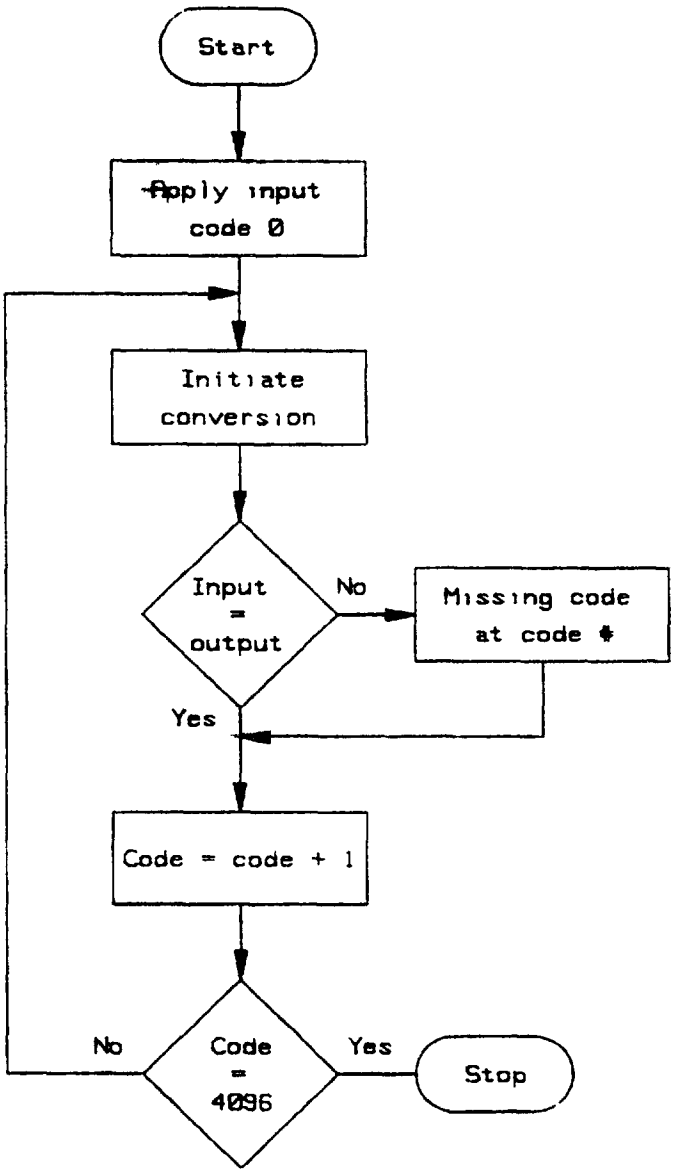


FIGURE 23. Missing code test algorithm for table XX.

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TABLE XXI. ADC offset voltage test.

Test cond	Parameter	Digital input code word MSB LSB	Digital code word (decimal)	Digital output code word MSB LSB	25 ± 5°C Initial limits Min	Max	Units	Figs
W1	Unipolar offset	0000 0000 0000	N = 0	0000 0000 000X	-1 5	2.5	LSB	21,22, 24
W2	Bipolar zero	0111 1111 1111	N = 2047	XXXX XXXX XXXX	-4 5	3.5	LSB	21,22, 24

NOTES

1. Vin = analog input voltage = Vzs + ((Vfs - Vzs)/4094)*N ± TR.
2. Vzs = zero scale voltage (0 to 1 transition).
3. Vfs = full scale voltage (4094 to 4095 transition).
4. N = input code word in decimal form
5. TR = voltage difference between the ideal transition voltage and the actual transition voltage (0 5 LSB max)
6. Transition voltage = analog input voltage required to cause a specific output transition between adjacent code words
7. X = transition point between two adjacent code words

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TABLE XXII. ADC gain error test.

Test cond	Parameter	Digital input code word MSB LSB	Digital code word (decimal)	Digital output code word MSB LSB	25 ± 5°C Initial limits Min	Max	Units	Figs
	#1	0000 0000 0000	N = 0	0000 0000 000X			V	21,22, 24
	#2	1111 1111 1110	N = 4094	1111 1111 111X			V	
X	Bipolar gain error				-0 3	0.3	percent FSR	

NOTES

1 Test condition X reading (percent FSR) = 10*(9.995-(#2-#1))

2. Vin = analog input voltage = Vzs + ((Vfs - Vzs)/4094)*N ± TR.

3. Vzs = zero scale voltage (0 to 1 transition)

4 Vfs = full scale voltage (4094 to 4095 transition).

5 N = input code word in decimal form.

6 TR = voltage difference between the ideal transition voltage and the actual transition voltage (0 5 LSB max).

7 Transition voltage = analog input voltage required to cause a specific output transition between adjacent code words.

8 X = transition point between two adjacent code words.

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TABLE XXIII. ADC linearity error test.

Test cond	Parameter	Digital input code word MSB LSB	Digital code word (decimal)	Digital output code word MSB LSB	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
Y1	Full scale end point	1111 1111 1110	N = 4094	1111 1111 111X	4.9564	5.0264	V	21,22, 24,25
Y2	Zero end point	0000 0000 0000	N = 0	0000 0000 000X	-5.0264	-4.9664	V	
Y3	Linearity error	0000 0000 0001	N = 1	0000 0000 00XX	-0.5	0.5	LSB	21,22, 24,25
Y4	Linearity error	0000 0000 0010	N = 2	0000 0000 001X				
Y5	Linearity error	0000 0000 0011	N = 3	0000 0000 0XXX				
Y6	Linearity error	0000 0000 0100	N = 4	0000 0000 010X				
Y7	Linearity error	0000 0000 0111	N = 7	0000 0000 XXXX				
Y8	Linearity error	0000 0000 1000	N = 8	0000 0000 100X				
Y9	Linearity error	0000 0000 1111	N = 15	0000 0000 XXXX				
Y10	Linearity error	0000 0001 0000	N = 16	0000 0001 000X				
Y11	Linearity error	0000 0001 1111	N = 31	0000 00XX XXXX				
Y12	Linearity error	0000 0010 0000	N = 32	0000 0010 000X				
Y13	Linearity error	0000 0011 1111	N = 63	0000 0XXX XXXX				
Y14	Linearity error	0000 0100 0000	N = 64	0000 0100 000X				
Y15	Linearity error	0000 0111 1111	N = 127	0000 XXXX XXXX				
Y16	Linearity error	0000 1000 0000	N = 128	0000 1000 000X				
Y17	Linearity error	0000 1111 1111	N = 255	000X XXXX XXXX				
Y18	Linearity error	0001 0000 0000	N = 256	0001 0000 000X				
Y19	Linearity error	0001 1111 1111	N = 511	00XX XXXX XXXX				
Y20	Linearity error	0010 0000 0000	N = 512	0010 0000 000X				
Y21	Linearity error	0010 1111 1111	N = 767	001X XXXX XXXX				
Y22	Linearity error	0011 0000 0000	N = 768	0011 0000 000X				
Y23	Linearity error	0011 1111 1111	N = 1023	0XXX XXXX XXXX				
Y24	Linearity error	0100 0000 0000	N = 1024	0100 0000 000X	-0.5	0.5	LSB	21,22, 24,25

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TABLE XXIII. ADC linearity error test - Continued.

Test cond	Parameter	Digital input code word MSB LSB	Digital code word (decimal)	Digital output code word MSB LSB	25 ± 5°C Initial limits Min	Max	Units	Figs
Y25	Linearity error	0100 1111 1111	N = 1279	010X XXXX XXXX	-0.5	0.5	LSB	21, 22, 24, 25
Y26	Linearity error	0101 0000 0000	N = 1280	0101 0000 000X				
Y27	Linearity error	0101 1111 1111	N = 1535	01XX XXXX XXXX				
Y28	Linearity error	0110 0000 0000	N = 1536	0110 0000 000X				
Y29	Linearity error	0110 1111 1111	N = 1791	011X XXXX XXXX				
Y30	Linearity error	0111 0000 0000	N = 1792	0111 0000 000X				
Y31	Linearity error	0111 1111 1111	N = 2047	XXXX XXXX XXXX				
Y32	Linearity error	1000 0000 0000	N = 2048	1000 0000 000X				
Y33	Linearity error	1000 1111 1111	N = 2303	100X XXXX XXXX				
Y34	Linearity error	1001 0000 0000	N = 2304	1001 0000 000X				
Y35	Linearity error	1001 1111 1111	N = 2559	10XX XXXX XXXX				
Y36	Linearity error	1010 0000 0000	N = 2560	1010 0000 000X				
Y37	Linearity error	1010 1111 1111	N = 2815	101X XXXX XXXX				
Y38	Linearity error	1011 0000 0000	N = 2816	1011 0000 000X				
Y39	Linearity error	1011 1111 1111	N = 3071	1XXX XXXX XXXX				
Y40	Linearity error	1100 0000 0000	N = 3072	1100 0000 000X				
Y41	Linearity error	1100 1111 1111	N = 3327	110X XXXX XXXX				
Y42	Linearity error	1101 0000 0000	N = 3328	1101 0000 000X				
Y43	Linearity error	1101 1111 1111	N = 3583	11XX XXXX XXXX				
Y44	Linearity error	1110 0000 0000	N = 3584	1110 0000 000X				
Y45	Linearity error	1110 1111 1111	N = 3839	111X XXXX XXXX				
Y46	Linearity error	1111 0000 0000	N = 3840	1111 0000 000X				
Y47	Linearity error	1111 0111 1111	N = 3967	1111 XXXX XXXX				
Y48	Linearity error	1111 1000 0000	N = 3969	1111 1000 000X	-0.5	0.5	LSB	21, 22, 24, 25

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TABLE XXIII. ADC linearity error test - Continued.

Test cond	Parameter	Digital input code word MSB LSB	Digital code word (decimal)	Digital output code word MSB LSB	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
Y49	Linearity error	1111 1011 1111	N = 4031	1111 1XXX XXXX	-0.5	0.5	LSB	21,22, 24,25
Y50	Linearity error	1111 1100 0000	N = 4032	1111 1100 000X				
Y51	Linearity error	1111 1101 1111	N = 4063	1111 11XX XXXX				
Y52	Linearity error	1111 1110 0000	N = 4064	1111 1110 000X				
Y53	Linearity error	1111 1110 1111	N = 4079	1111 111X XXXX				
Y54	Linearity error	1111 1111 0000	N = 4080	1111 1111 000X				
Y55	Linearity error	1111 1111 0111	N = 4087	1111 1111 XXXX				
Y56	Linearity error	1111 1111 1000	N = 4088	1111 1111 100X				
Y57	Linearity error	1111 1111 1011	N = 409'	1111 1111 1XXX				
Y58	Linearity error	1111 1111 1100	N = 4092	1111 1111 110X				
Y59	Linearity error	1111 1111 1101	N = 4093	1111 1111 11XX				21,22,
Y60	Linearity error	1111 1111 1110	N = 4094	1111 1111 111X	-0.5	0.5	LSB	24,25

NOTES

1. Vin = analog input voltage = Vzs + ((Vfs - Vzs)/4094)*N ± TR.
2. Vzs = zero scale voltage (0 to 1 transition voltage)
3. Vfs = full scale voltage (4094 to 4095 transition voltage)
4. N = input code word in decimal form
5. TR = voltage difference between the actual and ideal transitions (± 0.5 LSB max).
6. X = transition point between two adjacent code words

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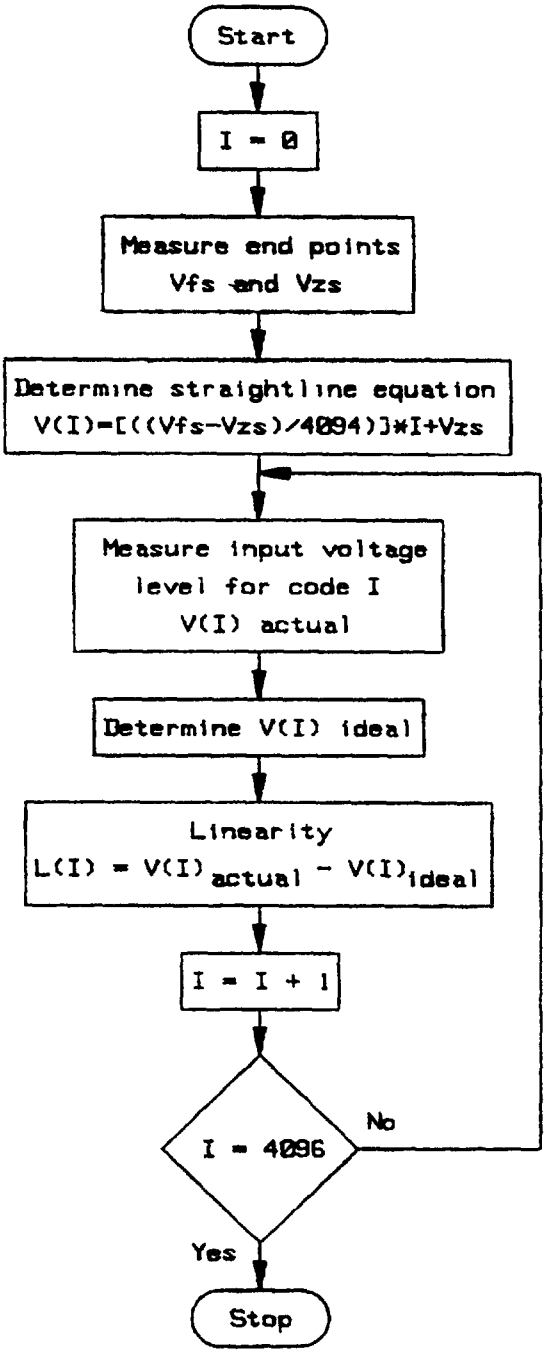
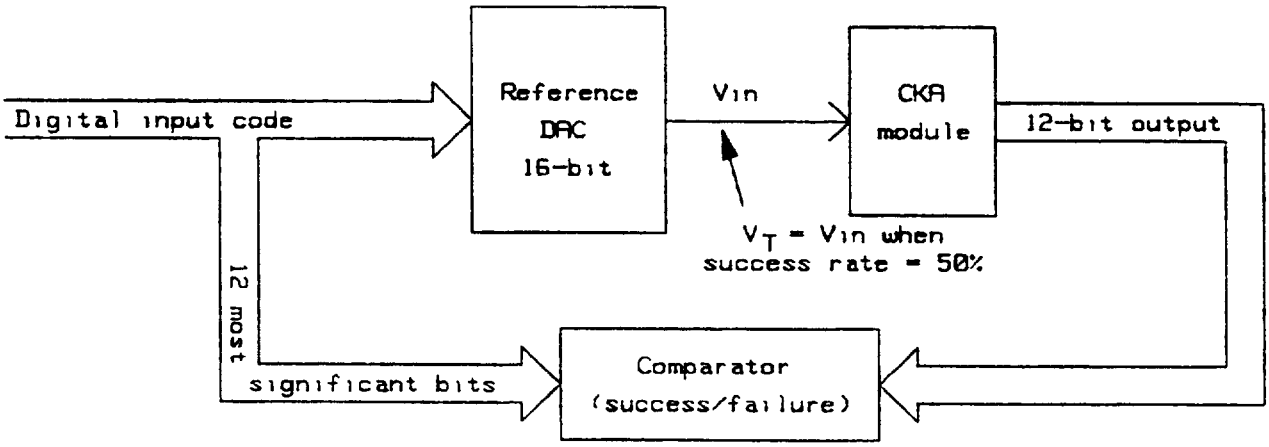


FIGURE 24. Linearity error test algorithm for tables XXI, XXII, and XXIII.

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Basic implementation for integral linearity test

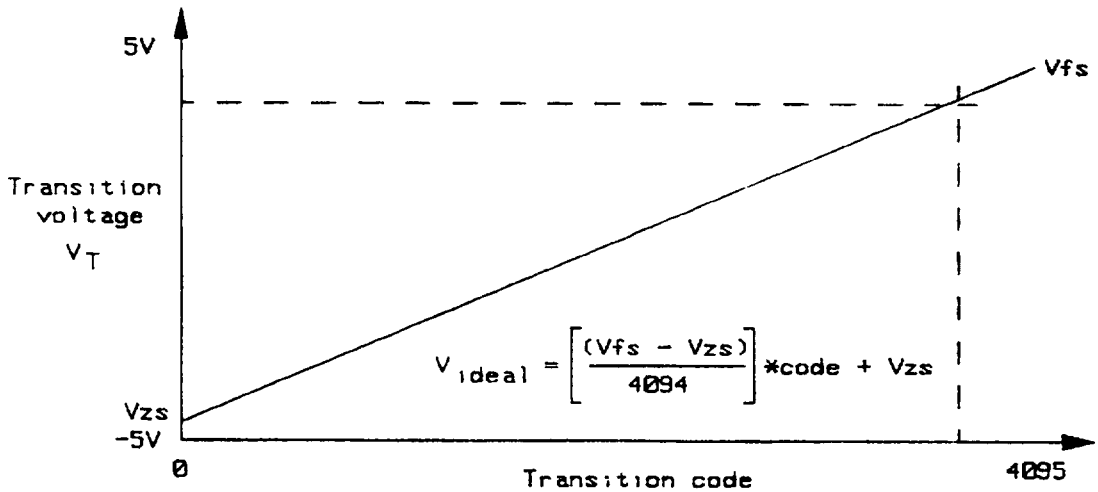


FIGURE 25. Endpoint straightline method for integral linearity.

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TABLE XXIV. ADC transition time test.

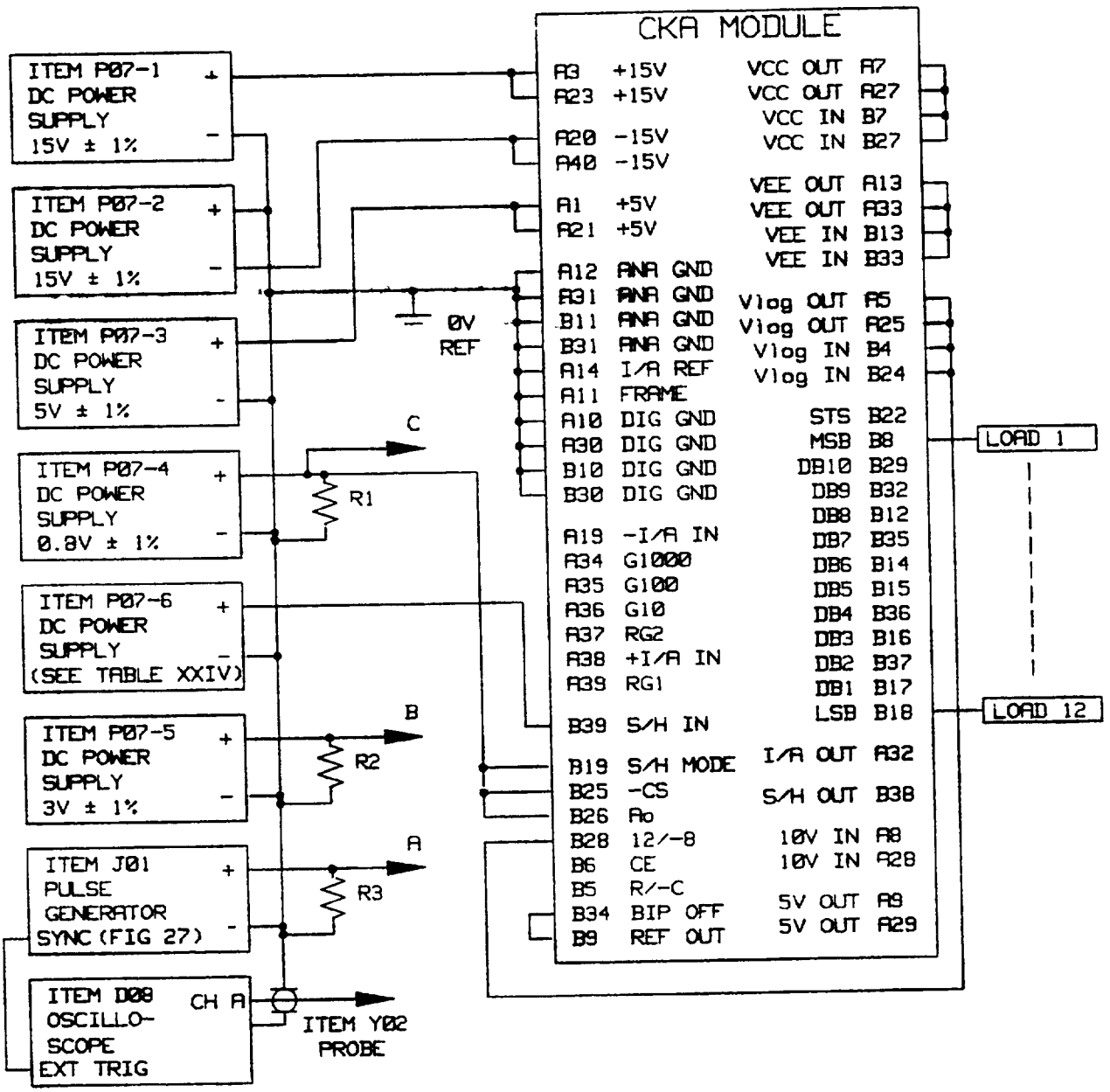
Test cond	Parameter	Item P07-6 voltage (V) ± 52 mV	Connect			25 ± 5°C Initial limits		Units	Fig
			Probe to pin	Pin B5 to	Pin B6 to	Min	Max		
AA1	tDSC	-5.25	B22	C	A	0	450	nS	26,27
AA2	tDD1	-5.25	B8	B		0	250	nS	26,27
AA3	tDD2	-5.25	B14						
AA4	tDD3	-5.25	B18						
AA5	tDD4	5.25	B8						
AA6	tDD5	5.25	B14						
AA7	tDD6	5.25	B18			0	250		
AA8	tHD1	-5.25	B8			25	INF		
AA9	tHD2	-5.25	B14						
AA10	tHD3	-5.25	B18						
AA11	tHD4	5.25	B8						
AA12	tHD5	5.25	B14						
AA13	tHD6	5.25	B18	B	A	25	INF	nS	26,27
AA14	tHS1	-5.25	B8	A	B	200	1000	nS	26,27
AA15	tHS2	-5.25	B14						
AA16	tHS3	-5.25	B18						
AA17	tHS4	5.25	B8						
AA18	tHS5	5.25	B14						
AA19	tHS6	5.25	B18			200	1000	nS	26,27
AA20	tHL1	-5.25	B8			0	250	nS	26,27
AA21	tHL2	-5.25	B14	A	B	0	250	nS	26,27

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TABLE XXIV CDC transition time test - Continued.

Test cond	Parameter	Item P27-5 voltage (V) ± 52 mV	Connect			25 ± 3°C Initial limits		Units	Figs
			Probe to pin	Pin B5 to	Pin B6 to	Min	Max		
AA22	tHL3	-5.25	B18	A	B	0	250	nS	26,27
AA23	tHL4	5.25	B8	↓	↓	↓	↓	↓	↓
AA24	tHL5	5.25	B14						
AA25	tHL6	5.25	B18			0	250	nS	26,27
AA26	tDS1	-5.25	B22			0	650	nS	26,27
AA27	tDDR1	-5.25	B8			0	250	nS	26,27
AA28	tDDR2	-5.25	B14						
AA29	tDDR3	-5.25	B18						
AA30	tDDR4	5.25	B8						
AA31	tDDR5	5.25	B14						
AA32	tDDR6	5.25	B19			0	250	nS	26,27
AA33	tHDR1	-5.25	B8			25	INF	nS	26,27
AA34	tHDR2	-5.25	B14						
AA35	tHDR3	-5.25	B18						
AA36	tHDR4	5.25	B8						
AA37	tHDR5	5.25	B14	↓	↓	↓	↓	↓	↓
AA38	tHDR6	5.25	B18	A	B	25	INF	nS	26,27

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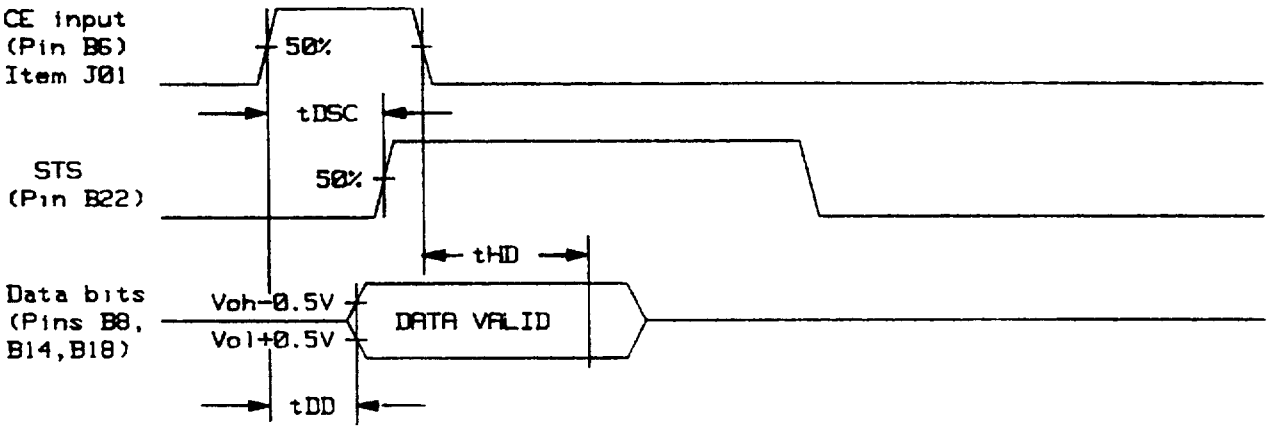


NOTES:

1. R1 = R2 = R3 = 51 Ohm ± 5 percent, 1 W
2. Loads 1 through 12 are as shown on figure 15.

FIGURE 26. Transition time test circuit for table XXIV.

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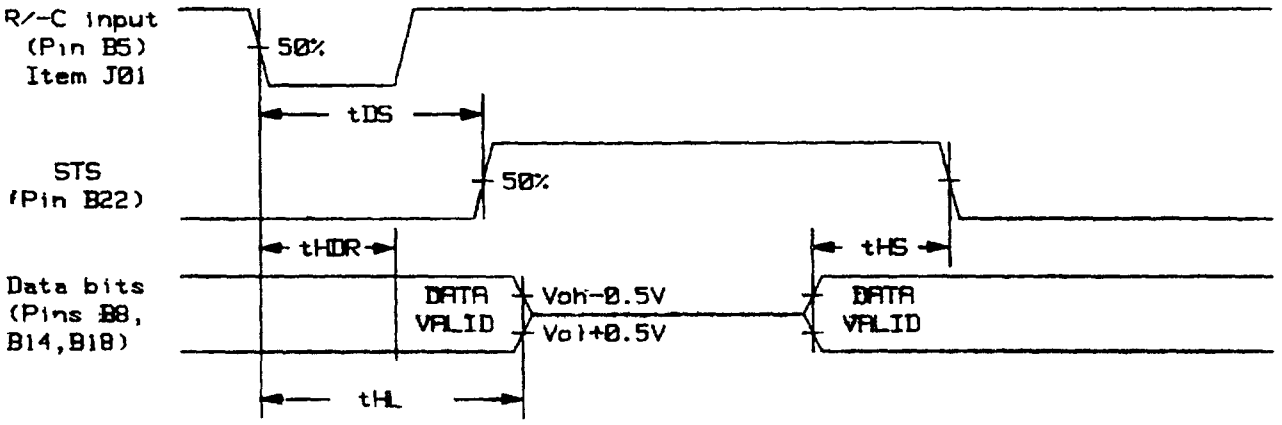


NOTE:

1. Item J01 setup:
   Single pulse output
   High pulse width = 350 ± 50 nS.
   Rise and fall times = 25 nS max.
   High level = 2.02 ± 0.02 V.
   Low level = 0 4 ± 0 4 V

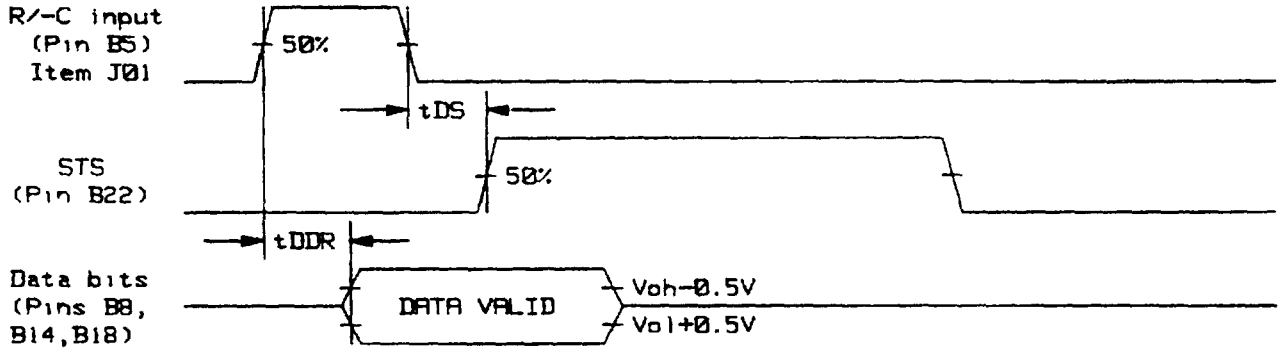
FIGURE 27a. Transition time test waveforms.

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NOTES:

1. Item J01 setup:
   Single pulse output.
   Low pulse width = 400 ± 50 nS
   Rise and fall times = 25 nS max
   High level = 2.02 ± 0 02 V.
   Low level = 0.4 ± 0.4 V



2. Item J01 setup:
   Single pulse output.
   High pulse width = 350 ± 50 nS.
   Rise and fall times = 25 nS max.
   High level = 2.0 V min, 5.5 V max.
   Low level = -0.5 V min, 0.8 V max.

FIGURE 27b. Transition time test waveforms - Continued.

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TABLE XXV. ADC transition error test.

Test cond	Parameter	Digital input code word MSB LSB	Digital code word (decimal)	Digital output code word MSB LSB	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
BB1	Midpoint error	0111 1111 1111	N = 2047	0111 1111 1111	-0 5	0.5	LSB	21,22, 28
BB2	Endpoint error	1111 1111 1110	N = 4094	1111 1111 1110	-0 5	0.5	LSB	

NOTES

1. Vin = analog input voltage = Vzs + ((Vfs - Vzs)/4094)*N ± TR.

2. Vzs = zero scale voltage (0 to 1 transition).

3. Vfs = full scale voltage (4094 to 4095 transition)

4. N = input code word in decimal form.

5. TR = voltage difference between the ideal transition voltage and the actual transition voltage (0 5 LSB max).

6. Transition voltage = analog input voltage required to cause a specific output transition between adjacent code words.

7. X = transition point between two adjacent code words

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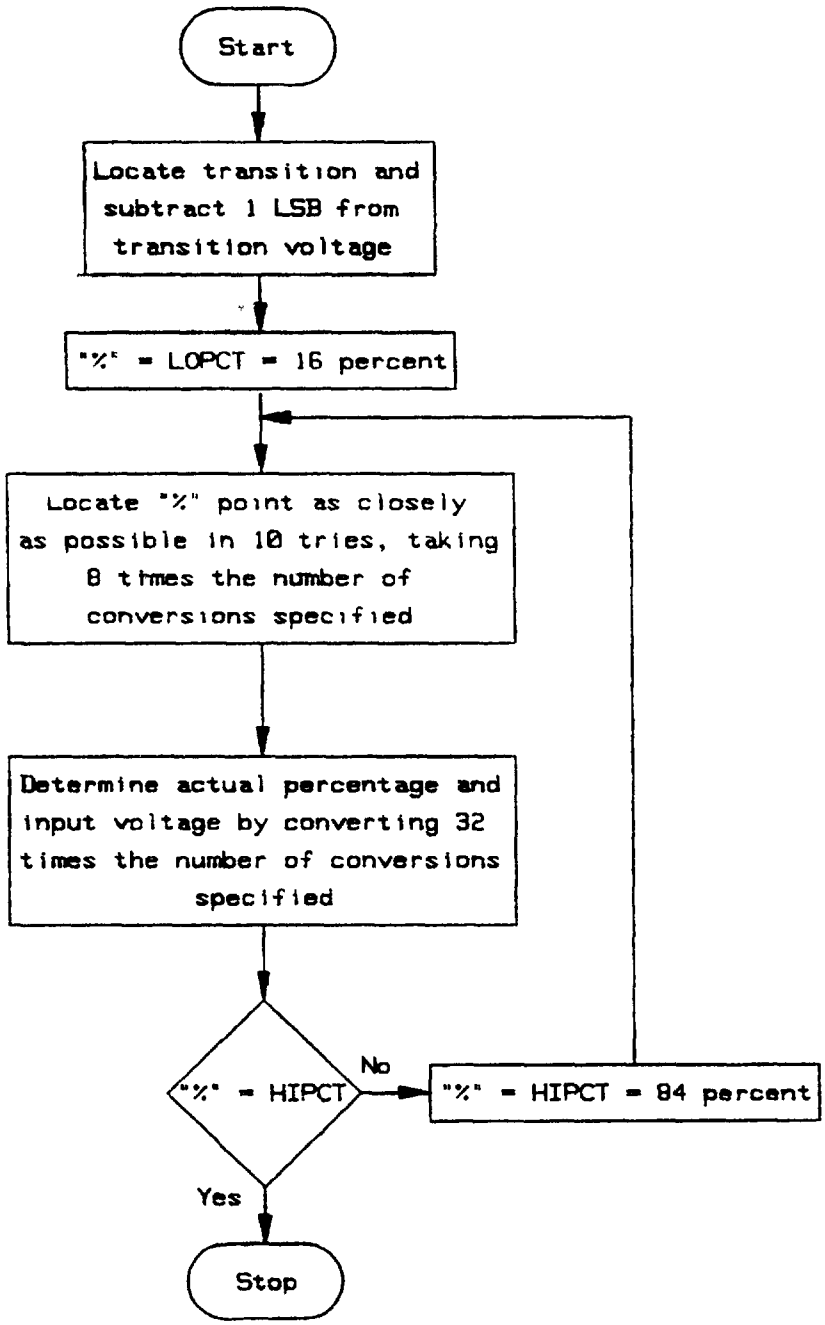


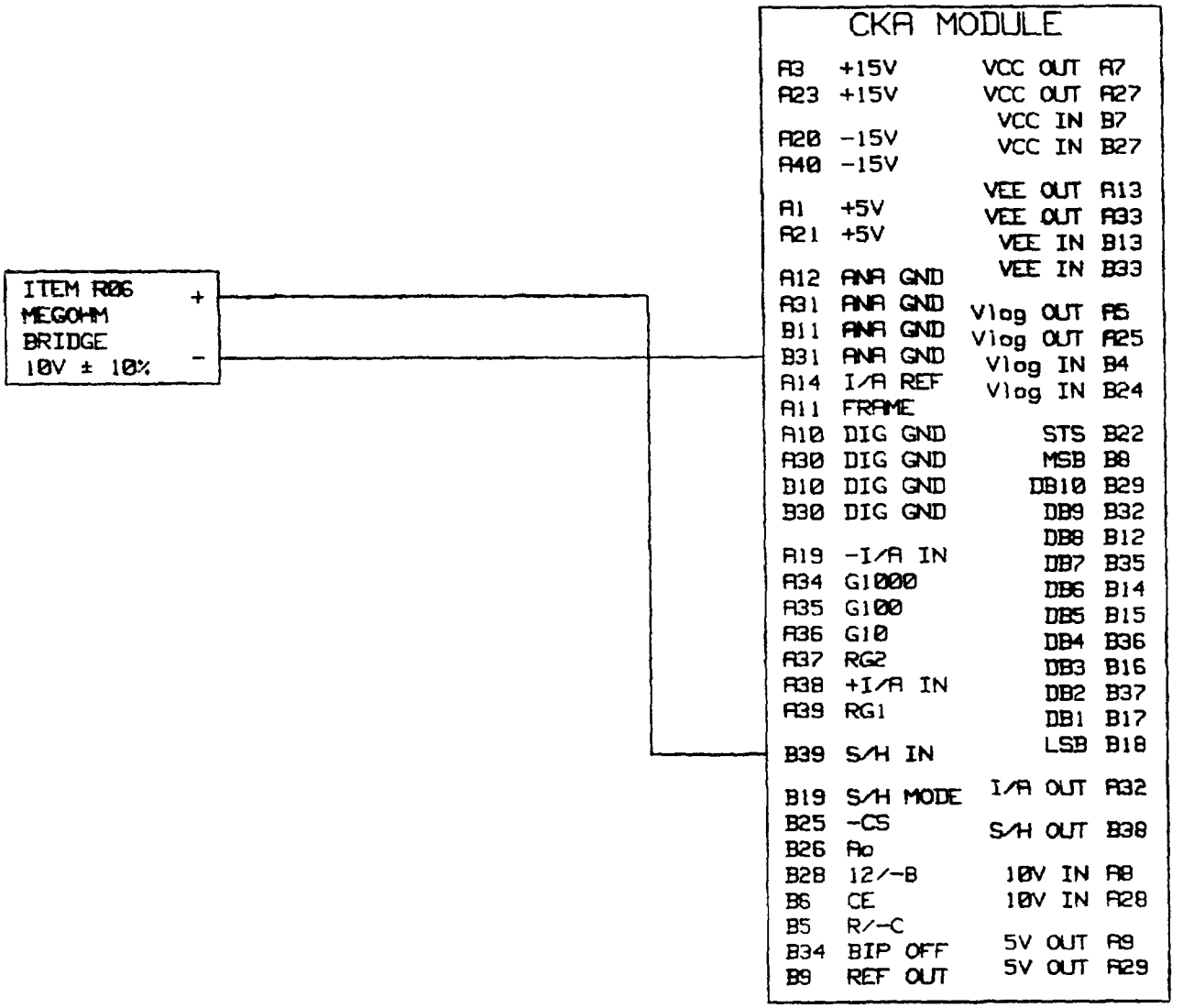
FIGURE 28. ADC transition error test algorithm for table XXV.

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TABLE XXVI. S/H input impedance test.

Test cond	Parameter	25 ± 5°C Initial limits		Units	Fig
		Min	Max		
CC	S/H input impedance	1	INF	MOhm	29

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NOTE:

1. Caution: The output of the megohm bridge shall not exceed 10V ± 10 percent.

FIGURE 29. S/H input impedance test circuit for table XXVI

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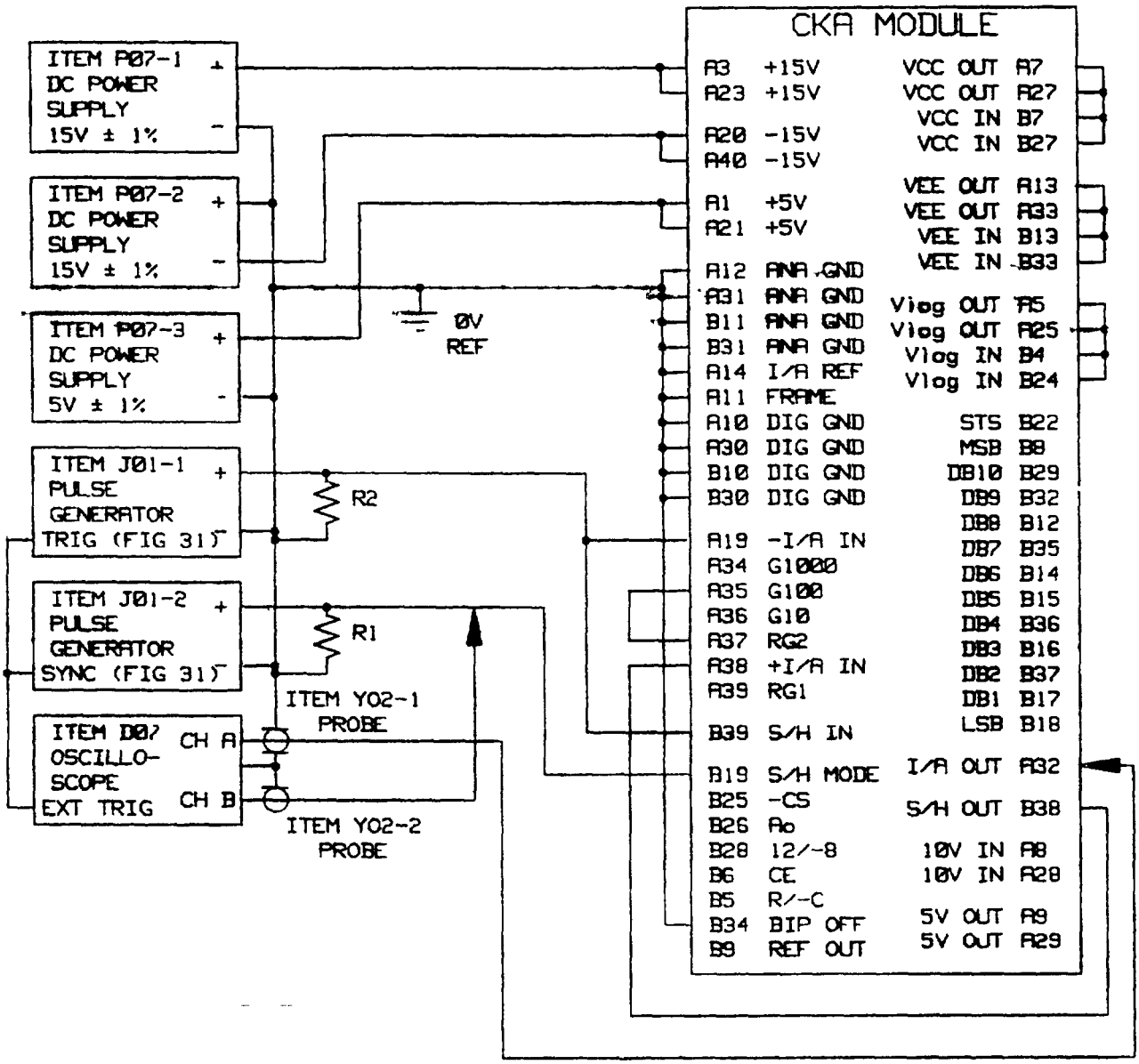
TABLE XXVII. S/H acquisition limit test.

Test cond	Parameter	25 ± 5°C Initial limits		Units	Fig
		Min	Max		
EE	S/H acquisition limit	-1	1	V	30,31

NOTE:

1. Successful completion of this test condition verifies a maximum acquisition time of 1.5 us.

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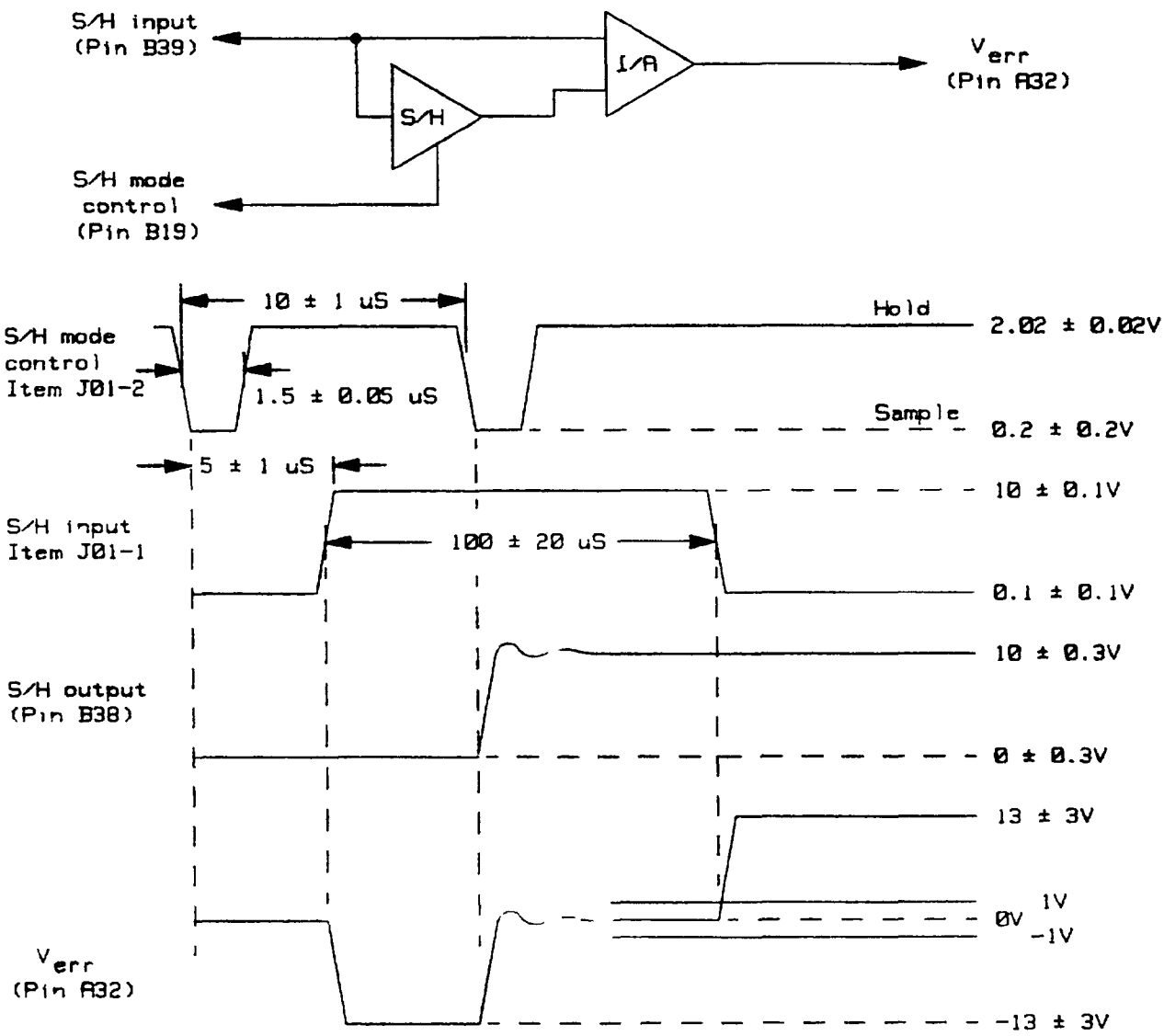


NOTE:

1. R1 = R2 = 51 Ohm ± 5 percent, 1 W

FIGURE 30. S/H acquisition time test circuit for table XXVII.

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NOTES

1. Item J01-1 setup:
   Single pulse output.
   Rise time = fall time = 100 nS max.

2. Item J01-2 setup:
   Double pulse output.
   Rise time = fall time = 25 nS max.

FIGURE 31. S/H acquisition time test waveforms for table XXVII.

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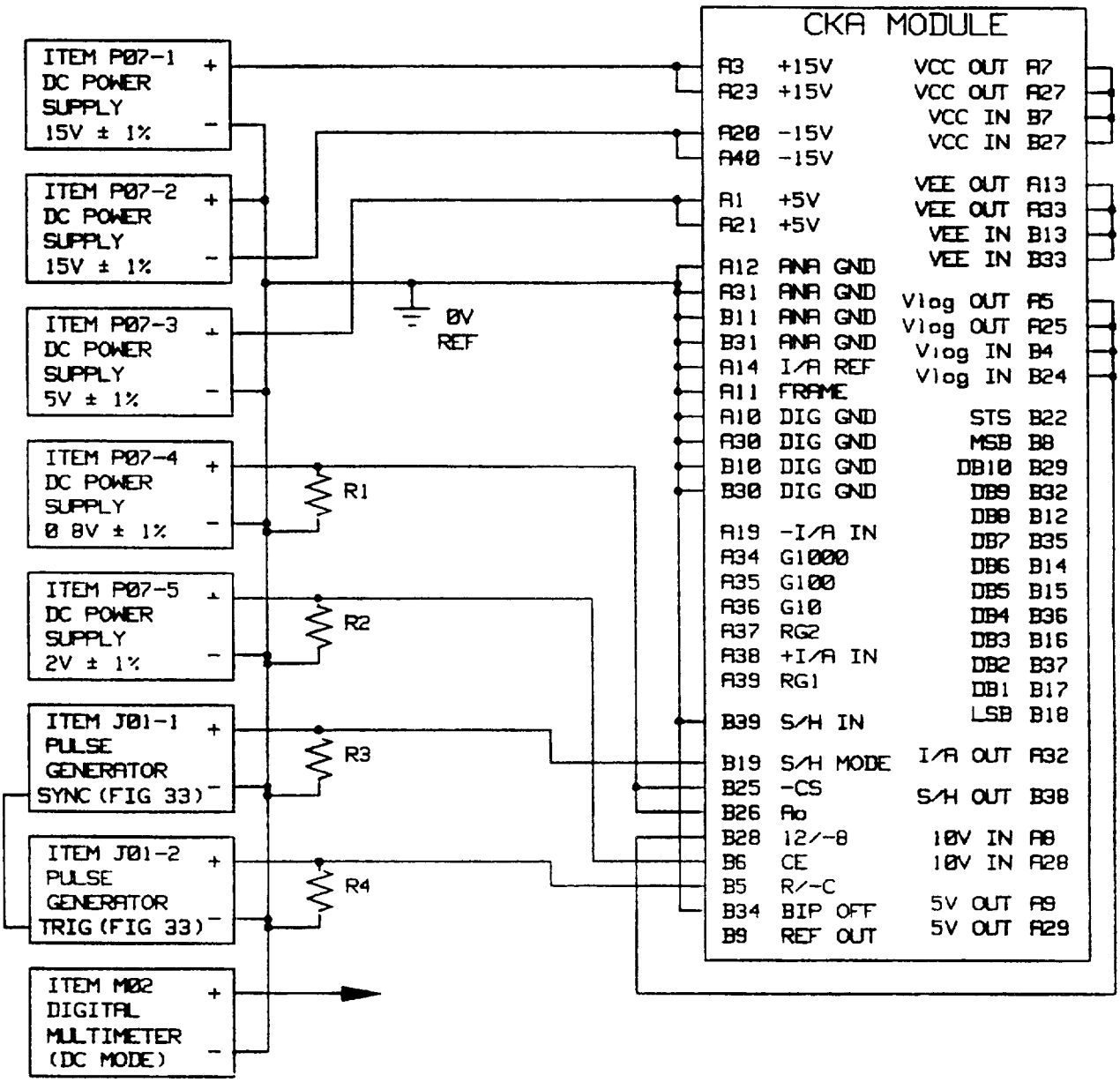
TABLE XXVIII. S/H droop rate test.

Test cond	Parameter	25 ± 5°C Initial limits		Units	Figs
		Min	Max		
FF	S/H droop rate	-100	100	mV	32,33

*NOTES:

1. Test condition FF reading (mV) = (2.44*N)+Vzs.
2. Vzs = zero scale voltage (0 to 1 transition).
3. N = output code word in decimal form.
4. Table I limits (uV/uS) = table XXVIII reading/tPD

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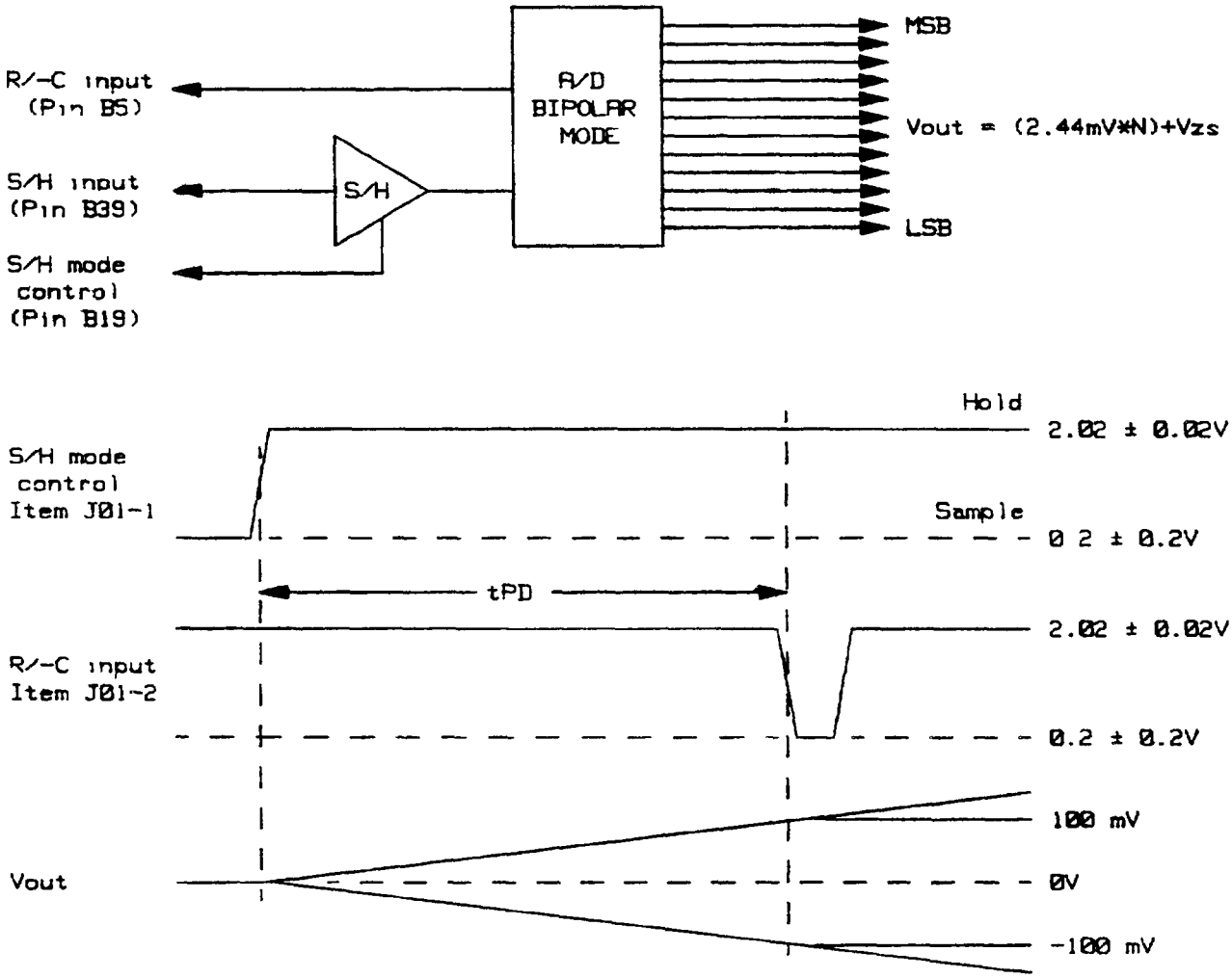


NOTES

1. R1 = R2 = R3 = R4 = 51 Ohm ± 5 percent, 1 W.
2. Measure the digital output code word using item M02.
3 Analog output voltage (mV) = (2.44*N)+Vzs

FIGURE 32. S/H droop rate test circuit for table XXVIII.

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NOTES:

1. Item J01-1 setup
   Single pulse output
   High pulse width = 500 ± 50 mS.
   Rise time = fall time = 1 uS max.

2. Item J01-2 setup:
   Single pulse output.
   Low pulse width = 500 ± 50 nS.
   Rise time = fall time = 25 nS max

3. tPD = 200 ± 5 mS (at 25°C) and 10 ± 0.1 mS (at 0°C and 60°C).

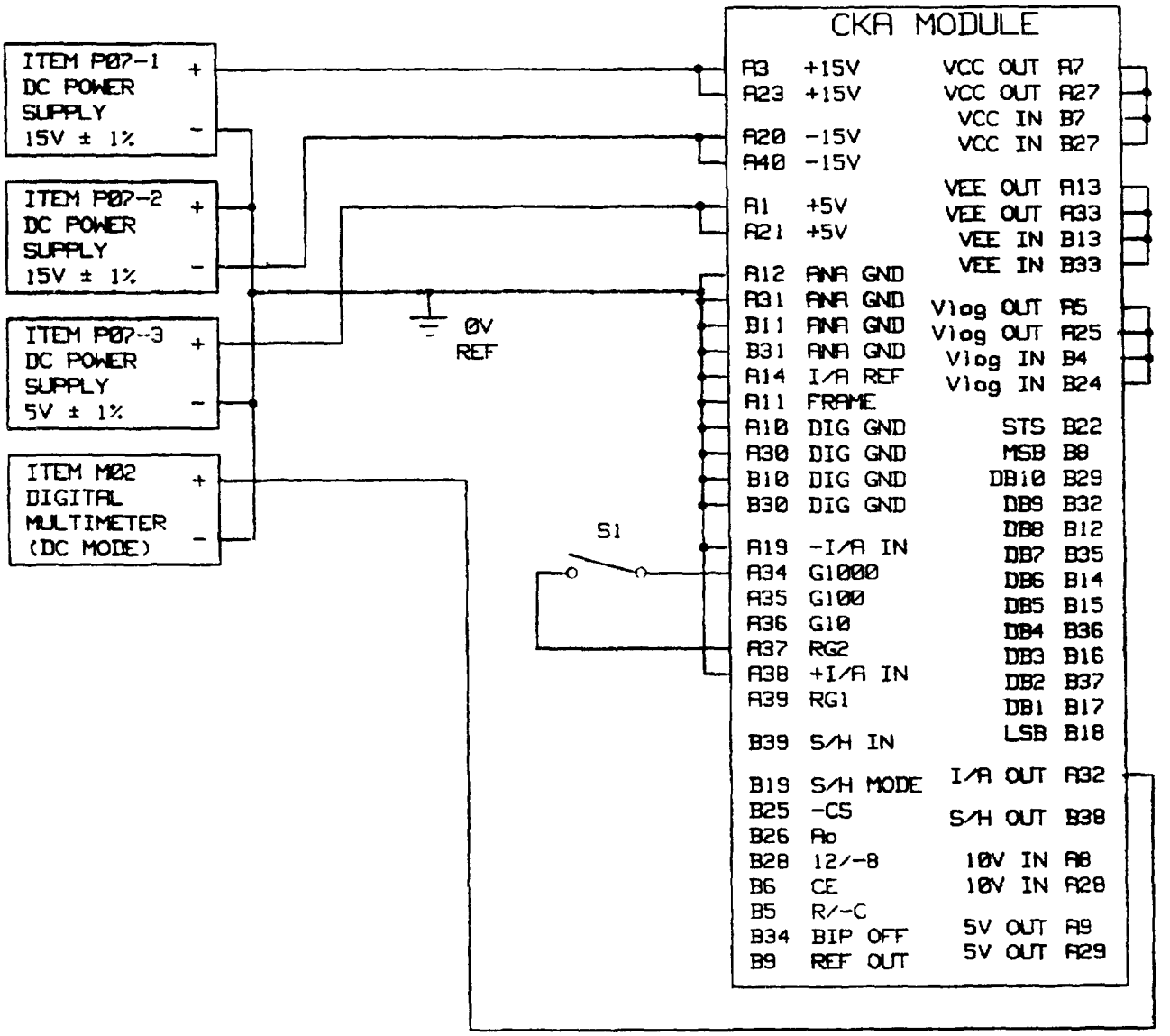
FIGURE 33. S/H droop rate test waveforms for table XXVIII

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TABLE XXIX. I/A offset voltage test.

Test cond	Parameter	S1 position	Measurement/ computation	25 ± 5°C Initial limits		Units	Fig
				Min	Max		
		Open	#1			V	34
		Closed	#2			V	34
GG1	Output offset voltage		GG1 = ((1000*#1)-#2)/999	-3.25	3.25	mV	34
GG2	Input offset voltage		GG2 = (#2-#1)/999	-0.11	0 11	mV	34

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NOTE:

1 S1 = SPST switch.

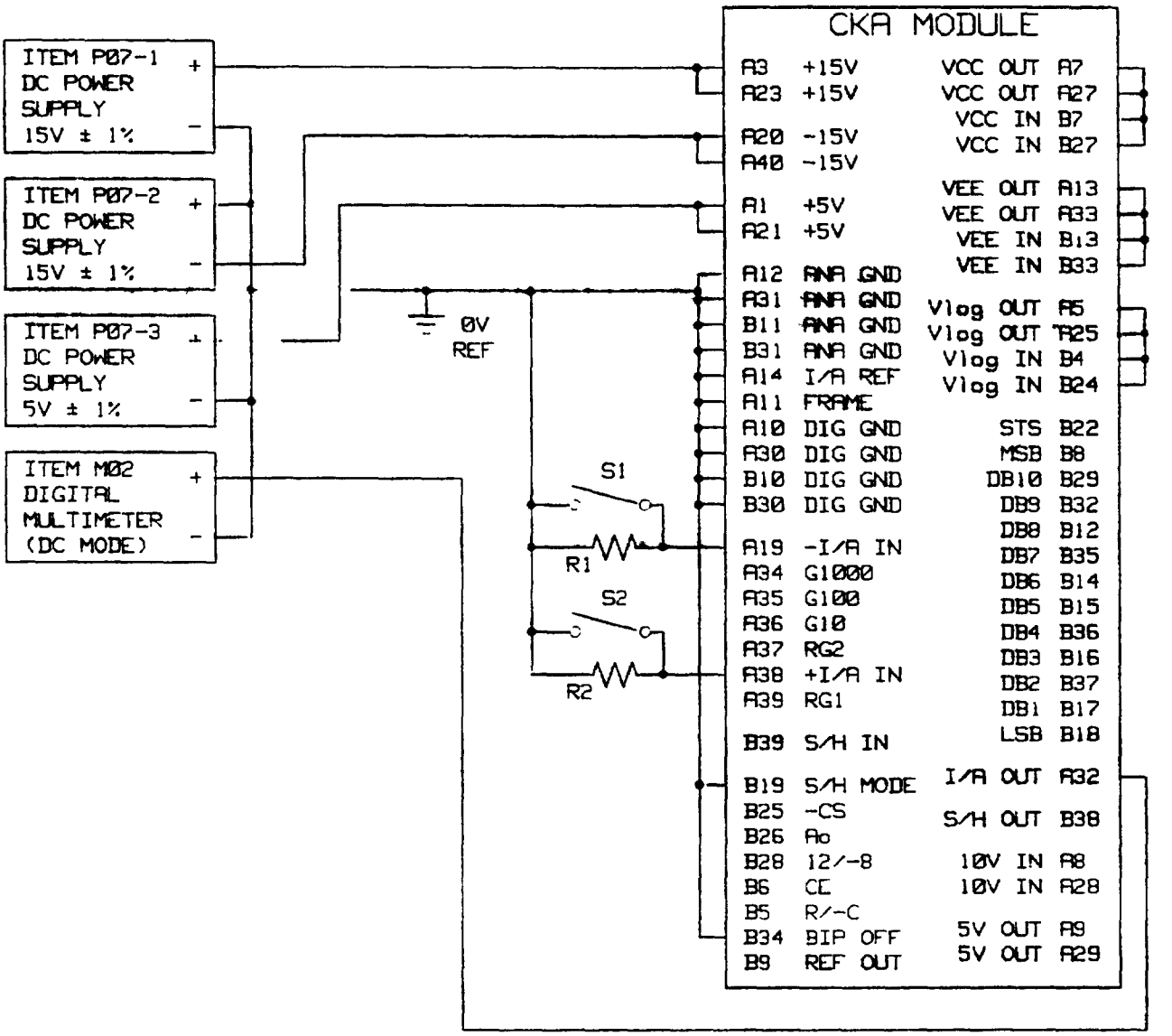
FIGURE 34. I/A offset voltage test circuit for table XXIX.

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TABLE XXX. I/A input bias and offset current test.

Test cond	Parameter	Switch positions		Measurement/ computation	25 ± 5°C Initial limits		Units	Fig
		S1	S2		Min	Max		
		Closed	Closed	#1			V	35
		Closed	Open	#2			V	35
		Open	Closed	#3			V	35
HH1	Input bias current (+)			HH1 = (#2-#1)/10^8	-50 5	50 5	nA	35
HH2	Input bias current (-)			HH2 = (#1-#3)/10^8	-50.5	50.5	nA	35
HH3	Input offset current			HH3 = HH2 - HH1	-35.5	35 5	nA	35

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NOTES:

1. S1 = S2 = SPST switch.

2. R1 = R2 = 100 kOhm ± 1 percent, 1/20 W.

FIGURE 35. I/A input bias and offset current test circuit for table XXX.

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TABLE XXXI I/A power supply sensitivity test

Test cond	Parameter	Supply voltages (V) ± 0.15 V		Connect A to pin	Measurement/ computation	25 ± 5°C Initial limits		Units	Fig
		P07-1	P07-2			Min	Max		
		16.5	16.5	Open	#1			V	36
		13.5	13.5	Open	#2			V	36
JJ1	PSS (G=1)				JJ1 = 20*log|6/(#1-#2)|	75	INF	dBV	36
		16.5	16.5	A36	#3			V	36
		13.5	13.5	A36	#4			V	36
JJ2	PSS (G=10)				JJ2 = 20*log|60/(#3-#4)|	95	INF	dBV	36
		16.5	16.5	A35	#5			V	36
		13.5	13.5	A35	#6			V	36
JJ3	PSS (G=100)				JJ3 = 20*log|600/(#5-#6)|	105	INF	dBV	36
		16.5	16.5	A34	#7			V	36
		13.5	13.5	A34	#8			V	36
JJ4	PSS (G=1000)				JJ4 = 20*log|6000/(#7-#8)|	110	INF	dBV	36

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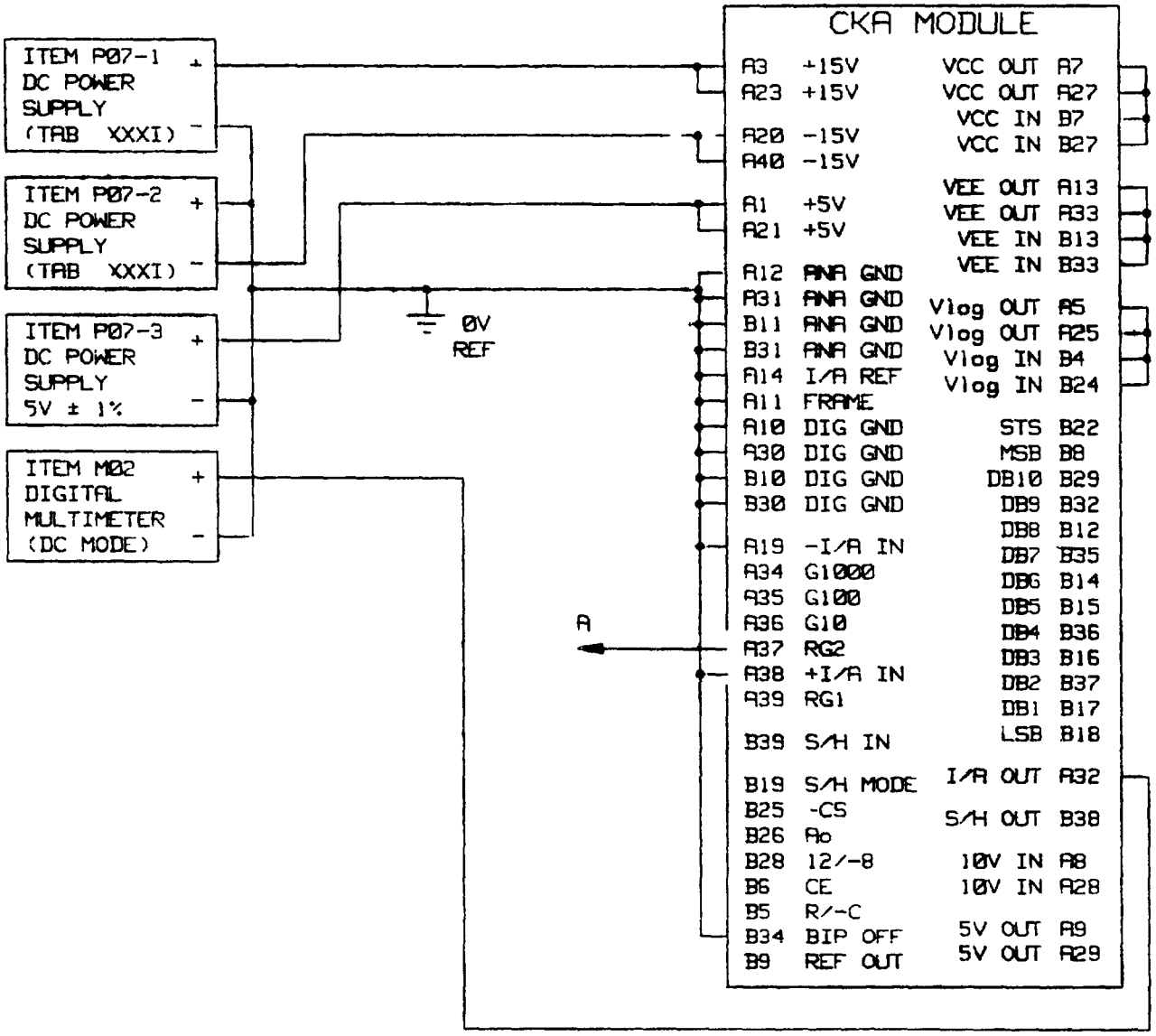


FIGURE 36. I/A power supply sensitivity test circuit for table XXXI.

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TABLE XXXII. I/A common mode rejection ratio test.

Test cond	Parameter	Connect			Measurement/ computation	25 ± 5°C Initial limits		Units	Fig
		A to pin	B to pin	C to pin		Min	Max		
		A19	A31	Open	#1			V	37
		A31	A19	Open	#2			V	37
kk1	CMRR (G=1)				kk1 = 20*log|20/(#1-#2)|	70	INF	dBV	37
		A19	A31	A36	#3			V	37
		A31	A19	A36	#4			V	37
kk2	CMRR (G=10)				kk2 = 20*log|200/(#3-#4)|	90	INF	dBV	37
		A19	A31	A35	#5			V	37
		A31	A19	A35	#6			V	37
kk3	CMRR (G=100)				kk3 = 20*log|2000/(#5-#6)|	100	INF	dBV	37
		A19	A31	A34	#7			V	37
		A31	A19	A34	#8			V	37
kk4	CMRR (G=1000)				kk4 = 20*log|20000/(#7-#8)|	110	INF	dBV	37

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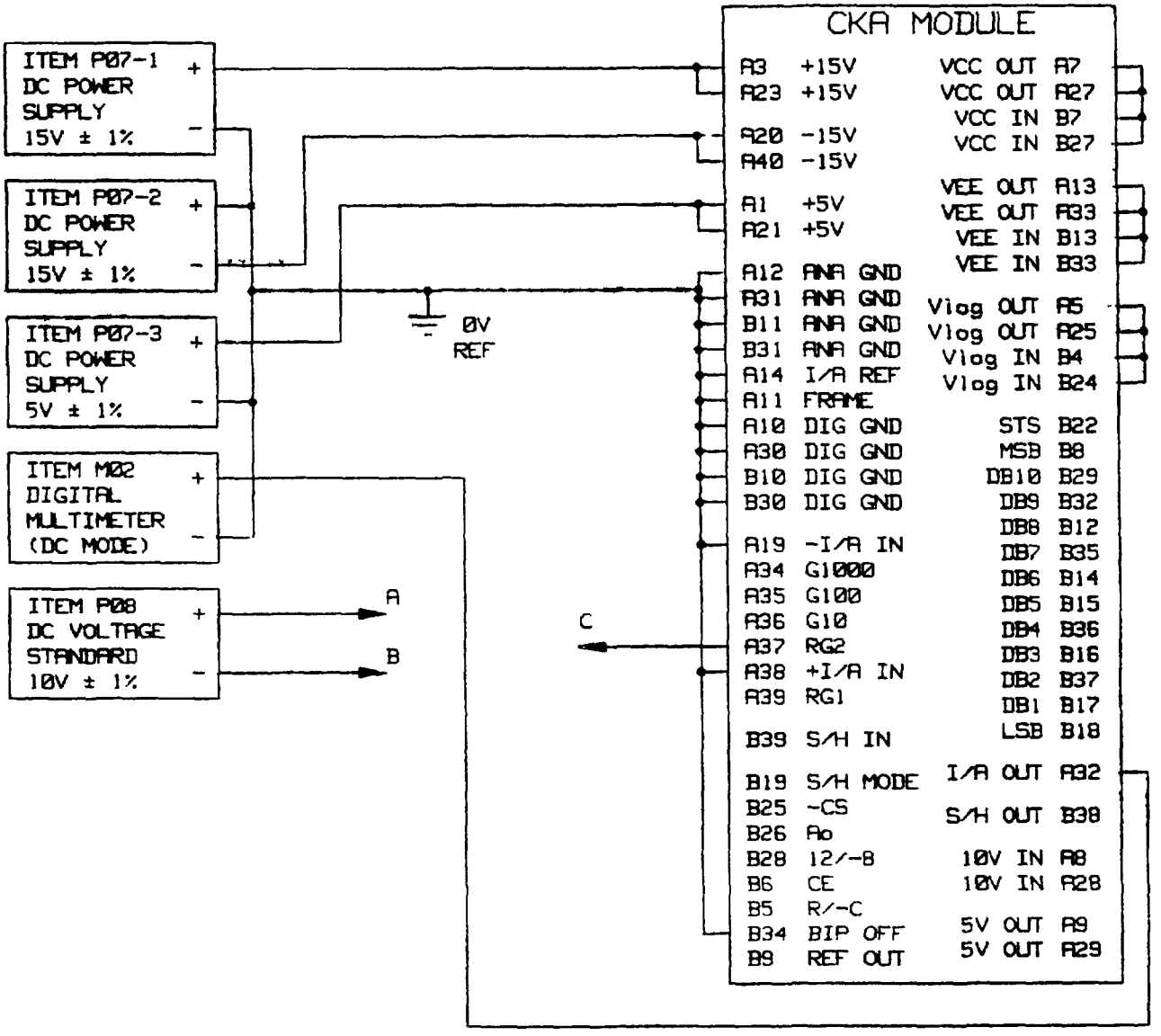


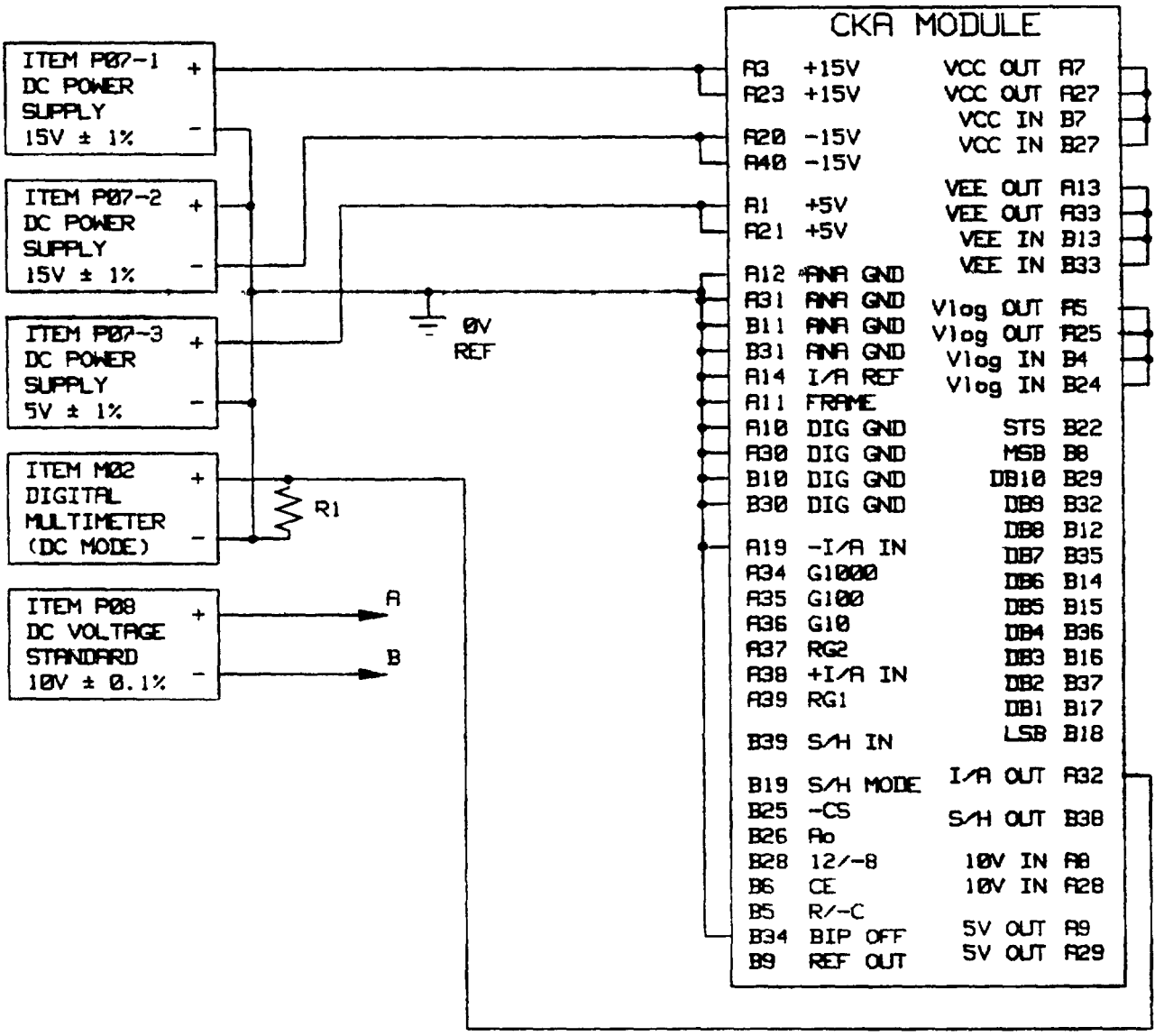
FIGURE 37. I/A common mode rejection ratio test circuit for table XXXII.

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TABLE XXXIII. I/A output swing test.

Test cond	Parameter	Connect		25 ± 5°C Initial limits		Units	Fig
		A to pin	B to pin	Min	Max		
LL1	Positive output swing	A38	A31	9 5	11.0	V	38
LL2	Negative output swing	A31	A38	-11 0	-9.5	V	38

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NOTE:

1. R1 = 2 kOhm ± 1 percent, 1/4 W.

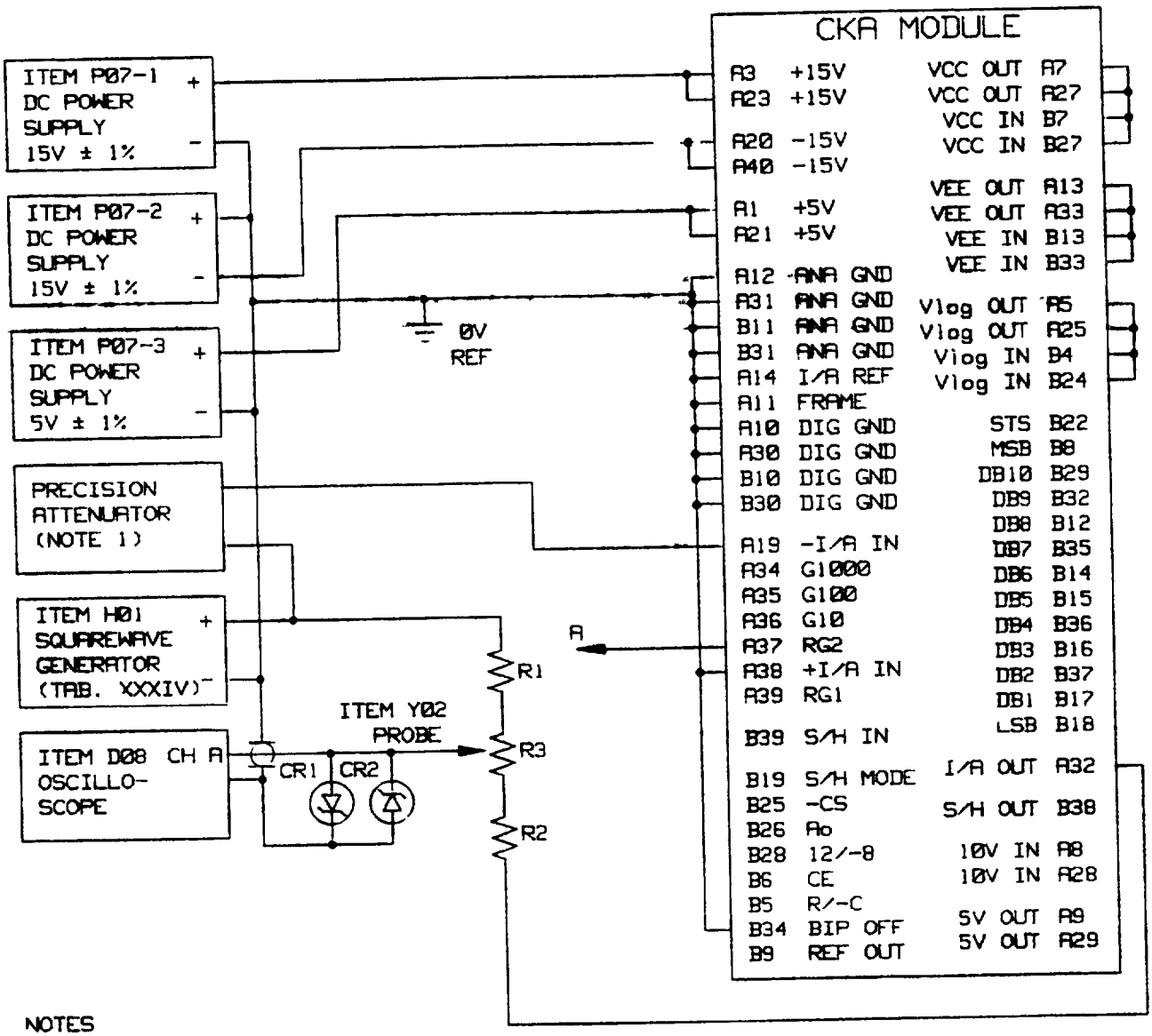
FIGURE 38. I/A output swing test circuit for table XXXIII.

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TABLE XXXIV. I/A settling time test.

Test cond	Parameter	Connect A to pin	Precision attenuator setting (g)	Item H01 voltage (Vpp) ± 5 percent	25 ± 5°C Initial limits		Units	Figs
					Min	Max		
MM1	Settling time (G = 1)	--	1.0	20	0	30	uS	39,40
MM2	Settling time (G = 10)	A36	0.1	20	0	30	uS	39,40
MM3	Settling time (G = 100)	A35	0.01	20	0	30	uS	39,40
MM4	Settling time (G = 1000)	A34	0.001	20	0	150	uS	39,40
MM5	Settling time (G = 1)	--	1.0	8	0	22	uS	39,40

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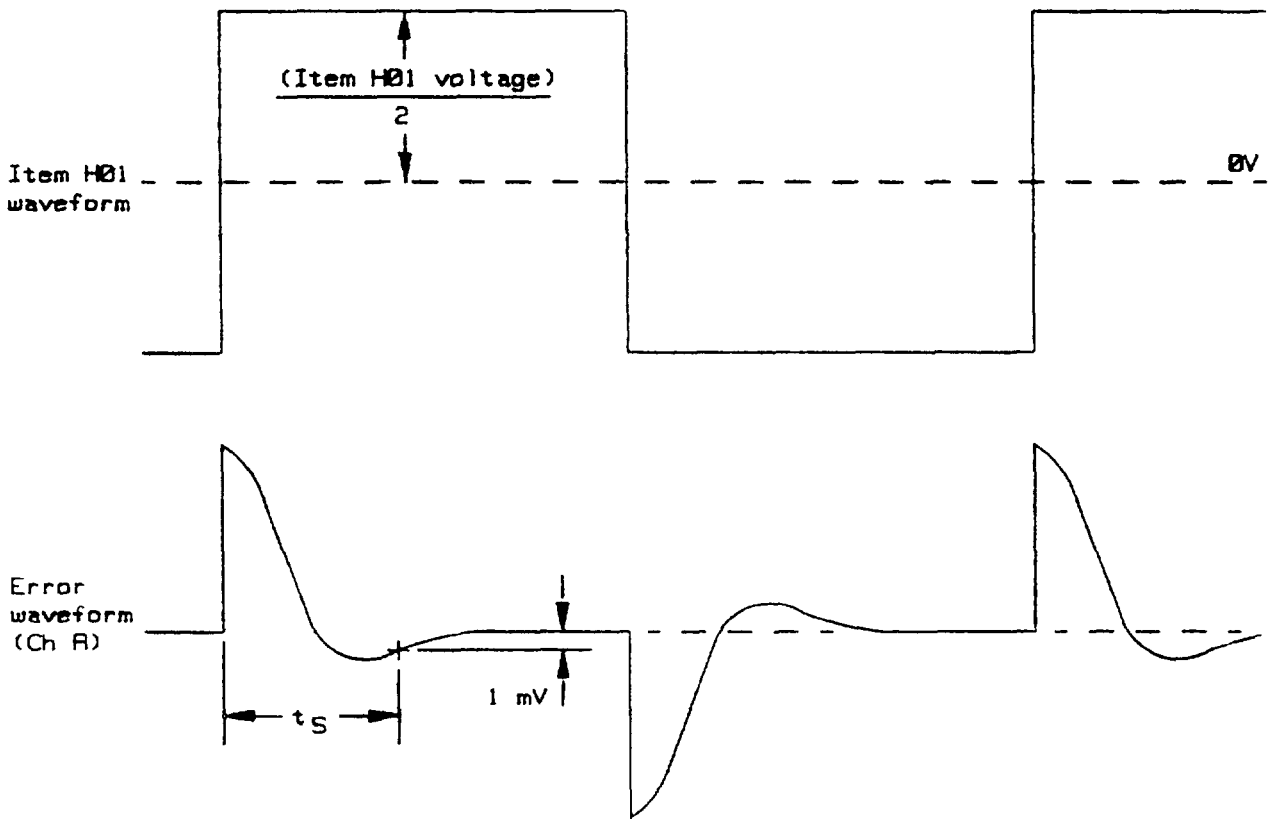


NOTES

1. Attenuator accuracy: g = 1.0, 0.1, 0.01, 0.001 (all settings ± 1 percent).
2. R1 = R2 = 10 kOhm ± 1 percent, 1/4 W.
3. R3 = potentiometer, 1 kOhm ± 5 percent.
4. CR1 = CR2 = Schottky diodes.
5. For procedure, see figure 40

FIGURE 39. I/A settling time test circuit for table XXXIV.

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NOTES

1. Item H01 setup:
   Frequency = 8.3 kHz ± 1 percent (MM1,MM2,MM3,MM5).
   Frequency = 1.6 kHz ± 1 percent (MM4).
   Amplitude as specified in table XXXIV.
   Rise time = fall time = 100 nS max

2 Adjust R3 so that the tails of the error on figure 40 align.

3. Determine the time it takes for the error waveform to come to within 1 mV of its final value, as shown on figure 40.

4. Settling time (tS) is defined as the time required for the output to reach and maintain within 0 01 percent of its final value (0 025 percent for MM5)

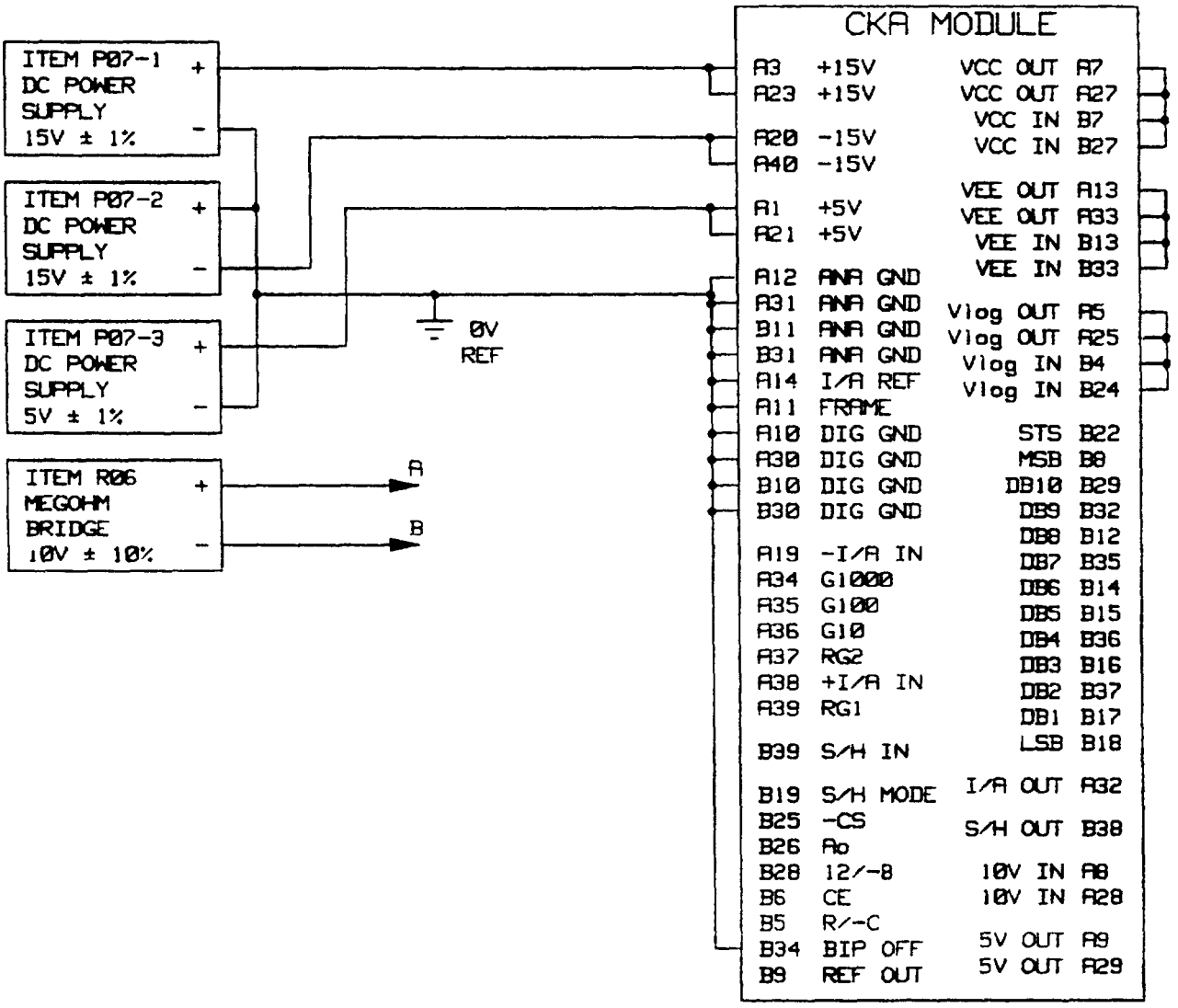
FIGURE 40. I/A settling time waveforms for table XXXIV.

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TABLE XXXV. I/A input resistance test.

Test cond	Parameter	Connect		25 ± 5°C Initial limits		Units	Fig
		A to pin	B to pin	Min	Max		
NN1	Common mode input resistance (-)	A19	A31	10	INF	MOhm	41
NN2	Common mode input resistance (+)	A38	A31	10	INF	MOhm	41
NN3	Difference mode input resistance	A38	A19	10	INF	MOhm	41

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NOTE:

1 Caution: The output of the megohm bridge shall not exceed 11 V.

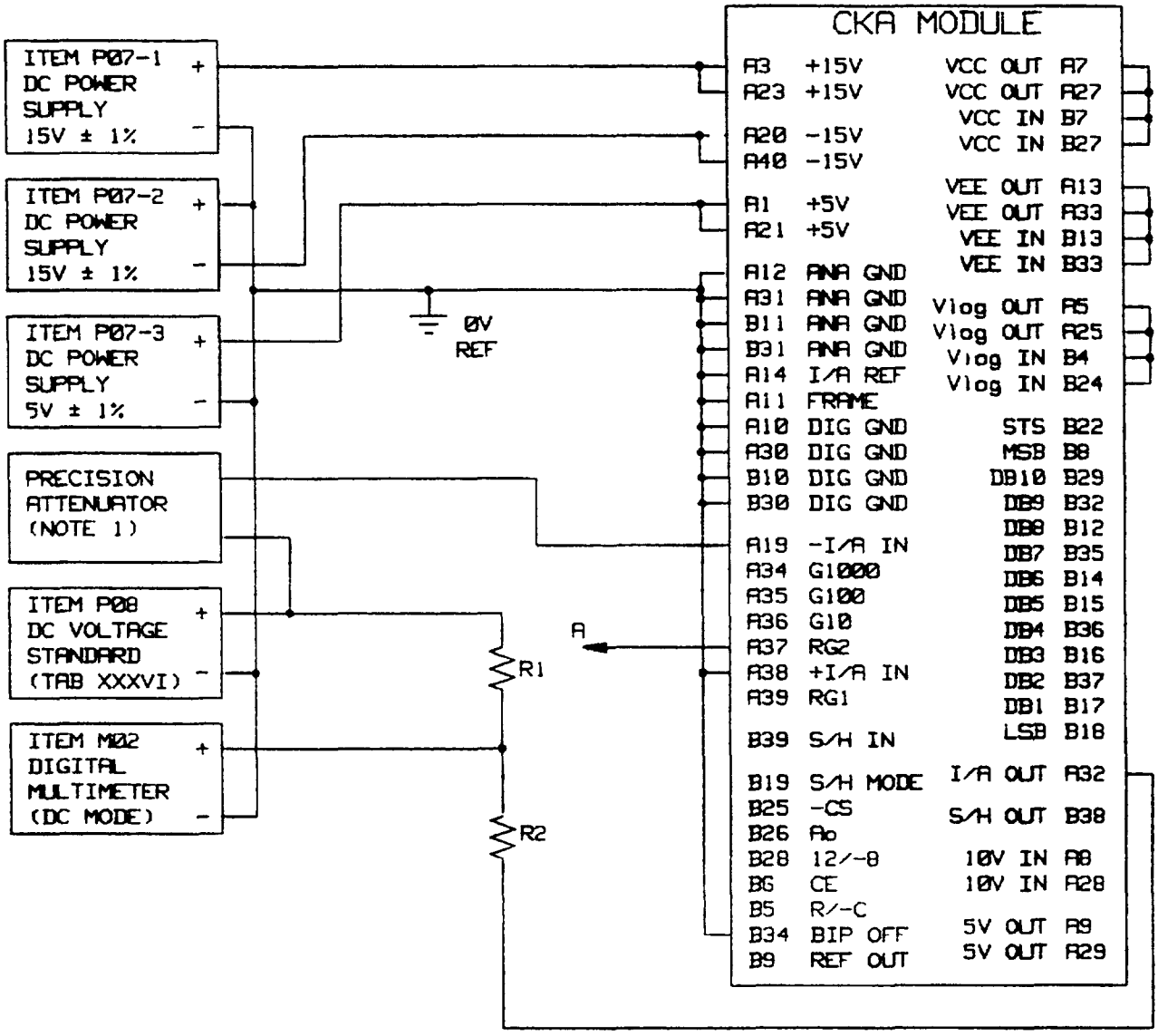
FIGURE 41. I/A input resistance test circuit for table XXXV.

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TABLE XXXVI. I/A gain accuracy test.

Test cond	Parameter	Precision attenuator setting (g)	Item P08 voltage (V) ± 0.1 mV	Connect A to pin	Measurement/ computation	25 ± 5°C Initial limits		Units	Fig
						Min	Max		
		1.0	0	Open	#1			V	42
		1 0	10	Open	#2			V	42
PP1	Gain				PP1 = $\frac{(10-2*(\#2-\#1))}{10}$	0.9985	1.0015	V/V	42
		0.1	0	A36	#3			V	42
		0.1	10	A36	#4			V	42
PP2	Gain				PP2 = $\frac{(10-2*(\#4-\#3))}{1\ 0}$	9.975	10.025	V/V	42
		0.01	0	A35	#5			V	42
		0.01	10	A35	#6			V	42
PP3	Gain				PP3 = $\frac{(10-2*(\#6-\#5))}{0.1}$	99.5	100.5	V/V	42
		0.001	0	A34	#7			V	42
		0.001	10	A34	#8			V	42
PP4	Gain				PP4 = $\frac{(10-2*(\#8-\#7))}{0.01}$	980.0	1020.0	V/V	42

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NOTES

1. Attenuator accuracy: g = 1.0, 0.1, 0.01, 0.001 (all settings ± 1 percent).
2. R1 = R2 = 49.9 kOhm matched to ± 0.005 percent, 1/4 W

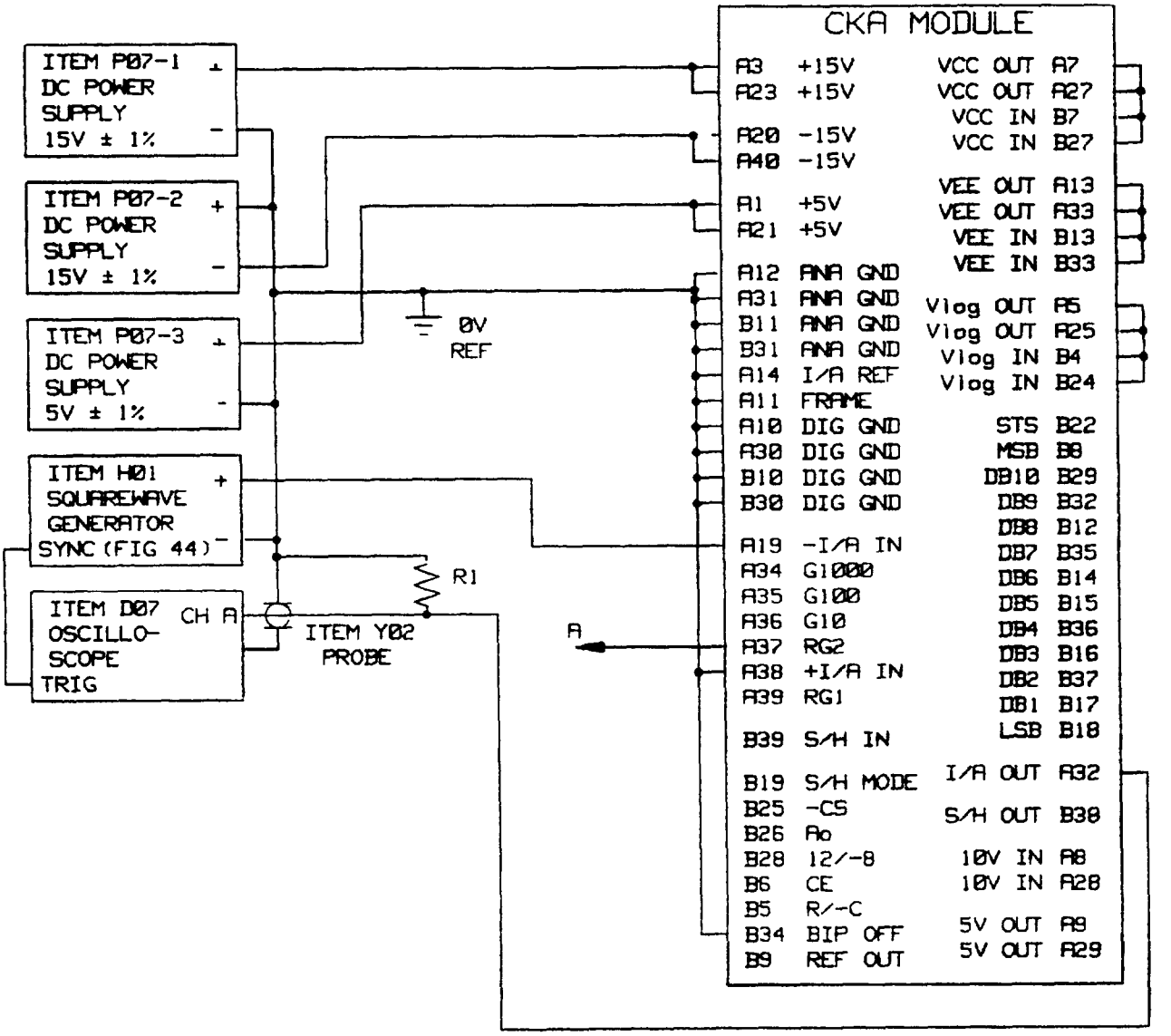
FIGURE 42. I/A gain accuracy test circuit for table XXXVI.

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TABLE XXXVII. I/A slew rate test.

Test cond	Parameter	Connect A to pin	Item H01 voltage (Vpp) ± 5 percent	25 ± 5°C Initial limits		Units	Figs
				Min	Max		
QQ1	Slew rate (G = 1)	--	20.0	3.0	INF	V/uS	43,44
QQ2	Slew rate (G = 10)	R36	2.0	↓	↓	↓	↓
QQ3	Slew rate (G = 100)	R35	0.2	↓	↓	↓	↓
QQ4	Slew rate (G = 1000)	R34	0.02	0.5	INF	V/uS	43,44

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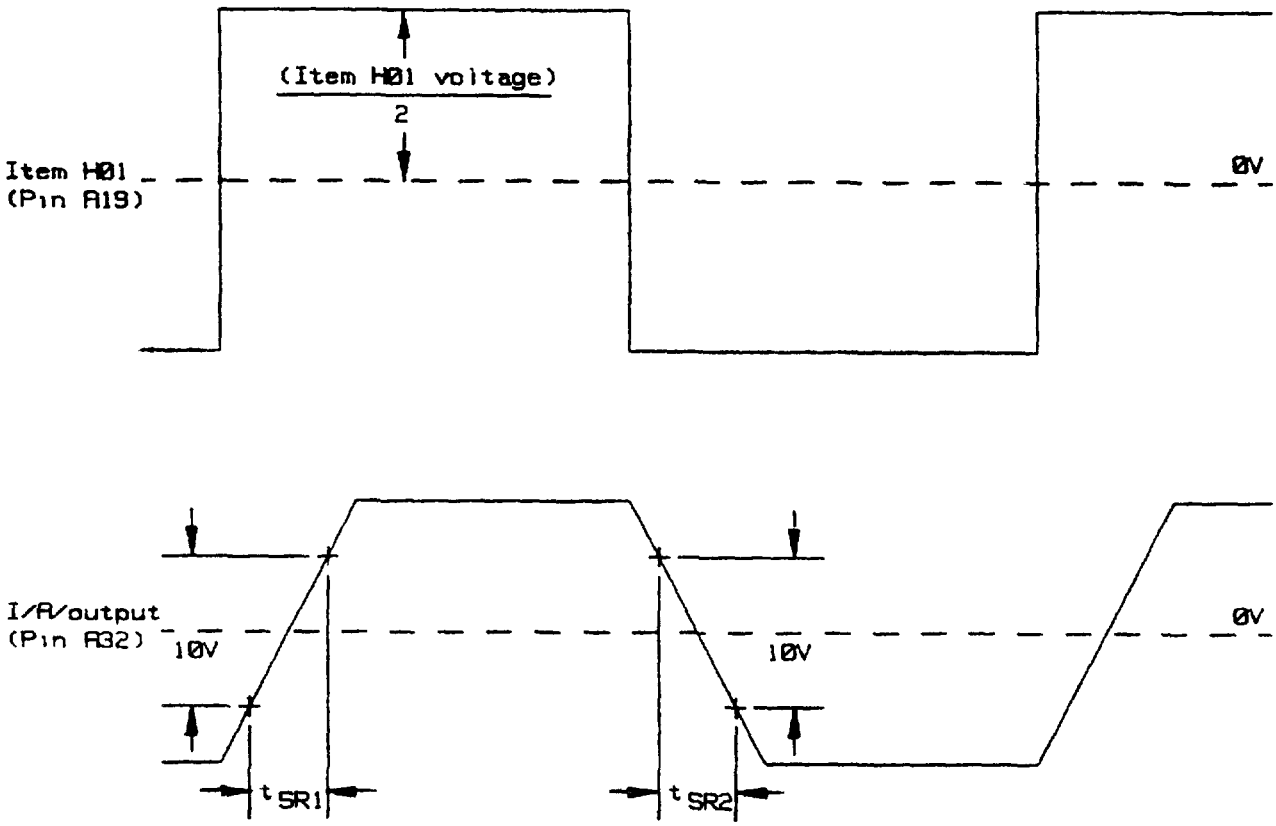


NOTE:

1. R1 = 2 kOhm ± 1 percent, 1/4 W.

FIGURE 43. I/A slew rate test circuit for table XXXVII.

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NOTES

1. Item H01 setup:
   Frequency = 60 kHz ± 1 percent.
   Amplitude as specified in table XXXVII.
   Rise time = fall time = 100 nS max.

2. Measure t_{SR1} and t_{SR2}. Both shall pass the limits in table XXXVII when calculated as shown below.

3. Slew rate = $\frac{10}{t_{SR}\ (\text{in uS})}$

FIGURE 44. I/A slew rate waveforms for table XXXVII.

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